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Foundations of Biolinguistics

A Primer for Cognitive Approaches

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I dedicate this book to Big Boy Andreas, an enthusiastic biolinguist in training, his cousin Baby Orestes, who impresses us all non-verbally, and his beautiful sister Béb  Alik  Maria, who's too small to care yet.

And to Lance, Tyler, and Zo  who learned the hard way that mastering their mother's tongue in a different country is certainly no child play.

PREFACE

Pivot was one of the two positions I played throughout the 1980s. I have since moved on to play in academia. My understanding of a pivot as an academic product is a key-point treatment of a concise subject matter. It could be expanded in the form of an introductory textbook, for example, or a short research monograph. It could thus make for a tremendous course guide. A pivot might also be the ideal platform to report on a series of related experimental studies or present a novel line of research. This book is neither, hence none of that. But it builds on and relates to both.

No, the present book is neither a textbook nor a research monograph. It is intended to be a ‘primer’, conceived as an introductory overview of selected issues in a growing field of research: *biolinguistics*. While I try to portray both sides of the apparent divide, I have been trained in generative grammar basically since (a week before) my first semester as an undergraduate student and worked within the developments of this framework ever since. So, this is where my second position as a former team handball player comes in: goalkeeper. As such I aim to defend a broadly understood view of biolinguistics, so that we can join forces in the offense and then tackle the many pertinent issues with, rather than against, one another.

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I was in the fortunate position of having spent a whole sabbatical year in Australia, made possible by the University of Cyprus—not at all a matter of course in the current climate, which I thus appreciate even more. There I mostly worked from our wonderful home but was lucky to have all my books in close proximity in the CBD of Adelaide at the University of South Australia, where I was a year-long visitor, and then have been appointed Honorary Adjunct Professor at the University of Adelaide (which both recently merged into Adelaide University).

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CONTENTS

1	Cognitive Approaches to Biolinguistics	1
1.1	<i>On the Biological Foundations of Language</i>	3
1.2	<i>Nativism in Two Minutes</i>	8
1.3	<i>Emergentism in Two Minutes</i>	9
1.4	<i>A Short Preview</i>	10
2	Lenneberg's Dream	13
2.1	<i>Five Foundational Questions</i>	14
2.2	<i>Four Questions from Ethology</i>	16
2.3	<i>Three Factors in Language Design</i>	18
2.4	<i>Two Cognitive Approaches to Linguistics</i>	20
2.4.1	<i>From the Strongest CLN View...</i>	20
2.4.2	<i>...to a Weaker CLB-Compatibility</i>	23
2.4.3	<i>Language and Cognitive Capacities</i>	25
2.5	<i>One Aim: Biolinguistics</i>	28
3	Humboldt's Problem	31
3.1	<i>The Nature of Linguistic Competence</i>	32
3.2	<i>Knowledge of Language</i>	34
3.3	<i>UG and the Language Faculty</i>	36
3.4	<i>Competence and Performance</i>	42
3.5	<i>I-Language and E-Language</i>	44
3.6	<i>Emergentist Perspectives</i>	45
3.7	<i>Summary</i>	49
4	Plato's Problem	51
4.1	<i>Language Acquisition and Development</i>	52
4.2	<i>The Logical Problem of Language Acquisition</i>	53
4.3	<i>The Genetic Endowment for Language</i>	59
4.4	<i>Developmental and Neural Evidence</i>	63
4.5	<i>Summary</i>	68

5	Descartes's Problem	71
5.1	<i>Linguistic Performance and Processing</i>	72
5.2	<i>The Production Problem</i>	75
5.3	<i>From Transformations to Merge</i>	78
5.4	<i>From Competence to Performance</i>	80
5.5	<i>Summary</i>	83
6	Broca's Problem	85
6.1	<i>Neural and Cognitive Bases of Language</i>	86
6.2	<i>The Language Network</i>	87
6.3	<i>Language and Thought</i>	93
6.4	<i>The Incommensurability Issue</i>	96
6.5	<i>Summary</i>	99
7	Wallace/Darwin's Problem	101
7.1	<i>Evolution of the Human Language Faculty</i>	102
7.2	<i>Language Evolution</i>	104
7.3	<i>Third Factors</i>	106
7.4	<i>Genetics and the Language-Ready Brain</i>	110
7.5	<i>The Mechanisms of Evolution</i>	112
7.6	<i>Summary</i>	115
8	Outlook	117
8.1	<i>Recap</i>	117
8.2	<i>Terminological Conundrums</i>	119
8.3	<i>Biolinguistics and Beyond</i>	121
8.4	<i>Where to from Here?</i>	124
	Bibliography	127
	Index	153

LIST OF TABLES

Table 1.1	Five foundational questions of biolinguistics (overview)	3
Table 2.1	Five foundational questions of biolinguistics (extended)	15
Table 2.2	Four fundamental questions of ethology	17
Table 2.3	Three factors of language design	19
Table 2.4	CLN–CLB commitment spectrum	23
Table 2.5	Fundamental hypotheses of CLB	24
Table 2.6	Tinbergen’s four questions for linguistics	27

Cognitive Approaches to Bilingualism

Abstract Bilingualism, to define it in a single phrase, is the study of language that takes its biological underpinnings seriously. This short book outlines the basic tenets of a bilingual approach framed in the form of five ‘foundational questions’. While often associated with generative investigations of language, bilingualism is not—and should not be—just another term for generative grammar; bilingual approaches to the scientific study of human language will thus be put in a richer context by inspecting and dissecting the foundational questions, with particular reference to cognitive approaches from generative and non-generative perspectives alike. The organizing distinction developed here is between ‘cognitive linguistics in the narrow sense’ (nativist, hence largely generative) and ‘cognitive linguistics in the broad sense’ (usage-based or emergentist, i.e. definitely non-generative). As such, the book can be seen as a primer for cognitive at-large approaches to the study of human language on bilingual foundations. This first chapter presents the basic notion and the dynamic field of bilingualism somewhat historically, provides relevant theoretical and conceptual backgrounds, introduces the five foundational questions and their eponymous problems, and sets the stage for what is to come.

Keywords Bilingualism • Cognitive linguistics (narrow/broad sense) • Nativism • Generative grammar • Emergentism • Usage-based

Bilingualism is the scientific study of human language that takes its biological underpinnings seriously. This or some general definition like it is a standard way to introduce the field. But what is it actually, and what is it not—a theory, an ideology, a framework?

Biolinguistics is none of these, of course, but has been mistaken, misused, and misunderstood, often even on purpose or with implicit intent. Much has been written about the origins of the term ‘biolinguistics’ and its substance, unabashedly generative by some and equally emphatically against by others. This is not a matter of claiming or re-claiming anything, with book-length treatments as diverse as ranging from Meader and Muyskens (1950) to Lenneberg (1967), and from Givón (1999) or Jenkins (2000) all the way to Pennisi and Falzone (2016) or Uriagereka (2018), for example; for relevant edited collections, see e.g. Boeckx & Grohmann (2013) or Benítez-Burraco et al. (2025).

The present book makes the attempt to selectively identify points of consensus, or at least mutual understanding, among researchers, perhaps even to make biolinguistics palatable to a larger group of scholars interested in language, cognition, and the mind–brain. In the spirit of full disclosure, I do this with a generative bias, not in small part due to the fact that I am a co-author of a widely used textbook on minimalist syntax (Hornstein et al. 2005)—yet any shortcomings, misinterpretation, or even factual errors committed along the way are based entirely on ignorance and not on malice.

As I said in the Preface, this book sets out to (re)visit biolinguistic research objectives from generative and non-generative perspectives. There are others who made similar observations. I am thinking of three in particular. De Vos (2015) lays out the nativist generative position on language variation by paying due respect to non-nativist, so-called ‘cognitive’ or ‘usage-based’, approaches; he specifically addresses the first three of the five foundational questions which build a frame around the present book. Pleyer and Hartmann (2019) also aim at reconciling generative (‘minimalist’) and usage-based (or generally, emergentist) approaches; their emphasis lies on the last two of the five foundational questions. Grohmann and Kambanaros (2021), finally, and the updated Grohmann (2027) survey all five questions as representative of a biolinguistic enterprise in handbooks on cognitive linguistics; these two chapters lay the foundations of this pivot. The structure follows Jenkins’ (2000) cue, with a chapter dedicated to each question. The foundational questions themselves were expanded from our original exposition (Boeckx & Grohmann 2007), were further discussed in Boeckx (2010), and have received considerable attention since.

1.1 ON THE BIOLOGICAL FOUNDATIONS OF LANGUAGE

Biolinguistics, in its most intuitive connotation, explores what Eric Lenneberg termed the title-giving *biological foundations of language* (Lenneberg 1967). As that book appeared around the time when generative grammar had gained a firm standing in the scientific study of language, it can be seen as an ideal companion to the research agenda of transformational grammar (Chomsky 1965). It also connected well to the philosophical footing of generative grammar (e.g., Chomsky 1966). Chap. 2 builds on this relationship in more depth (see also Chap. 5).

A provocative framing sets the stakes: Fodor and Piattelli-Palmarini (2010) argue that Darwin’s theory of natural selection and Skinner’s operant conditioning are “virtually identical” (p. 3)—both propose the same selection mechanism for the same kind of function. If Skinnerian learning is now universally rejected, their argument goes, why should Darwinian selection be accepted as an explanation for every biological trait, including language? Whatever one makes of this argument (see especially Chap. 7), it sharpens the central question of both nativist and emergentist approaches: What counts as a satisfactory explanation of why language has the properties it has? By Chap. 8 we should be better equipped to find good answers.

Throughout this book, I will introduce and further expound on the biolinguistic research agenda via five *foundational questions*, which go back to decades of Chomsky’s writings (cf. Chomsky 1986: 3, 33–44; 1988: 3, 166–170; 1995: 14–15 [= Chomsky & Lasnik 1993: 508]). They can be formulated as in Table 1.1 (Grohmann 2027; cf. Jenkins 2000: 228, Boeckx & Grohmann 2007: 1), which shows the eponymous problem next to each of the five foundational questions.

Table 1.1 Five foundational questions of biolinguistics (overview)

<i>Foundational question</i>	<i>Eponymous problem</i>
1. What is knowledge of language?	Humboldt’s problem
2. How is that knowledge acquired?	Plato’s problem
3. How is that knowledge put to use?	Descartes’s problem
4. How is that knowledge implemented in the brain?	Broca’s problem
5. How did that knowledge emerge in the species?	Wallace/Darwin’s problem

Each foundational question relates to a ‘problem’ denoting its intellectual origin or connection (as discussed in e.g. Boeckx 2009: 44–45, Leivada 2012: 35–36, or Grohmann & Kambanaros 2021: 690). As a rough guide, consider the language-related aspects of Humboldt’s problem to roughly correspond to philosophy of language (with argumentation from formal syntax), of Plato’s problem to language acquisition (itself built on theoretical linguistics), of Descartes’s problem to psycholinguistics on theoretical grounds (including pragmatics), of Broca’s problem to neuro- and patholinguistics (as well as at least cognitive neurobiology), and of Wallace/Darwin’s problem to evolutionary linguistics (though this may not be a familiar, traditional sub-discipline, at least not in generative grammar). As Chap. 2 will show, the five questions have a striking structural analogue in Tinbergen’s (1963) four fundamental questions of ethology, a parallel that situates biolinguistics even tighter within the broader framework of biological inquiry.

This book adopts a terminological distinction that aims to cut across the generative/non-generative divide without presupposing it. Wen et al. (2014) distinguish between what they call ‘macro-cognitive’ and ‘micro-cognitive’ linguistics: approaches that treat language as a mental phenomenon, prioritizing hierarchical structure and the autonomy of syntax, versus approaches that emphasize meaning, context, and the grounding of language in general cognition and use. Inspired by Hauser et al.’s (2002) understanding of the faculty of language and Ménéndez-Giró’s (2025b) extension to biolinguistics (cf. Chap. 2), I refer instead to macro-cognitive as *cognitive linguistics in the narrow sense (CLN)* and to micro-cognitive as *cognitive linguistics in the broad sense (CLB)*. CLN thus encompasses the generative tradition in its various instantiations insofar as all of these treat language as a cognitive object internal to the mind–brain with its own dedicated architecture. CLB, by contrast, designates the tradition more commonly referred to in the field as ‘Cognitive Linguistics’: construction grammar, cognitive grammar, usage-based approaches, and related frameworks. The distinction is not a judgment of scope or importance; both CLN and CLB are cognitive, both are scientific, and both will be taken seriously here. What differs is the theoretical scaffolding each brings to the five questions.

These two broad orientations, *nativism* (as instantiated in generative grammar) and usage-based *emergentism* (as instantiated in cognitive grammar and related frameworks), have defined the landscape of theoretical linguistics for decades. They differ profoundly on what language

is, how it is acquired, and what kind of explanation counts as satisfactory. On the emergentist side, this is not merely a difference of explanatory taste but an empirical wager: Evans and Levinson (2009) argue that linguistic diversity runs so deep, at every level of structure, that the substantive universals *Universal Grammar (UG)* is meant to encode largely dissolve, leaving cross-linguistic variation, rather than an innate common core, as the fundamental fact a theory of language must explain. Yet both take human language seriously as an object of scientific inquiry, and both have made genuine progress on at least some of the five foundational questions. The chapters that follow are intended to trace that progress, and some of the disagreement, in turn.

In a nutshell, the first three questions in Table 1.1 were directly addressed, though not all equally explicitly (see also e.g. Jenkins 2013 and McGilvray 2013 for historical and philosophical retrospectives): What is the nature of language in the human organism, what are the mechanisms underlying language acquisition, and what are the relevant properties of language? The core of the familiar answers is denied wholesale in emergentist approaches: a domain-specific faculty in the mind–brain, the Language Acquisition Device, and a domain-specific computational system encapsulating the autonomy of syntax, to be brief. These are not merely technical disagreements about linguistic analysis; they reflect fundamentally different answers to Humboldt’s problem. For CLB, linguistic knowledge is not a discrete mental state that children arrive at by triggering an innate grammar but an open-ended repertoire assembled incrementally through communicative experience. This difference ramifies through all questions, as the following chapters will show.

It may be worth considering the questions themselves in some more detail as the guiding inquiries for research into language as a species-unique capacity. Perhaps a slight reformulation without the accompanying rhetoric, which arguably contributed to the polemics that Taylor (2007) dissected, allows different perspectives or at least appreciation. Stripped of their theoretical affiliations, the five questions reduce to something any linguist, regardless of framework, might recognize as legitimate: What does it mean to know a language? How do children come to know it? How do speakers deploy that knowledge in real time? What neural mechanisms support it? And how did the capacity for language come to exist in our species at all? Framed this way, the questions are not the exclusive property of any single tradition—they are the shared agenda of the scientific study of language.

As Boeckx and Grohmann (2007: 2–3) recap, the term ‘biolinguistics’ first appears as part of a book title published well before the advent of generative grammar. The book advocates (as the authors put it) a modern science of biolinguistics, whose practitioners “look upon language study [...] as a natural science, and hence regard language as an integrated group of biological processes [...] seek[ing] an explanation of all language phenomena in the functional integration of tissue and environment” (Meador & Muyskens 1950: 9). Some 25 years later ‘biolinguistics’ resurfaces as part of a report on an interdisciplinary meeting on language and biology (Piattelli-Palmarini 1974), attended by the microbiologist Salvador Luria and Noam Chomsky. It was organized by Massimo Piattelli-Palmarini under the sponsorship of the Royaumont Center for a Science of Man (see also Chomsky 2009).

Shortly after, Lyle Jenkins and colleagues attempted to launch a journal bringing together language and biology. They even received support from pre-eminent biologists (documented by three letters reproduced in the appendix of Boeckx & Grohmann 2007). Even though the journal never materialized (see Jenkins 2000), the issues discussed at the time did not disappear. As a matter of fact, all these topics, many of which incidentally anticipated in Lenneberg (1967), came back on the agenda of linguists and other cognitive scientists. The journal *Biolinguistics* eventually did materialize, launched in 2007 as an open-access, peer-reviewed venue for exactly the kind of interdisciplinary work the field had long called for (see Boeckx & Grohmann 2007 for details).

Chomsky has always been very clear that the study of language is essentially a part of human biology whereby linguistics explores the eponymous ‘biological foundations of language’ (Lenneberg 1967); see e.g. Chomsky (1976) for the biological basis of language capacities. The consequence of this position is that linguistics becomes the study of an internal property of the human organism, which Chomsky (1986) identifies as I-language (internal, individual, intensional), as distinct from E-language (external, socially shared, extensional). This internalist turn is what gives biolinguistics its distinctly biological flavor: It is not language as a cultural artifact or social institution that is under investigation (themselves entirely appropriate areas of investigation), but language as a real state of the mind–brain.

Likewise, there is a strong cognitive basis to language, made most explicit in Fodor’s (1983) *modularity of mind*: The mind is organized into informationally encapsulated, domain-specific input systems (i.e.

modules), of which language is a paradigm case. While generative grammar and Fodorian modularity are not identical commitments, they are natural allies: Both treat language as a bounded cognitive system with its own architecture, different from general reasoning and perception. This convergence strengthens the biolinguistic case that language is not merely a cultural overlay on general intelligence but a genuine biological specialization (see also Chaps. 2 and 3).

As such, “[l]inguistics is simply that part of psychology that is concerned with [...] the cognitive structures that are employed in speaking and understanding” (Chomsky 1975: 160) as it is of the cognitive sciences more generally. Biolinguistics occupies a privileged position within this interdisciplinary landscape. It is where the formal study of linguistic structure meets the empirical study of the mind–brain, where acquisition data bears on evolutionary hypotheses, and where clinical observations, from developmental language disorder to aphasia, constrain theoretical models. The field’s breadth is not a weakness but a feature: The five foundational questions are deliberately chosen to map onto this full range of inquiry, from philosophy and formal syntax at one end to neurobiology and evolutionary anthropology at the other.

Indeed, generative grammar has often been referred to as the “second cognitive revolution” (among many others, Pinker 1994: 21, Jenkins 2000: 2, Boeckx & Piattelli-Palmarini 2005: 448, Hinzen et al. 2011: 274, Roberts & Watumull 2015: 211, also Baars 1986, Chomsky 1992, Miller 2003) with the first cognitive revolution going back to 17th and 18th century philosophers, and ultimately the Cartesian approach (Chomsky 1966). If generative grammar constitutes the second cognitive revolution, then the rise of usage-based and emergentist approaches to language in the 1980s and 1990s might be argued to constitute a third wave—not a rejection of the cognitive turn but a redirection of it, away from formal representation and towards embodied, experience-driven accounts of linguistic structure. The tension between these waves is real, but so is the shared commitment to language as a mental phenomenon. It is this shared commitment that justifies bringing them together under the umbrella of the CLN–CLB distinction suggested here.

And just as cognitive science can be characterized as “the interdisciplinary study of mind and intelligence, embracing philosophy, psychology, artificial intelligence, neuroscience, linguistics, and anthropology” (Thagard 2005: ix; see also Nadel & Piattelli-Palmarini 2003), it is obvious that “[t]he study of language is pertinent to many fields of

inquiry” (Lenneberg 1967: 1). The following takes up that challenge in a specific way: by using the five foundational questions as a common frame through which both CLN and CLB approaches can be assessed, compared, and, where possible, brought into productive dialogue.

1.2 NATIVISM IN TWO MINUTES

Against the background of over seven decades of generative grammar (Chomsky 1955), this pivot cannot do justice to presenting and defending biolinguistics as the field it has grown into. This holds especially for the present attempt that embraces ‘CLN–CLB’ as a working dichotomy, perhaps even as a spectrum. What I can do here is sketch the core commitments that define the nativist position, before revisiting them in more detail through the foundational questions.

The nativist view holds that humans are born with a biologically endowed capacity specifically dedicated to language; a *language faculty* in the mind–brain. This innate endowment, often referred to as UG, constitutes the initial state of the language faculty: the set of principles and properties that every child brings to the task of language acquisition prior to any experience. The logical problem of language acquisition makes this endowment necessary. After all, children acquire their native language rapidly, uniformly, and on the basis of input that is far too impoverished to explain the resulting grammatical competence. This is known as the ‘poverty of the stimulus’ argument (Chomsky 1980), even if not without its critics (Chap. 4).

Our object of inquiry is what Chomsky (1986) termed I-language (Chap. 3): an internal, individual, and intensional system of knowledge. This stands in opposition to E-language, which refers to language as an external, socially shared object. The internalist perspective is what makes generative grammar genuinely psychological and biological: It studies a real property of the human mind–brain, not an abstraction (see also Chap. 6), namely “structures, not strings” (Everaert et al. 2015).

This is the line of reasoning Chomsky has long pursued and which we will unfold in the remainder: the *Basic Property*, the nature of UG, and the biological endowment that underlies both. The Basic Property of human language, in its most prominent formulation (Berwick & Chomsky 2016), is the ability to construct a discrete infinity of hierarchically structured expressions from a finite lexicon (Chaps. 3 and 5)—

an operation identified as *Merge* (Chomsky 1995 *et seq.*, Chomsky et al. 2023, Marcolli et al. 2025). It is this (basic) property, unique to *Homo sapiens* (Chap. 7), that biolinguistics seeks to explain.

1.3 EMERGENTISM IN TWO MINUTES

Obviously, the emergentist, usage-based tradition starts from a fundamentally different premise: Language is not a discrete, domain-specific module; it emerges from the interaction of general cognitive capacities (such as attention, categorization, analogy, pattern recognition, social cognition) operating over repeated communicative experience. There is no need, on this view, to posit an innate, language-specific endowment; the structures of language are shaped by, and ultimately reducible to, the structures of cognition and use.

The intellectual roots of this tradition run deep. Cognitive Grammar (Langacker 1987), one of the foundational frameworks, treats grammar as an inventory of symbolic units (form–meaning pairings) learned through usage and stored as constructions. Construction Grammar (e.g., Goldberg 1995) extends this logic: Even apparently abstract syntactic patterns are constructions, directly meaningful and acquired from input. Tomasello (2003) provides the developmental complement: Children acquire language through intention-reading and pattern-finding over the linguistic data they encounter—and not through triggering of an innate grammar. What emerges is a picture of language as a structured set of habits shaped by frequency, analogy, and communicative function.

Where nativism emphasizes the uniformity of acquisition outcomes across variable input, emergentism emphasizes the sensitivity of those outcomes to the statistical properties of the input itself. The debate is genuine and ongoing. Christiansen and Chater (2016b), for example, provide an account of why language has the properties it does (Chap. 3), optimized for rapid sequential processing (Chaps. 4–6).

But as Pleyer and Hartmann (2019) have shown in the domain of language evolution, and as the present book aims to demonstrate across all five foundational questions, the two traditions share more explanatory targets than their rhetoric sometimes suggests. The CLB perspective will perhaps resurface most directly in Chaps. 5 and 7 (Christiansen & Chater 2016a, among others), where usage-based accounts of linguistic structure and language evolution, respectively, come into focus.

1.4 A SHORT PREVIEW

Following on from my remarks in the Preface, this pivot is neither a research monograph (of which many will be referenced throughout) nor a textbook (for a great example, even if slightly old and not mentioning the term ‘biolinguistics’, see Boeckx 2010); it is more of an overview—a reading guide through the questions, offering some paths towards answers and presenting further overall discussion. For this purpose, I will cite a lot of sources (and miss many more), but the length restrictions of a pivot do not always allow for elaboration.

Naturally, this short book cannot summarize 70+ years of formal inquiry into language, nor should it try to; accessible overviews already exist for the interested reader (for entry points into each tradition, see e.g. Adger 2019 for CLN and Goldberg 2019 for CLB). What it can do is sketch the core nativist commitments that underpin the biolinguistic research agenda and that will recur across the foundational questions. The same applies to usage-based and emergentist approaches, which have their own extensive literature and internal debates. Hamawand (2020), among others, provides an orientation across different schools of thought. What matters here is not comprehensiveness but positioning: From my vantage point, I aim for getting across an understanding of what CLN brings to each foundational question, and where the conversation with CLB may become possible.

Somewhat counterintuitively perhaps, I follow the Husserl-inspired suggestion that “we should get rid of all prejudices and confront the thing itself rather than what’s behind it” (Wen et al. 2014: 159). This is the spirit in which CLB is approached here. Instead of treating usage-based and emergentist frameworks as rivals to be refuted, the present book takes them as serious interlocutors, that is, as frameworks that ask important questions about meaning, acquisition, and use that the generative tradition has not always prioritized; see Nefdt (2025) for an excellent philosophical companion that develops this stance systematically.

This reference guide serves as a primer for cognitive approaches to biolinguistics. Consider it an invitation to traverse the five foundational questions from multiple theoretical vantage points, in the intended attempt to do so without presupposing allegiance to any single framework. The aim is not synthesis for its own sake but honest engagement: identifying where CLN and CLB genuinely converge, where they talk past each other, and where the questions themselves might be sharpened

by the confrontation. My hope is that the reader does not finish this book needing to echo the late Mary Midgley, paraphrasing: “It seems that bumper stickers have been seen [...] bearing the message ‘Just Say No to [Biolinguistics]’” (Midgley 2018: 7).

After putting the core of biolinguistic inquiry on the table in the form of the *five foundational questions* (Chap. 2), I revisit each one, starting with the three best known (Chaps. 3–5), which may also be the hardest to bridge across the front lines, and the two more recent ones (Chaps. 6–7), which may in turn offer more easily fruitful collaboration among the CLN–CLB axis. Note that the first two problem chapters are also the longest, by design: Humboldt’s and Plato’s problems introduce the core machinery and background that the later chapters lean on rather than restate. The pivot closes with a brief outlook (Chap. 8).

Lenneberg's Dream

Abstract This chapter presents the five foundational questions of biolinguistics, anchoring them in Eric Lenneberg's original framework. It sets them within their historical and theoretical contexts in considerable detail. By so doing, it also situates them within two additional broader frameworks: Niko Tinbergen's four fundamental questions of ethology and Noam Chomsky's three factors in language design. Drawing on the CLN–CLB distinction used throughout the book, the chapter sketches the core commitments of both usage-based and emergentist approaches, and explores where productive dialogue between the two traditions is most plausible.

Keywords Foundational questions of biolinguistics • Questions from ethology • Three-factors framework • Cognitive approaches (CLN vs. CLB) • Biolinguistics (strong/broad vs. weak/narrow)

Eric Heinz Lenneberg (1921–1975) was a German-born American scientist who initiated the biological approach to the study of the human language capacity, together with linguists Morris Halle and Noam Chomsky as well as the cognitive scientist George A. Miller. After a PhD in linguistics and psychology, he studied neuroscience and subsequently held academic positions in the United States. He also frequently lectured on psychology, neurobiology, and related topics in the study of mind, brain, and language around the world. In the early 1960s, he argued for the biological innateness of psychological capacities, including language, and formulated the critical period hypothesis.

2.1 FIVE FOUNDATIONAL QUESTIONS

The title of this chapter is borrowed from Wexler (2003), who used the phrase to capture Lenneberg’s vision of a biologically grounded science of language development, one in which typical language acquisition and its pathological deviations would be understood within a unified biological framework (also endorsed by e.g. Chomsky 2005).

Boeckx and Longa (2011) offer a sympathetic but careful reassessment of this legacy, distinguishing what Lenneberg actually argued from what subsequent biolinguistic work attributed to him. Lenneberg’s (1967: 28) own framing was notably inclusive: “There are biological limits to its future behavioral repertoire, and these express themselves as species-specificities. Such specificities are ever present and there is no behavior (including language) that is exempt from them.” Longa (2018) has since questioned whether the customary nativist framing accurately reflects Lenneberg’s actual theoretical commitments, arguing his views on language development and evolution were considerably more nuanced, and in some respects more compatible with CLB positions, than the standard CLN appropriation suggests.

Either way, Lenneberg (1967) remains the foundational reference point, the work that first gave biological seriousness to the scientific study of language—and this chapter’s title donor. It already laid out four dimensions of a biological approach to language: (i) the *ontogenetic development* of language as a form of biological behavior; (ii) the *genetic foundations* of that behavior; (iii) its *neurological correlates*, especially as revealed by language pathologies; and (iv) the *evolutionary framework* for human language. As testimony to the coherence and foresight of Lenneberg’s original work, these four dimensions anticipate the five foundational questions that organize this pivot.

Biolinguistics, the open-access journal, was founded 40 years later on the premise that linguistic theory is still engaged in providing satisfactory answers to what we call the five ‘foundational questions’. These can be provided in a more detailed way than Table 1.1 in Table 2.1. Each question is dubbed a specific ‘problem’ intended to denote their intellectual origin or connection, something discussed at length by Boeckx (2009: 44–45) and Leivada (2012: 35–36), for example. However, more can be said on the more recent eponymous namegivers (Broca, Wallace, and Darwin) and I also provide a short caption for each question, which will resurface throughout the remainder of this book.

Table 2.1 Five foundational questions of biolinguistics (extended)

<i>Foundational question (with a basic caption)</i>	<i>Eponymous problem (original reference)</i>
1. What is knowledge of language? (the nature of linguistic competence)	Humboldt's problem (Chomsky 1965)
2. How is that knowledge acquired? (language acquisition and development)	Plato's problem (Chomsky 1986)
3. How is that knowledge put to use? (linguistic performance and processing)	Descartes's problem (Chomsky 1997)
4. How is that knowledge implemented in the brain? (neural and cognitive bases of language)	Broca's problem (Boeckx 2009)
5. How did that knowledge emerge in the species? (evolution of the human language faculty)	Wallace/Darwin's problem (Chomsky 2010/Fujita 2002)

The first three—the better known and hence, in a sense, original—'problems' were coined by Chomsky himself: *Humboldt's problem* (Chomsky 1965), *Plato's problem* (Chomsky 1986), and *Descartes's problem* (Chomsky 1997). Each of these will be separately addressed in the next three chapters (Chaps. 3–5).

Broca's problem was named as such by Boeckx (2009), who is sometimes cited as the name-giver of *Darwin's problem* (also suggested at the same time by Hornstein 2009, though). However, Fujita (2002) had already introduced it in his review of Jenkins (2000). Moreover, the term "has been in frequent use, also outside the issue of language emergence" and "the earliest occurrence of the phrase is [arguably] found in Jewett (1914)" (Leivada 2012: 36, fn. 24). The attribution of 'Darwin's problem' is not only contested in terms of who first coined the phrase in a linguistic context but also whether Darwin is indeed the most appropriate intellectual ancestor of the question of language evolution. *Wallace's problem*, suggested by Chomsky (2010) but really made popular by Bickerton (2014), could constitute a viable alternative. It can be further decomposed into two sub-questions, the 'How' (mechanism of language emergence) and the 'Why' (adaptive advantages conferred), whose answers diverge significantly between CLN and CLB.

These problems will be addressed in Chaps. 6 and 7, respectively. The naming conventions matter less than the questions themselves, but they serve a useful function: They anchor each problem in a broader intellectual tradition and signal that biolinguistic inquiry did not arise in a vacuum but inherits centuries of philosophical and scientific

reflection on the nature of mind, knowledge, and life. A useful formulation makes the point more sharply (Berwick & Chomsky 2016): “Languages change, but they don’t evolve. What evolves is the language capacity of users of language, i.e., human beings” (Noam Chomsky in Bambini et al. 2012: 222). This keeps the biological capacity distinct from its cultural products and prevents a conflation that has generated unnecessary confusion in discussions of language evolution.

The most important ingredient of answers to any of the foundational questions in Table 2.1 is the assumption that language is made possible by a domain-specific faculty of language unique to *H. sapiens*, which allows for the identification and investigation of an innate mental capacity (‘knowledge of language’), sometimes referred to, and mixed up with (as Leivada 2020 suggests), UG. The conflation of ‘knowledge of language’ with ‘Universal Grammar’ is, as Leivada (2020) documents, one of the most persistent terminological confusions in the field (cf. Chap. 8). UG, strictly speaking, refers to the initial state of the language faculty (what the child brings to acquisition before any experience), while ‘knowledge of language’ denotes the steady state attained after acquisition. The two are related but distinct, and keeping them apart matters for how one frames both Humboldt’s and Plato’s problems.

2.2 FOUR QUESTIONS FROM ETHOLOGY

Generative research has argued that, at a suitable level of abstraction, the study of the language faculty can really be viewed as a branch of biology, a point raised even before ‘biolinguistics’ became popular (cf. Chomsky 1976; other early relevant references include Chomsky 1975, Chomsky & Walker 1978, Von Eckardt Klein 1978, and Caplan & Chomsky 1980). The analogy that follows, between the foundational questions of biolinguistics and Niko Tinbergen’s four fundamental questions of ethology, is one of the most productive in the field because it situates the study of language within the broader framework of biological inquiry without reducing it to any particular theoretical commitment. After all, the five foundational questions can be seen as variants of Tinbergen’s (1963) four fundamental questions in his seminal contribution to the biology of animal behavior (Boeckx & Grohmann 2007: 2; for extensive discussion, see e.g. Boeckx 2010, Fitch 2010b, or Reboul 2017). Tinbergen’s questions are listed in Table 2.2.

Table 2.2 Four fundamental questions of ethology

-
1. What stimulates the animal to respond with the behavior it displays, and what are the response mechanisms?
 2. How does an organism develop as the individual matures?
 3. Why is the behavior necessary for the animal's success and how does evolution act on that behavior?
 4. How has a particular behavior evolved through time? Can we trace a common behavior of two species back to their common ancestor?
-

While biolinguistics may be characterized as the scientific inquiry into language that aims to highlight the commitment of the generative enterprise to the biological foundations of language vis-à-vis the faculty of language, the emphasis on the necessarily interdisciplinary character of such an enterprise should also allow for a more inclusive understanding. An inclusive biolinguistics, in this sense, is one that takes the five foundational questions as its organizing frame without presupposing that the answers must be generative ones.

The questions themselves—about the nature, acquisition, use, neural implementation, and evolutionary origin of language—are framework-neutral, even if the answers proposed within the generative tradition have historically dominated the field. CLB approaches, emphasizing meaning, context, and general cognition, have their own answers to offer, and the ethological framing helps make visible exactly where those answers converge and where they may diverge from CLN responses.

What lies behind the questions from ethology is a new weight given to the comparative method, beyond its commonly construed linguistic meaning towards the more familiar methodology employed in the biological sciences. In linguistics, 'comparative' traditionally refers to the comparison of languages—their structures, histories, and typological variation. In biology, it means something different; it denotes the systematic comparison of species, or of traits across species, in order to understand the functional and evolutionary significance of those traits.

Biolinguistics imports this second sense of comparison into the study of language. This means asking not only how languages differ from one another, but how human language differs from animal communication systems, how the language faculty relates to other cognitive faculties in the same organism, and how the developmental trajectory of language in *H. sapiens* compares to analogous developmental processes in other

species (see Chap. 7). These are questions that no framework focused exclusively on cross-linguistic variation, however sophisticated it may be, is positioned to answer on its own.

2.3 THREE FACTORS IN LANGUAGE DESIGN

The generative study of UG thus aims to “construct a biologically based natural science of language—of its development in the individual (growth) and the human species (evolution)” (McGilvray 2017: 77). As far as I am aware, these answers would be categorically rejected by all micro-cognitive linguists, that is, within the framework of *Cognitive Linguistics* and related derivatives, including *Construction Grammar* (collectively referred to here as CLB).

A goal of the ensuing presentation is to put the five foundational questions, and possible ways of tackling them, into a wider perspective. One way of widening that perspective within the generative paradigm is through the framework of the *three factors in language design*, which Chomsky (2005) proposed as a principled decomposition of the forces shaping the language faculty. This framework is notable not only for what it says about language, but also for how it opens the door to create a dialogue between CLN and CLB.

For starters, the five foundational questions of biolinguistics are couched within a larger framework of language design, larger than a full-blown UG under any name (see especially Chaps. 3 and 6). With the rise of the *Minimalist Program* (Chomsky 1993 *et seq.*), much effort has been put on reducing (the number and content of) innate language-specific concepts—an aim its proponents state in as many words: to “share the desire to reduce any language-specific innate endowment, ideally to a logical minimum” (Berwick et al. 2011: 1210).

The minimalist reduction of UG is significant for the present discussion because it narrows the gap between CLN and CLB. If the language-specific component of the first factor is lean, perhaps reducible to a single operation like Merge (or the ‘Basic Property’), then much of what shapes language falls under the second and third factors, namely environmental input as well as general principles of computation and cognition. These are then precisely the domains where CLB approaches have the most to contribute, and where a productive cross-framework dialogue may become most plausible.

The interplay of nature and nurture in language acquisition, and the role of cognition at large, can be captured by the 'three factors' as in Table 2.3 (Chomsky 2005: 6). Indeed, as Fitch (2017b: 445–446) observes, Lenneberg (1967) had already transcended the nature/nurture dichotomy that has since bedeviled the field, recognizing the species-typical biological components but also the culturally specific, learned character of language.

Table 2.3 Three factors in language design

-
- | | |
|----|--|
| 1. | Genetic endowment, apparently nearly uniform for the species, which interprets part of the environment as linguistic experience [...] and which determines the general course of the development of the language faculty. |
| 2. | Experience, which leads to variation, within a fairly narrow range, as in the case of other subsystems of the human capacity and the organism generally. |
| 3. | Principles not specific to the faculty of language [...]: (a) principles of data analysis that might be used in language acquisition and other domains; (b) principles of structural architecture and developmental constraints that enter into canalization, organic form, and action over a wide range, including principles of efficient computation [...]. |
-

The first factor relates to the part of the human genome that is dedicated to language (say, UG), while the second factor expresses the epigenetic environment (introduced as 'primary linguistic data' in Chomsky 1965). Third factors include general principles of computation and cognition (not language-specific).

For our purposes, the significance of the three-factor framework cannot be overstated. From a CLN viewpoint, the first factor is where biolinguistics lives most comfortably: It is the domain of UG, innateness, and the species-specific endowment that makes language possible. The second factor, experience and its role in shaping the steady-state grammar, is where language acquisition research, including work on heritage languages, multilingualism, and cross-linguistic variation, does its most important work; this is terrain that CLN and CLB researchers genuinely share, even if they interpret their findings differently.

The third factor is perhaps the most interesting one from a dialogic perspective: Principles of efficient computation, structural economy, and developmental canalization are not the sole property of any single framework. Minimalists raise them to explain why UG has the shape it does; usage-based researchers invoke related notions, such as iconicity,

frequency, or analogical extension (e.g., Tomasello 2003, Christiansen & Chater 2016a), to explain why linguistic structures have the properties they do. The overlap is real, even if the theoretical superstructures differ (but see e.g. Newmeyer 2021). A biolinguistics capacious enough to accommodate both CLN and CLB approaches must, at minimum, acknowledge that the third factor is contested ground worth mapping carefully—with the caveat, as Vitor Nóbrega (p.c.) observes, that the first factor is less prominent on the CLB side, if relevant at all.

As Hornstein (2024) argues most precisely, echoing his earlier monograph (Hornstein 2009), the tension of a richer UG assumed in earlier generative frameworks that would solve Plato’s problem but worsen Wallace/Darwin’s problem is the deepest motivation for the minimalist program: Deriving those earlier results from simpler axioms would simultaneously address learnability and evolvability, as a simpler endowment is both more easily acquired and more evolutionarily tractable.

2.4 TWO COGNITIVE APPROACHES TO LINGUISTICS

2.4.1 *From the Strongest CLN View...*

Boeckx and Grohmann (2007) associate a ‘weak’ and a ‘strong’ sense with biolinguistics. In its *weak sense*, biolinguistics means ‘business as usual’ for linguists, so to speak, to the extent they are seriously engaged in discovering the properties of grammar. In effect, then, this continues the research program Noam Chomsky initiated in *Syntactic Structures* (Chomsky 1957). The *strong sense* of the term relates to attempts to provide explicit answers to questions that necessarily require the combination of linguistic insights and insights from further relevant disciplines, some traditionally more related than others (such as evolutionary biology, genetics, or neurology).

This strong/weak distinction maps onto something important for the present book’s agenda. Biolinguistics in the weak sense is essentially CLN work with biological awareness; biolinguistics in the strong sense requires collaboration across disciplinary lines that CLB approaches, with their emphasis on usage, context, and general cognition, might in some respects be better positioned to pursue. As mentioned in Chap. 1, I use the CLN–CLB dichotomy after Mendivil-Giró’s (2025b) reformulation of *biolinguistics in the narrow sense* (‘weak’) to denote the

“scientific core” of the field and *biolinguistics in the broad sense* (‘strong’) to refer to the necessary “interdisciplinary work” (p. 20).

However, one hope the *Biolinguistics* journal editors harbored from the outset (see also Grohmann 2015) is that ‘generative grammar’ and ‘biolinguistics’ are *not* taken to be synonyms, or that the theoretical perspective espoused in ‘Chomskyan’ generative grammar is the only valid underpinning of biolinguistic investigations; see Martins and Boeckx (2016) for extended discussion. Yet, it is fair to say (e.g., Di Sciullo & Boeckx 2011) that the recent “resurgence of activity surrounding biolinguistics” (Berwick 2011b: 461) is due in large part to the advent of the minimalist program in linguistic theory (Chomsky 1993 *et seq.*; for textbooks, see e.g. Adger 2003, Hornstein et al. 2005, Koenenman & Zeijlstra 2017, and for handbook collections Boeckx 2011b, Roberts 2017b, Grohmann & Leivada 2026, Leivada & Grohmann 2026b).

The relationship between the minimalist program and biolinguistics is one of productive but asymmetric overlap: Minimalism provides the most elaborated CLN framework for addressing the five foundational questions, but it neither exhausts those questions nor commands exclusive rights to their answers. To illustrate, and clarify the terminology used throughout, two axes are in play and worth keeping apart: The ‘CLN/CLB’ distinction proposed here marks a difference of *tradition*, whereas the ‘narrow/broad’ contrast (and the ‘weak/strong’, in Boeckx & Grohmann 2007) marks the *scope* of the biolinguistic enterprise, from its scientific core to its interdisciplinary reach. While the two often travel together, they are independent; CLN can pursue the broad, interdisciplinary sense, just as CLB may keep to the narrow one.

Be it as it may, at the heart of the minimalist program, to continue with the best worked out generative approach, lies the question of how much of the architecture of the language faculty can be given a principled explanation. Specifically, minimalism asks how well the engine of language meets design requirements imposed by the cognitive systems it subserves. Inevitably, linguists working in the context of the minimalist program are forced to address and sharpen questions of cognitive specificity, ontogeny, phylogeny, and so on, in order to even begin to understand the design requirements imposed on the language faculty.

This is not to say that previous generations of linguists were not interested in such issues. But in practice, biological issues had little effect on empirical inquiry into questions of so-called descriptive and explanatory adequacy (Chap. 4). Minimalism changed this by making

biological plausibility not merely a hoped-for consequence but a real design criterion; biological plausibility is an explicit desideratum of linguistic theory. This is most clearly stated in the quest to go “beyond explanatory adequacy” (Chomsky 2004, 2007), which in a sense subsumes the other levels of adequacy that emerged along the way, including evolutionary adequacy. A grammar that posits elaborate, language-specific mechanisms not reducible to general computational principles becomes a less adequate grammar, not merely a more complex one. This shift in explanatory standards is what makes minimalism the natural internal locus of biolinguistic inquiry within the generative tradition.

Likewise, it is important to stress that biolinguistics is also independent of the minimalist program. As Eric Lenneberg’s work already made clear, biolinguistic questions can be fruitfully addressed outside of a minimalist context (e.g., Lenneberg 1967). But such a context may certainly facilitate, if not necessitate, inquiry into the biological foundations of language. From this perspective, linguistic minimalism is an approach to language that is largely independent of theoretical persuasion. It is thus an aspect of cognitive linguistic research that can be shared by virtually all existing frameworks of linguistic theory. In this sense, the minimalist aspiration, to show that language is what it is because of general principles of efficient computation operating over a minimal biological endowment, is not foreign to CLB. Usage-based frameworks also appeal to general cognitive principles; they differ from minimalism not in rejecting a principled explanation but in locating that explanation outside a dedicated language faculty altogether.

It bears emphasizing that ideally CLN and CLB mark positions on a spectrum rather than the boundaries of two sealed camps—a way of situating approaches for which there is an empirical precedent (Butler & González-García 2014; also Sect. 2.4.2). The distinction is best understood as a shorthand for a cluster of related, but partially independent, commitments and a researcher can simultaneously sit at different points on different commitments.

A concrete illustration of why these commitments come apart, also directly echoing the excursion on ‘tradition’ and ‘scope’ above, is the *ecological-developmental (eco-devo)* program of Balari et al. (2020). On syntactic architecture they remain firmly on the CLN side, retaining a computational system and reading the Chomsky hierarchy as a space of possible ‘computational phenotypes’ (as in Balari & Lorenzo 2009,

2013). Yet on innateness and acquisition they side with CLB, rejecting the picture on which the language faculty is blueprinted in the genotype and merely matured, in favor of one on which even a core property like the lexicon–syntax interface is actively constructed through the child's interaction with a culturally supplied environment. A researcher can, in other words, sit at the CLN end of one commitment and at the CLB end of another, which is what the framing of a spectrum captures nicely.

Table 2.4 maps these commitments onto the foundational questions that organize this book. The chapters that follow are structured around the points where CLN and CLB diverge most sharply on each question.

Table 2.4 CLN–CLB commitment spectrum

<i>Commitment</i>	<i>CLN form (strong)</i>	<i>CLB form (weaker)</i>	<i>Chap.</i>
Innateness	rich, domain-specific UG	minimal first factor; strong third-factor effects	3
Acquisition	poverty of stimulus; parameter-setting	statistical learning plus bootstrapping; eco-devo	4
Syntax	autonomous; generative	emergent from use and meaning	5 (+ 3)
Neural basis	dedicated language circuitry	distributed networks recruited from general cognition	6
Evolution	single mutation; discontinuity	gradual mosaic of pre-adaptations; continuity	7

2.4.2 ...to a Weaker CLB-Compatibility

Cognitive linguistics does not denote a single or specific theory of language, “but rather a cluster of broadly compatible approaches [...] in which natural language is studied as a mental phenomenon [...] where cognitive refers to the crucial role of intermediate informational structures in our encounters with the world” (Geeraerts & Cuyckens 2007: 3–5). By now, the framework has grown into a multifaceted field of research from its inception starting at least with Langacker (1987) but originating in the aftermath of generative semantics (e.g., Lakoff 1971).

Since the first textbook, originally published in 1989 (Taylor 2003), the field has also seen an incredible influx of introductory books and collected volumes (Wen & Taylor 2021a). The theoretical commitments of CLB can be distilled into a set of fundamental hypotheses to the

study of language that stand in direct contrast to the CLN position, and which will serve as a recurring reference point here. These may be summarized as provided in Table 2.5 (Wen et al. 2014: 156).

Table 2.5 Fundamental hypotheses of CLB

-
1. Language is not an autonomous cognitive faculty but a main part of cognition.
 2. Human languages are a list of symbolic units in which forms are conventionally paired with meanings.
 3. Meaning is what language is all about.
 4. Meaning is conceptualization.
 5. Knowledge of language emerges from language use.
-

Comparing Table 2.1 with Table 2.5 from a CLB perspective, there is no language faculty, the autonomy of syntax is rejected, the fundamental role of the computational system of human language is denied, a conceptual-intentional system does not play a role, and innate capacities are not assumed to deal with poverty-of-stimulus issues. Yet, CLN and CLB approaches are, ultimately, *cognitive* (perhaps as also in the cognitive-functional sense of Butler & González-García 2014; for early retrospectives of the term, see de Gelder 1985)—with the crucial difference that in CLN, language is the product of a domain-specific faculty in the human mind/brain coupled with additional factors (Table 2.3), while in CLB, “[t]he hallmark [...] is the relationship between language and cognition, how humans conceptualize the world around them, how they create and represent meaning, and how this is reflected in language” (Siewierska 2013: 485).

A useful distinction here is that drawn by Adger and Svenonius (2015) between a faculty of language that is (i) domain-specific, (ii) domain-specialized, or (iii) domain-general. A *domain-specific* faculty is dedicated exclusively to language and applies only to linguistic objects; a *domain-specialized* faculty, in turn, employs mechanisms that originate in or are shared with other cognitive domains but have been recruited and refined for linguistic purposes. And in stark contrast, a *domain-general* faculty draws entirely on cognitive resources that are not proprietary to language in any sense.

The CLN position is standardly associated with domain-specificity, though the minimalist reduction of UG may eventually push it towards domain-specialization: If Merge is the core of the language faculty (as

in Chomsky et al. 2023), and if Merge is a general operation of hierarchical combination (the Basic Property of Berwick & Chomsky 2016), then even the most austere CLN position may perhaps be closer to being domain-specialized than domain-specific. By contrast, CLB is committed to domain-generalty: The structures of language emerge from the same cognitive machinery that underlies categorization, attention, and social cognition. Adger and Svenonius' (2015) taxonomy thus serves not to adjudicate between CLN and CLB—but to sharpen what exactly is at stake in their disagreement and suggest that the distance between them may be smaller, at least on the domain-specificity question, than the polemics of past decades implies.

As Geeraerts and Cuyckens (2007: 5) put it, “the two approaches differ with regard to the epistemological role of natural language”: Both concur “that there can be no knowledge without the existence of a mental representation that has a constitutive, mediating role in the epistemological relationship between subject and object” (pp. 5–6). It is just that “natural languages precisely embody such categorial perspectives onto the outside world” in micro-cognitive linguistics, while generativists hold “natural language as the object of the epistemological relationship, rather than as the intermediate link between subject and object” (p. 6). This epistemological difference has a direct bearing on how each approach frames the foundational questions. If language is the object of inquiry (CLN), then the questions are about the properties of that object—its structure, its acquisition, its neural realization, and its evolution. If language is the medium through which the world is conceptualized (CLB), the questions shift: They become questions about how meaning is constructed, how experience is organized, and how communication shapes the systems speakers develop.

Both sets of questions are legitimate; neither exhausts what there is to say about human language. Rather than beginning our inquiry with the Basic Property and working outward to acquisition, use, and evolution, CLB starts out from language use and works inward, from attested patterns of meaning and communication towards whatever structural generalizations those patterns support. Framed this way, the five questions look somewhat different, as the remainder will make explicit.

2.4.3 Language and Cognitive Capacities

The next five chapters will thus address each ‘problem’ from the

perspective of the five foundational questions. But something needs to be said for a viewpoint of CLB, including other functional approaches, which concentrate on “language as a means of communication and how this communicative function has structured grammar both synchronically and diachronically” (Siewierska 2013: 485). What CLB brings, in particular, is a sustained focus on Descartes’s problem and on the relationship between use and structure that the CLN tradition has historically treated as a matter for performance rather than competence. From a CLB perspective, this separation is itself a theoretical choice, not a natural fact, and one that the foundational questions, taken seriously, should force us to scrutinize.

Let me start with the first. If the (macro-)cognitive linguistic object of study understands natural language as a species-specific mental phenomenon, the foundational questions of biolinguistics, coupled with the fundamental questions of ethology, may be phrased in more general terms to capture the investigation of any (mental) cognitive capacity, as Boeckx (2010) suggests in closing his textbook. “For a given cognitive capacity *X*” (p. 187), he asks: “What are the rules and structures involved in *X* that are necessary to account for what we tacitly know about *X*?” And he follows this up with the other foundational questions, how *X* develops in the individual, how *X* is put to use, how *X* is implemented in the brain, and how *X* evolved.

Yet, from a CLB perspective, one might object to assigning language a (domain-)specific cognitive capacity, such as the faculty of language. Here Adger and Svenonius’ (2015) taxonomy introduced above may become relevant. CLB might accept *domain-specialization* (that certain cognitive resources be preferentially deployed in linguistic contexts) without accepting *domain-specificity*, as that would require a dedicated faculty CLB categorically rejects. This leaves open the possibility of a rapprochement: A biolinguistics that treats the faculty of language as domain-specialized rather than domain-specific may be one that CLB researchers find less foreign, and that CLN researchers can still accept without abandoning their core commitments—if only for the purpose of collaborative dialogues instead of animosity. Arguably, this requires shared theoretical vocabularies, but perhaps efforts like this pivot can contribute to establishing such (see also Sect. 8.2).

From a generative vantage point, Boeckx’s (2010) generalized take on “a given cognitive capacity *X*” is nicely captured by Fodor’s (1983) modularity of mind (Sect. 1.1; see Fodor 2000a for a retrospective),

where ‘X’ is such a module within the cognitive system—the language faculty. This faculty arises from the first factor (Chap. 3), interacts with the linguistic environment through the second factor (Chap. 4), and ties in with general principles of human computation and cognition via third factors (Chap. 5). We should bear in mind, though, that “Fodor’s claims are about processing; Chomsky’s are about analytic/competence units” (Allott & Smith 2021: 536; cf. Chap. 3); see also Chomsky (2018) for clarifications on these two different views of modularity.

But there might be an alternative integration of Tinbergen’s insights into a research agenda for a more inclusive biolinguistics paradigm, something like “Biolinguistics 2.0” (Boeckx & Benítez-Burraco 2014b) which reflects Boeckx and Grohmann’s (2007) intention for the ‘strong’ sense of biolinguistics (or better, with Mendívil-Giró 2025b: its broad sense, for interdisciplinary research, as he has himself pursued since at least Mendívil-Giró 2019).

Since the fundamental questions of ethology can be said to essentially concern *mechanism* (proximal cause), *function* (ultimate cause), *ontogeny* (development), and *phylogeny* (evolution), we may consider Reboul’s (2017: 2, 223) reformulation in a more straightforward application to the study of language; this is listed in Table 2.6.

Table 2.6 Tinbergen’s four questions for linguistics

-
1. What are the mechanisms of language?
 2. What are the functions of language?
 3. What is the ontogeny of language?
 4. What is the phylogeny of language?
-

Reboul does not pursue the analogy with the foundational questions any further; or rather, she exploits the four questions in a different direction, which is in a sense already compatible with both CLN and CLB approaches. What her reformulation does make possible is a version of the ontogeny/phylogeny distinction that does not presuppose a domain-specific faculty. For CLB researchers, questions about how language develops in the individual and how it emerged in the species can be raised without invoking UG, and Reboul’s formulation leaves that door open.

One can discern *Plato’s problem* (ontogeny) and *Wallace/Darwin’s*

problem (phylogeny). The mapping is not perfect, as Tinbergen's four questions do not carve the biolinguistic terrain at exactly the same joints as the five foundational questions; but the overlap is sufficiently substantial to serve as a bridge. *Descartes's problem* pertains to the second question (functions), whereas *Humboldt's problem* forms a substantive part of the first (mechanisms)—except that it does not make explicit reference to an implicit domain-specific faculty, module, or organ.

Finally, *Broca's problem* can be seen as the connecting link: It refers to the brain structures and processes required for implementation. This connecting role is what makes Broca's problem strategically central to the whole enterprise; progress on a neural implementation of language bears directly on both the nature of the faculty (Humboldt's problem) and its evolutionary origin (Wallace/Darwin's problem). It is where the strong or broad sense of biolinguistics would do its most demanding, and most genuinely interdisciplinary, work.

2.5 ONE AIM: BIOLINGUISTICS

Biolinguistics in the 'strong' sense comes out clearest in the last two foundational questions. These are possibly the most exciting problems to tackle; at the same time, they are also the hardest to address. The depth of that difficulty has historical illustration: In 1866, the *Société de Linguistique de Paris* prohibited inquiries into the origin of language in its founding statutes, followed by a similar ban shortly after from the *Philological Society of London*. This effectively removed such questions from academic debate for nearly a century (an early statement in modern times can be found in Chomsky 1968). That they have returned with rigorous empirical traction is itself one of the more striking developments in the history of linguistics (Nóbrega 2022). Jenkins (2000) cites Maynard Smith and Szathmáry (1995) on the issue, but one of the first public lamenters was already Max Müller, someone who has worked on language evolution himself (Müller 1861, 1887). Indeed, Adornetti et al. (2019: 7) begin their recount: "The topic of language origin and evolution has been considered for a long time as a difficult question to address scientifically because of poverty of empirical data and limitations in methodology (Müller, 1861)."

The other reason why the last two foundational questions are so difficult to face is two-fold. On the one hand, there are clear obstacles

along the way for both questions, more obviously so than for the first set of questions and independently of one's philosophical or theoretical persuasions (think of dichotomies such as rationalist/empiricist, nativist/behaviorist, formalist/functionalist, macro-/micro-cognitive, and so on). And on the other, cognitive (bio)linguists cannot address them on their own—they require multidisciplinary collaboration. Moreover, Wallace/Darwin's problem simply “cannot be solved in isolation from” the neural implementation of Broca's problem (Trettenbrein 2015: 6).

Biolinguistics in the ‘weak’ sense may be viewed as the continuation of the ‘traditional’ generative research agenda. In the first modern book featuring ‘biolinguistics’ in its title, which subsequently kickstarted the emergence of this newly named field, we can read this footnote:

Chomsky often refers to the first question [...] as *Humboldt's problem*, to the second question [...] as a special case of *Plato's problem*, and to the third question [...] as *Descartes' problem*, to highlight the fact that the modern study of these problems has a long and rich historical tradition. In what follows we will be primarily focusing on a part of the ‘second cognitive revolution’, the modern study of biolinguistics; i.e., on work going back to the early 1950s.

(Jenkins 2000: 2, fn. 3)

For CLN, this Chomsky-inspired “second cognitive revolution” (cf. Sect. 1.1) could be thought of as ‘nativism with a cognitive twist’; for CLB, it means cognition all the way down, with no language-specific module required.

What this chapter has attempted to demonstrate is that the first three foundational problems, however they have been developed within the CLN tradition, are not the exclusive property of generative grammar. Whether approached from Chomsky's three-factor model, Tinbergen's four questions for ethology, or Reboul's reformulation in four questions for linguistics, for example, a broad-minded inquiry opens each of them to engagement from CLB perspectives. The following chapters pursue that engagement in detail, beginning with Humboldt's problem in Chap. 3. Reboul's reformulation (Table 2.6) serves as a recurring CLB touchstone throughout the remainder—a way of asking, for each problem, what a non-nativist answer that starts out from the fundamental CLB hypotheses (Table 2.5) may look like in its most fully developed form.

Humboldt's Problem

Abstract This chapter presents the first foundational question: *What is knowledge of language?* It recaps the main arguments advanced in favor of an innate capacity for language. Universal Grammar is a theory to describe this faculty of language, while emergentist approaches do not subscribe to this nativist perspective. Yet, Humboldt's problem can, and must, receive treatment in non-nativist frameworks as well. Several suggestions are provided here. Also relevant for the discussion is the distinction into competence and performance, which, once again, can be viewed from the original nativist position (CLN) as well as from other cognitive, usage-based and emergentist, perspectives (CLB).

Keywords Knowledge of language • Universal Grammar • Language faculty • I-language • E-language • Competence • Performance • Basic Property • Merge • Recursion

Friedrich **Wilhelm** Christian Karl Ferdinand **von Humboldt** (1767–1835) was a Prussian scholar, diplomat, and statesman whose contributions spanned anthropology, education, literature, philosophy, political thought—and linguistics. Key founder of the University of Berlin, which was later named after him, he profoundly influenced 19th-century German linguistics. Linguists the world over find important insights in his work on philosophy of language and ethno-linguistics. Humboldt emphasized the creative aspect of the human linguistic capacity, suggested the existence of universal properties across languages, and developed the concept of *Weltansicht* (often utterly misconstrued). All three themes continue to generate productive debate in contemporary (bio)linguistics.

3.1 The Nature of Linguistic Competence

Where CLN asks what the mind must contain for language to be possible, CLB asks what patterns of use must look like for the illusion of a grammar to emerge (cf. Sect. 2.4). Noam Chomsky articulated what has since become widely known as *Humboldt's problem* as: “What constitutes knowledge of language?” (Chomsky 1986: 3). As he remarks very early on (e.g., Chomsky 1964, 1965), Humboldt's insistence on the generative, creative dimension of human linguistic capacity (see also Chap. 5) anticipates the spirit of the generative enterprise, hence Wilhelm von Humboldt is the appropriate historical namegiver.

Regardless of his views on (the nature of) language and its philosophical underpinnings, Humboldt's impact and insights on linguistics were tremendous (Mueller-Vollmer & Messling 2023). This conception is important not only technically but also strategically: By defining language in terms of what it computes rather than what it contains, the *Basic Property* shifts the focus from the inventory of a language (its words, sounds, constructions) to the generative mechanism that underlies all languages. This is where CLN and CLB diverge most sharply—and where the dialogue between them may be most productive.

According to Chomsky, the generative enterprise intends to seek “a real understanding of how a language can (in Humboldt's words) ‘make infinite use of finite means’” (Chomsky 1965: 8; cf. Humboldt 1836: 70). The fundamental divide between CLN and CLB approaches concerns whether the infinity property truly results from domain-specific properties of the mind/brain or emerges from domain-general processes. The trickiest part of addressing Humboldt's problem will arguably be making an underlying *knowledge* of language palatable to CLB or non-nativist approaches—or at least reformulating the problem in such a way that it is still meaningful in the context of biolinguistics beyond a mere “What is language?” (see especially de Vos 2015, but also the discussion of Reboul's reformulation in Chap. 2).

From a CLN perspective, Humboldt's problem naturally assumes the existence of *knowledge* of language, as typically encapsulated by the Innateness Hypothesis: Humans are genetically endowed with a capacity for language, the language faculty, which unfolds through interaction with environmental input. Regarding how finite linguistic means generate infinite expressions, the generative answer to the question involves the computational system of human language, particularly

recursion as yielded by the operation Merge. More recently, also in the context of language evolution (Chap. 7), this has become known as the Basic Property of language (Berwick & Chomsky 2016; cf. Sect. 3.2).

CLB approaches, in contrast, do not assume domain-specific knowledge of language vis-à-vis an innate faculty: The framework initiated by Langacker (1987) views language as emergent from general cognitive processes. Linguistic knowledge is constructed from patterns of use, frequency effects, and analogical reasoning—without being specific to a language domain or faculty. In the words of Wen & Taylor (2021a), “[k]nowledge of language basically emerges from language use” (p. 2). Usage-based approaches generally hold that language is shaped by experience, interaction, and cultural transmission; this may even include ecological factors (e.g., Mühlhäusler 2003).

More precisely, while CLB certainly denies the first clause of the Basic Property, that the generative mechanism is a dedicated, domain-specific computational operation, it may be more acceptable to go with the observation that linguistic knowledge enables the productive combination of meaningful units. The disagreement is not about whether language is generative in some sense, but about what kind of mechanism generates it—and especially whether that mechanism is proprietary to language or an instance of more general cognitive operations.

The above portrayal of CLB approaches to language has two immediate consequences. On the one hand, while potentially complementary at first glance, they do stand in opposition to Chomsky's (2005) three factors in language design that guide the study of language acquisition and processing; these are genetic endowment, experience, and domain-general cognitive principles (Table 2.3). The first factor relates to UG, the initial state of the human language faculty. Despite appearance, the second and third factors differ from usage-based conceptions: Experience refers to the epigenetic environment, introduced as ‘primary linguistic data’ in Chomsky (1965), leading to variation found across the world's languages, while domain-general factors are supplementary.

On the other hand, micro-cognitive conceptions of language allow no place for a domain-specific language faculty (or other related innate constraints). Usage-based approaches also challenge modularity by viewing language as continuous with other cognitive domains and as a complex adaptive system. In sharp contrast, CLN approaches adopt the Modularity Hypothesis (e.g., Crain et al. 2017), which holds that the mind/brain consists of specialized modules, each with its own distinct

properties and functions. The faculty of language is considered such a specialized module, characterized by domain-specificity—meaning, it processes information unique to language and operates independently of other cognitive systems.

Admittedly, considerable confusion has arisen among linguists and non-linguists alike, even among generative linguists, about some of these terms—about ‘universal’ in UG, about ‘grammar’ in both UG and generative grammar more generally, about the nature of the faculty of language, and so on. For useful clarifications (see also Sect. 8.2), see recent discussions by Leivada (2020) or Leivada and Murphy (2021) as well as Trettenbrein (2015), Grohmann and Leivada (2020), Leivada and Grohmann (2026a), but also Watumull and Chomsky (2020).

Adding to this, Hauser et al. (2002) introduced a further refinement of the language faculty which was already mentioned earlier in passing. They distinguish between the *faculty of language in the broad sense (FLB)*, ubiquitous in biological organisms, and the *faculty of language in the narrow sense (FLN)*, by assumption unique to *H. sapiens*. This distinction has led to enormous speculation and discussion over the past two decades (e.g., chapters in Larson et al. 2010) to pinpoint the Basic Property as capturing the essence of knowledge of language (Merge, recursion, and so on). More recent efforts build on the strong association between advances in neuroscience and the scientific study of language; for example, Arbib (2002) refers to the “language-ready brain” instead and Murphy (2025) proposes a ‘Universal Neural Grammar’ (both intimately related to Broca’s problem, discussed in Chap. 6).

The following will move from the Basic Property (often employed synonymously with the operation Merge, but see Sect. 3.2 right below) and nativism to the central distinctions of the two pairs *competence vs. performance* and *I- vs. E-language*. It will recapitulate arguments put forth for an innate knowledge of language from language acquisition (more in Chap. 4), on the one hand, and provide a comparison with alternative approaches and explanations on the other, yet within the spirit of addressing Humboldt’s problem.

3.2 KNOWLEDGE OF LANGUAGE

Nóbrega (2022) reviews four core properties that characterize linguistic knowledge as *tacit* rather than *explicit* (Polanyi 1966; see Davies 2015

for general overview and Hornstein 2005 for linguistic knowledge). It is tacit in a 'strict sense'; speakers know things they do not know that they know. It is 'complex'; linguistic knowledge underwrites subtle judgments about structure and meaning that no explicit instruction conveyed. It is 'untutored'; the vast bulk of linguistic knowledge is acquired without formal teaching. And it is attained in the face of 'impoverished input'; the data available to the learner radically underdetermine the rich, uniform competence that is eventually attained.

Taken together, these four properties define what any adequate theory of knowledge of language must explain. When we ask what constitutes knowledge of language, this question, of course, presupposes that there is some *knowledge* of language that we can talk about to begin with. Chomsky's introduction of the term couched within generative grammar points to an innate capacity to deal with language—to acquire it, without instruction (Plato's problem), and to use it (Descartes's problem)—which connects to our brain circuitry (Broca's problem;) and also forms a part of human evolution (Wallace/Darwin's problem;).

Before addressing some core issues, let me provide an understanding of *language* in the generative sense. With Berwick and Chomsky (2016: 1) we can take "the Basic Property of language" to be that "a language is a finite computational system yielding an infinity of expressions, each of which has a definite interpretation in semantic-pragmatic and sensorimotor systems (informally, thought and sound)." It is this CLN conception of language, as a computational system, that most crucially separates generative grammar from most other approaches to human language; this 'Basic Property' will also play a role in the following.

De Vos (2015: 3) rightly suggests to approach Humboldt's problem in two ways: (i) what "we, as linguists and cognitive scientists, know to empirically constitute language" (but see Waciewicz et al. 2020) and (ii) "what theoretical tools, systems and structures are required to explain those facts". These two sub-questions map neatly onto the CLN–CLB divide: CLN prioritizes (ii), the theoretical architecture required to explain linguistic knowledge, and infers the nature of that knowledge from the architecture—which, as pointed out by Vitor Nóbrega (p.c.), constitutes a system of knowledge that organizes thought. In turn, CLB prioritizes (i), the empirical facts of language use, and builds theoretical structures only as far as those facts require.

Concerning such macro- vs. micro-cognitive outlooks, "[w]hether these systems are considered to be 'knowledge of language' in an

intrinsic or extrinsic sense depends on whether one considers them to be real or not” (*ibid.*). De Vos (2015: 34) correctly adds that “[t]he differences between the theoretical clusterings characterized as ‘generative’ or ‘functionalist’ do not necessarily hinge on formalism vs functionalism, but rather on their commitments to a range of issues such as modularity, representation and empiricism/nativism which, in turn, can be seen as choices prompted by Humboldt’s and Plato’s problems.” This is not a minor terminological point. It means that the deepest disagreements between CLN and CLB are not about method or formalism but about prior ontological commitments; Humboldt’s problem is precisely where those commitments first become visible.

Whatever one’s take on the nature of knowledge of language, there are a number of core empirical phenomena that any approach must be able to account for: the well-known, systematic ambiguity of sentences like *I shot an elephant in my pajamas* (where the prepositional phrase attaches differently for different interpretations), but also the contrast between ambiguous *Marty didn’t sell his bike because the gears broke* and unambiguous *Because the gears broke, Marty didn’t sell his bike* (Johnston 1994; hence structure plays a bigger role than non-generative frameworks admit), sensitivity of grammaticality judgments to structure rather than linear order (as in island constraints), and many more, including productivity and systematicity of morphological paradigms or intricate dependencies that hold across long distances in a sentence.

3.3 UG AND THE LANGUAGE FACULTY

The nativist view holds that this innate capacity, the genetic endowment for language, is UG. It constitutes the first of the three factors (cf. Table 2.3), the initial state of the human language faculty, whose growth and further development depends on the other two factors as well.

UG is neither a language nor a set of grammatical properties; it is “[t]he theory of the genetically based language faculty” (Chomsky 2017: 3). Part of the above-mentioned confusion surrounding UG stems from its use in at least three distinct senses that are rarely kept apart (Navarro 2024): UG as a theory of the capacity for language (what the organism brings to acquisition), UG as a theory of the knowledge of language (what the mature speaker knows), and UG as a genetic factor in the biological sense (what is encoded in the genome). Trettenbrein

(2015) and Mendivil-Giró (2018) document this oscillation carefully. Keeping the three senses distinct matters because critics of UG may be right about one reading while wrong about another—and defenders of UG may be defending a different sense than the one under attack.

Adger and Svenonius' (2015) taxonomy of domain applicability (see Sect. 2.4.2) may be a useful method for locating the minimalist UG. A grammar reducible to Merge and interface conditions is less obviously domain-specific (dedicated exclusively to language) than the rich rule systems of Government-Binding theory (Chomsky 1981a, Chomsky & Lasnik 1993), as put in perspective by Berwick (2011a). If Merge is a general hierarchical combination operation ('Basic Property', but also Hornstein 2009), the minimalist language faculty may in fact be closer to being domain-specialized: mechanisms originally shared with other cognitive domains, recruited and refined for linguistic purposes. This pushes CLN and CLB closer than the standard diatribes suggest—and it is worth noting that even the most austere CLN position may be more compatible with domain-specialization than with domain-specificity.

Perhaps a more amenable, bridge-building name for the starting point to address Humboldt's problem could be Arbib's (2002) notion of the "language-ready brain" (see also Boeckx & Benítez-Burraco 2014a and Chap. 6), which carries less terminological, even theory-specific baggage. In the generative paradigm, the human faculty of language takes center stage. Originally, the term itself made clear that this is a domain-specific entity unique to the species, though this is becoming less clear with Hauser et al.'s (2002) distinction into FLB vs. FLN.

The FLB/FLN distinction is significant for the present discussion in several ways. FLB encompasses all cognitive and communicative capacities that humans share with other species (vocal learning, recursion in non-linguistic domains, theory of mind, and so on). FLN is whatever remains after subtracting FLB; by Hauser et al.'s original conjecture (see also Fitch 2010a), this amounted to recursion alone, though that claim has since been extensively debated.

What makes the distinction useful for bridging CLN and CLB is that FLB is, by definition, domain-general: It consists of capacities that are not specific to language. Only FLN is genuinely language-specific—and if FLN is lean (perhaps just Merge as the Basic Property), then the language-specific component of the first factor is correspondingly small. This maps directly onto the domain-distinction: A rich FLN corresponds to domain-specificity, a lean one to domain-specialization.

CLB, which is generally committed to domain-generality, may now find in this trajectory of minimalist reduction something it can engage with—if not agreement, then at least genuine proximity. Domain-specificity, domain-specialization, and domain-generality are hence three distinct empirical positions, making different predictions about what neural and developmental evidence should look like. Keeping them distinct prevents the common conflation of ‘generative linguistics’ with ‘strong modularity’ and opens space for precisely the kind of dialogue this book strives for. A domain-specialized perspective, in this reading, is the productive middle ground, one that CLN can inhabit without abandoning its commitment to a biologically endowed faculty, and that CLB can approach without conceding a fully modular, encapsulated language organ.

The debate over the content of FLN remains one of the most productive in contemporary biolinguistics, precisely because it forces the field to be explicit about what, if anything, is genuinely unique to human language—and therefore what requires a specifically biolinguistic explanation as opposed to one invoking general cognitive or evolutionary mechanisms.

Much effort has subsequently been put in so as to give each of these faculties *substance*, including which aspects of human language must be part of FLN. The literature on both sides is rich. The substantive question (‘What belongs to FLN?’) has generated competing proposals ranging from Merge (recursion; Hauser et al. 2002, also Fitch 2010a) or Merge plus Labeling (determining which element projects; Chomsky 2008, Hornstein 2009) to Merge without Labeling (an assumed third factor; Chomsky 2013) or Labeling only (Merge-only as concatenation shared with other species; Murphy 2015) to Minimal Search instead (under a third-factor interpretation; Ke 2019, 2024) or a cluster of other properties that may include displacement, hierarchical structure, and discrete infinity. The *Strong Minimalist Thesis* (Chomsky 1998, 2004; see also e.g. Freidin 2021, Chomsky et al. 2023) allows us to sharpen the point further: The language-specific endowment reduces to little more than recursion together with interface conditions—in other words, “Interfaces + Recursion = Language” (Chomsky 2010: 46, guiding the title of the volume collected by Sauerland & Gärtner 2007).

This UG-reframing has a historical shape. The shift from 1980s GB-era generativism to minimalism can be captured as a reversal of the guiding question (Al-Mutairi 2014, Smith & Allott 2016). Where the

1980s asked, “How much must be attributed to UG to account for language acquisition?” (Chomsky 2007: 4), the 2000s ask instead: “How little can be attributed to UG while still accounting for the variety of I-languages attained?” (*ibid.*). Such a reversal (i.e., the ‘lean-versus-rich question’ above) is not a merely technical change but a shift in explanatory strategy. It is consequential well beyond the theory of grammar: How one answers the question determines what kind of evolutionary story one needs to tell for Wallace/Darwin’s problem (Chap. 7), and what kind of neural implementation one expects to find for Broca’s problem (Chap. 6).

The original suggestion was that what makes human language different from all other communication systems is recursion, in order to “make infinite use of finite means” (but see Fitch 2010a for clarifications). A vivid illustration is the ‘anti-missile series’ (from Lasnik 2000: 15): *anti-missile*, *anti-missile-missile*, *anti-anti-missile-missile-missile*. Formally, this can be captured as ‘antiⁿ missileⁿ⁺¹’. Each step is grammatical; there is no longest member. Crucially, tracking what came earlier requires memory—a simple finite-state device cannot generate the series once it grows. This is precisely the step from Type 3 (finite-state, no memory) to Type 2 (context-free, push-down memory) in the formal language hierarchy (see Hunter 2021 for overview): Natural language requires at minimum the computational power to track non-adjacent dependencies, which in turn requires recursive hierarchical organization.

And the same hierarchy (Chomsky 1956, 1959) extends upward: Type 1 (context-sensitive) grammars require multiple memory stacks, and some constructions involving long-distance dependencies arguably push natural language to that level. It reappears in Chap. 7 as a methodological tool for evaluating complexity in the archaeological record. The recursion hypothesis was provocative not because it was wrong, but because it was too simple. The relevant question for FLN is not whether recursion exists in other species—but whether the specific deployment of recursive functions of specific implementations in the service of unbounded semantic composition is unique to humans; and here the evidence remains genuinely contested.

‘Recursion’ figures in three distinct denotations in the literature, as reviewed by Lobina (2017), among others (e.g., Sauerland & Trotzke 2011); conflating them has generated much unnecessary controversy. As a property of algorithms, recursion is a process “by which the value of a function is defined by the application of the same function applied

to smaller arguments” (Dean & Naibo 2025: 1). As a property of syntactic representations, recursion means self-embedding (a structure of type *X* appearing within a larger structure of the same type). And as a property of cognitive processes, recursion refers to the parser calling itself during execution. Lobina argues that the grammar is recursive in the first sense, some syntactic representations are recursive in the second sense, but the parser operates iteratively in the third sense, building hierarchical constituents through a push-down stack rather than through recursive self-calling.

Keeping these three senses apart dissolves several apparent conflicts. Whether starlings process recursive sequences (e.g., Gentner et al. 2006) or whether Pirahã lacks recursion (as argued by Everett 2005, but see Pesetsky et al. 2009) are questions that only become manageable once the relevant sense of recursion is specified. The equivocation at the heart of the Pirahã debate has been clearly diagnosed: Recursion is a property of the language faculty (the meta-capacity to acquire grammars), not a requirement on every acquired grammar (Hornstein 2012, 2016). After all, a Pirahã grammar without recursive structures is fully compatible with a Pirahã speaker whose faculty of language can acquire any recursive grammar.

However, it is not only the computational property of recursion (which could be implemented in language through Merge, the structure-building operation), it is crucially the fact that these concatenated or merged building blocks of language are labeled (‘Labeling Algorithm’); see also Hornstein (2009). Labeling is what gives syntactic objects their interpretive properties. It is what allows a phrase to be identified as a noun phrase, verb phrase, and so on, and thereby to receive a compositional interpretation at the conceptual-intentional interface.

The mechanism is worth making explicit. On Hornstein’s (2024) account, labeling does more than annotate a structure: It closes the combinatorial operation in the domain of the lexical atoms, mapping each complex expression to an atom and so forming equivalence classes of expressions with lexical atoms as their moduli—the formal reconstruction of classical constituency, in which a phrase behaves, for the purposes of further combination, as a unit of a finite, named type (noun phrase, verb phrase, and so on). It is this closure that yields recursive hierarchy: Since a labeled phrase can be treated as if it were a lexical atom, the output of one application of Merge becomes an eligible input to the next. Bare combination alone would deliver ever-larger sets and

labeling lets the system re-enter its own outputs, hence generate the unbounded hierarchical structures that constitute the Basic Property.

On this view, the linguistically specific ingredient of the narrow faculty is not the combinatorial operation, which is cognitively generic, but the labeling that renders its products re-usable. I refer the reader to Hornstein (2009), expanded further in Hornstein (2024), for a very clear exposition of the issues and an overview of the literature. He develops a generative biolinguistic framework that links to the other foundational questions as well (see also e.g. Murphy 2015, Blümel 2026).

At the center of that framework is a strong claim about the reach of the operation. Hornstein's 'Fundamental Principle of Grammar' holds that every grammatical dependency is established by Merge; there is, on this view, no other way to couple two syntactic expressions grammatically. The 'Extended Merge Hypothesis' says that the grammatically significant relations are all mediated by Merge, so that what must be innately specified reduces, in the limit, to Merge itself: "It's Merge all the way down and all across the grammar" (Hornstein 2024: 210).

It is the working-out of that principle across the construal phenomena; not only structure-building and displacement but also binding, control, case, and θ -assignment are treated as instances of Merge (as the labeling-inclusive operation just laid out) rather than as outputs of separate modules or rule types. The pay-off is the very reduction this chapter has been tracking: If dependencies that earlier frameworks distributed across many sub-theories all reduce to one operation, the language-specific endowment shrinks accordingly, and the question of what must be innate becomes correspondingly sharper. Hornstein's framework is notable for its integrative ambition: By deriving syntactic phenomena from a minimal set of operations, it connects Humboldt's problem directly to Plato's and Broca's problems (Chaps. 4 and 6).

This falls within the (minimalist) architecture of the Basic Property, which connects the computational system of human language (which concatenates or merges lexical items, concepts, and formal features) to the interpretive and externalizing interface components, the conceptual-intentional system (feeding off an expression's Logical Form or LF for short) and the sensorimotor system (feeding off an expression's Phonetic/Phonological Form or PF), respectively. This 'Y-model' architecture (cf. Chomsky 1993, Adger 2021), so called because the syntactic derivation splits at a branching point into LF and PF, is where Humboldt's problem connects most directly to Descartes's problem (Chap.

5): The mapping between the computational system and the sensorimotor system is what language use in real time requires.

3.4 COMPETENCE AND PERFORMANCE

Returning to the notion of ‘knowledge of language’, I will next sketch some additional notions which divide cognitive approaches to language and which will accompany us for the remainder of this book. The first is the competence–performance distinction, introduced in Chomsky (1965); it is an important concept in CLN theorizing, though in early CLB it had already been labeled as one of the “false dichotomies” (Lan-gacker 1987: 154). The rejection is not merely terminological. For Lan-gacker, the competence/performance distinction rests on a commitment to the existence of an abstract, context-independent grammar that usage-based linguistics denies. If what speakers know is inseparable from how they use the language, then there is no level of ‘pure’ competence insulated from the pressures of performance.

It is intimately tied to Humboldt’s problem as it explicitly refers to (one aspect of) *knowledge* of language. However, in CLB, “knowledge of language is derived from patterns of language use, and further, that knowledge of language is knowledge of how language is used” (Evans & Green 2006: 108), which makes it hard to address Humboldt’s problem in these terms. One way to retain something like it within a CLB framework is to ask not ‘What do speakers know?’ but ‘What structured representations do speakers build up through use, and how do those representations enable novel production and comprehension?’ This re-framing preserves the spirit of the question while grounding the answer in usage rather than in an innate grammar.

While *competence* refers to speakers’ tacit knowledge of their language, *performance* reflects language understood as a set of actual utterances produced by native speakers (see Crocker 1996 for a formal but neutral definition). A distinct but closely related dichotomy (and, by assumption, equally rejected in CLB as “false”) is the second notion, that of ‘I-language’ versus ‘E-language’. The parallel rejection is telling: Both dichotomies draw a boundary between an internal, abstract system and its external manifestation. CLB refuses that boundary at both levels, insisting that the ‘external’ phenomena are not noise to be filtered out but the substance from which linguistic knowledge is built.

From a CLN perspective, what both accounts are describing, however differently, is a finite means of generating potentially infinite linguistic structures: Merge builds hierarchical syntactic objects recursively; constructions, at higher levels of abstraction, are themselves schematic patterns that combine in order to generate novel utterances. The question is not whether there is a generative mechanism, but what kind it is and how much it is shared with other cognitive domains.

In the hypothesized context, set up by Chomsky (1965) simply for illustration, of the “ideal speaker–listener, in a completely homogeneous speech community” (p. 3), speakers know their (speech community’s) language perfectly. Thus, temporary conditions such as memory limitations, distractions, or production errors can be ignored for purposes of linguistic analysis in applying a speaker’s knowledge of this language in actual productions; they are mere performance factors that are not relevant for grammaticality or acceptability. Linguistic competence denotes speakers’ knowledge of their language or grammar, the abstract rule system they have mastered to produce and understand an infinite number of sentences, and to recognize grammatical mistakes and ambiguities. The idealization to a homogeneous speech community is, of course, exactly that—an idealization. CLB approaches take the heterogeneity of real speech communities seriously as a feature of language rather than a complication to be abstracted away from; from a CLN perspective, the idealization is methodologically justified because it isolates the abstract computational system from confounding factors.

Construction Grammar (Goldberg 1995, 2006), possibly the most fully developed CLB alternative to rule-based generative grammar, approaches this divide differently. Rather than positing an abstract competence system insulated from use, it treats constructions (form–meaning pairings at all levels of abstraction) simultaneously as units of grammatical knowledge and as units of processing: What speakers know is what they use, and vice versa. From a CLN perspective, this collapses a distinction that is necessary for explaining why speakers produce and comprehend sentences they have never encountered before; from a CLB view, that explanation is available without positing a separate competence level, through the generativity of schematic constructions.

Performance refers to language seen as a set of specific utterances produced by native speakers, for example, as encountered in a corpus. The utterances of performance will contain features irrelevant to the abstract rule system (e.g., hesitations and unfinished structures), arising

from the various psychological and social difficulties acting upon the speaker (e.g., lapses of memory), or from biological limitations (e.g., as pauses being introduced through the need to breathe). The CLN view is that these features are theoretically uninteresting for the purposes of grammar description: They tell us about the human organism as a physical and social creature but not about the computational system that underlies linguistic knowledge. CLB, by contrast, finds in precisely these features—the pauses, the repairs, the incomplete structures—evidence about how language is organized in the mind and deployed in real time.

3.5 I-LANGUAGE AND E-LANGUAGE

A more fine-grained distinction can be drawn between two further conceptions of the nature of language, the above-mentioned distinction of ‘I-language’ versus ‘E-language’. This distinction (Chomsky 1986) is, in many ways, a more theoretically precise version of the competence/performance divide. Where competence and performance are primarily epistemological distinctions about what we know versus what we do, I-language and E-language are ontological: They concern what kind of thing language is.

Whereas *I-language* is used to refer to the language faculty (with ‘I’ denoting ‘intensional’, ‘internal’, ‘individual’), *E-language* seems to be “a covering term for a loose collection of theories that take language to be a shared social object, established by convention, and developed for purposes of communication” (Ludlow 2011: 45). It “is part of the outside world [...] and it includes the kinds of things that a child might hear”—but E-language “does not reflect any single system.” After all, “[n]o two children hear exactly the same things and no two people speak exactly the same way” (Lightfoot 2006: 12). E-language, on this view, is not a unified object at all; it is an abstraction over the linguistic behavior of a community that masks the internal states of individual speakers. This is why CLN insists that E-language cannot be the object of scientific linguistic inquiry: Scientific explanation requires a well-defined object, and a community-level abstraction does not qualify.

As Roberts (2017a) sums it up, “UG is a more general notion than I-language; indeed, it can be defined as ‘the general theory of I-language’ (Berwick and Chomsky 2016: 90)” (p. 7). And Isac & Reiss (2013) characterize this as “the biolinguistic approach that focuses on the study

of I-language, language as knowledge that is encoded in the human mind, and not on a notion of language as a corpus of texts or as a set of observed behaviors” (p. 91). This is a strong claim, and CLB researchers have tried challenging it on empirical grounds: Corpus data reveal systematic patterns—collocations, frequency effects, constructional generalizations—that appear to be psychologically real and that any account of linguistic knowledge must accommodate. The CLN response is that these patterns belong to performance and use, not to the grammar proper.

But from a CLB perspective, one might not be so much interested in ‘knowledge of language’ and also reject the I-/E-language dichotomy, and identify other patterns provided by a language corpus. In this case, “what differs is whether these patterns are assumed to be rooted in biology or not, whether they are assumed to be dependent or independent of psychological states of individuals” (Isac & Reiss 2013: 70). This is, ultimately, an empirical question as much as a theoretical one—and one where progress may be made from both sides: CLN approaches taking seriously the role of frequency and statistical learning in parameter setting are moving towards the kind of data CLB approaches privilege; CLB approaches developing explicit computational models of construction learning are moving towards the precision CLN demands.

3.6 EMERGENTIST PERSPECTIVES

One could argue that much of the picture sketched in Sect. 3.3 describes the (generative) linguist’s bread and butter. If so, the properties of this system can be studied disregarding biological considerations (generativism without a biolinguistic commitment) or it can be continued to be investigated with an appreciation for the biological underpinnings of language (biolinguistics in the narrow sense). From this vantage point, that ‘bread and butter’, the description and explanation of syntactic phenomena across languages, constitutes itself a contribution to biolinguistics, whether or not the linguist doing it frames it in those terms.

This approach seems to stand in stark contrast to CLB approaches, where language is investigated, as Evans and Green (2006) put it, “for its own sake” in an “attempt to describe and account for its *systematicity*, its *structure*, the *functions* it serves and how these functions are realised by the language system” (p. 5).

Cognitive linguists assume further “that language reflects patterns of thought” and as such “offers a window into cognitive function, providing insights into the nature, structure and organisation of thoughts and ideas” (*ibid.*). This position aligns CLB with a broadly neo-Whorfian tradition, in which the structure of language shapes (or at least reflects) the structure of thought. CLN, by contrast, treats the conceptual base—that is, ‘thought’—as largely language-independent: The conceptual-intentional system exists prior to and independently of the linguistic system that interfaces with it.

This CLN side of the contrast is worth stating more directly. The link between language and thought is of a different kind altogether than the neo-Whorfian one just described. Chomsky takes the language faculty to be, in the first instance, a ‘language of thought’: a system whose primary function is the construction of structured thought, with externalization a secondary, ancillary process. This view is stated with increasing emphasis from Chomsky (2007, 2013) to its most explicit statement (Chomsky 2016), and glossed by Hinzen (2011) as the claim that “the computational system of language [...] equates with having a ‘language of thought’, with externalization being a derivative affair” (p. 499).

Note that the claim here is not that a particular language would mold its speakers’ cognition, but that the generative procedure yielding linguistic structure just *is* the engine of structured thought. Thus the conceptual atoms it combines may be largely language-independent, but their unbounded hierarchical combination is not. This understanding of *language of thought* should be kept apart from Fodor’s (1975), where the phrase names a distinct innate symbolic medium underlying cognition (which he calls ‘Mentalese’); for Chomsky, it characterizes the design and primary function of the language faculty itself.

The empirical backbone of the CLB position is the sheer diversity of the world’s languages. Evans and Levinson (2009) make the point most forcefully. In their view, the search for substantive language universals has largely failed, and what is most striking about language, biologically speaking, is not what languages share but how radically they differ—at every level, from phonology to syntax to the semantics of space and number. On this account, languages are ‘bio-cultural hybrids’ and as such shaped by cultural-historical descent rather than read off a fixed UG. It is this diversity, not a hypothesized universal core, that cognitive science most needs to explain. A complementary observation comes from the brain, to be developed in Chap. 6).

Here a CLN reply is to relocate the target. After all, what Evans and Levinson document is variation in substantive, surface-level patterns (so-called Greenbergian universals), whereas the minimalist faculty has been pared down precisely to abstract away from such patterns. If all that is universal is the capacity to generate hierarchically structured expressions (such as Merge plus interface conditions), deep surface diversity is not an embarrassment but an expectation: A lean UG and rich cross-linguistic variation are, on this view, perfectly compatible; the disagreement is over whether anything law-like survives the variation, and at what level of abstraction (we return to this in Chap. 7, where the same diversity facts bear on what exactly must have evolved).

A further consequence of the thought-first conception is that the core computational system is modality-independent, whether language is externalized as speech or as sign, a point we return to at the close of this chapter. For neural evidence, there is a recent review (Trettenbrein et al. 2025), an fMRI meta-analysis (Trettenbrein et al. 2021), and a short overview (Trettenbrein & Friederici 2026).

However, even Geeraerts and Cuyckens (2007) agree in the ultimate goal: “As cognitive enterprises, Cognitive Linguistics and Generative Grammar are similarly interested in those mental structures that are constitutive of knowledge” (p. 6). The major difference breaks down to where these mental structures lie (in “natural language itself” or in “genetic endowment of human beings that enables them to learn the language”) and what kind of knowledge we are talking about (knowledge of the world vs. knowledge of language). This is, in a sense, the deepest version of the CLN–CLB divide. For CLN, what theoretical linguistics investigates is knowledge of language, a specific, bounded cognitive object with its own architecture. For CLB, it is how language encodes and transmits knowledge of the world.

These are not competing answers to the same question; they are answers to different questions. Chomsky and Fodor operate at different analytical levels and are complementary rather than competing:

Fodor's claims are about processing; Chomsky's are about analytic/competence units. The language faculty as generativists conceive it is a discrete system that is accessed by both input (parsing) and output (production) systems, and is itself neither an input nor output module: as Chomsky says, no one speaks only Japanese and understands only Swahili.

(Allott & Smith 2021: 536)

This fits the broader minimalist picture: On the three-factor view, the language module is “less encapsulated and domain-specific than was previously thought and less so than a full-fledged Fodorian module” (Roberts 2017a: 18); a real concession, given that informational encapsulation was for Pylyshyn (1999) the signature property of a module.

Geeraerts & Cuyckens (2007) sum up the epistemological tension as follows: “Whereas Generative Grammar is interested in knowledge of the language, Cognitive Linguistics is so to speak interested in knowledge through the language” (p. 6). Sperlich (2006: 59) puts this a little longer: “Various schools of thought, such as functional and pragmatic linguistics, take much more store in communication as the driving force of language. Cognitive linguistics, although closer to Chomskyan linguistics, claims that language competency derives more directly from general cognitive functions, thus not stipulating a separate interface such as Chomsky’s UG.” His formulation captures the CLB position with useful precision: It is not that CLB denies the relevance of cognitive mechanisms to language but that it insists those mechanisms are not language-specific. The dispute is therefore not about whether cognition matters for language (both traditions agree it does)—but about whether there is a level of cognitive organization dedicated exclusively to language.

By so doing, “language is assumed to reflect certain fundamental properties and design features of the human mind” (Evans & Green 2006: 5), as opposed to reflect the fundamental properties of a dedicated computational system. Chomsky has always held that “one must deplore the common tendency to insist that the mechanisms of language must be special cases of ‘generalized learning strategies’ or general cognitive mechanisms of some sort” (Chomsky 1976: 23), even though at the time his resentment was mostly directed at psychologists. This resistance was not arbitrary: It was motivated by the poverty-of-stimulus argument (Chap. 4). If language were simply the product of general learning strategies operating over experience, the uniformity of grammatical knowledge across children, despite variation in input, would be difficult to explain. This remains the strongest CLN argument against CLB, and one that CLB approaches continue to actively contest.

Yet this is not to say that such strategies and mechanisms do not exist or even play a role for language; they do, but in an ancillary way as co-opted resources. These are third-factor considerations. A point for collaboration between CLN and CLB biolinguists would be to investigate

these computational and cognitive processes, whether held as the driving force behind language or as third factors, in order to pinpoint the cognitive (neuro)biological basis of language *use* at least. Yang's (2016) work is particularly instructive here. By developing formal models of how children identify productive morphological patterns within the input they receive, he shows that the boundary between 'innate' and 'learned' is a continuum shaped by the interaction of the child's computational capacities with the statistical structure of the input—third-factor territory where CLN–CLB collaboration is most plausible.

3.7 SUMMARY

In sum, the first factor from Table 2.3 (Chap. 2) assumes a species- and domain-specific genetic endowment for language (UG), equipping us with a language-ready brain, which is one aspect of how to address Humboldt's problem. Nativist biolinguists take this to constitute part of the human language faculty (FLN), but for language to develop and be used, other factors play important roles as well, including cognitive procedures. The picture is thus one of biological endowment interacting with environmental input and general cognitive principles to produce the specific grammar of a language—a picture that is in principle richer and more flexible than the stereotype of a rigid, innate grammar sometimes attributed to CLN, and one that creates genuine space for engagement with CLB on the second and third factors.

For CLB, the specificity of a faculty or module within the cognitive system dedicated to language is rejected. The language-ready brain may be conceived as the general cognitive system that subsumes language as well. The object of study is "natural language itself". What is species-specific is not a language faculty but a configuration of general capacities—joint attention, imitation, analogical reasoning, social cognition—that together enable the acquisition and use of language.

Humboldt's problem is then just as relevant for CLB as for CLN, but the road to an answer looks different: "What is knowledge of language?" could be trivially replaced by "What is language?"—perhaps a better way to approach this problem, however, would be via Reboul's (2017) reformulation of Tinbergen's four questions for language (Table 2.6, Chap. 2): "What are the mechanisms of language?" Such a reformulation is more productive because it does not presuppose a domain-

specific faculty, and therefore would more easily invite answers from both CLN and CLB. The mechanisms of language, on a CLN account, include Merge, possibly plus the Labeling Algorithm and/or Minimal Search, and the interfaces with LF and PF; on a CLB account, they include categorization, schematization, analogy, and the entrenchment of patterns through use. Both sets of answers are in principle empirically evaluable—and the dialogue between them is what makes biolinguistics, in the strong sense, genuinely interdisciplinary.

The answers cognitive linguists will reach might look very different from generative efforts, but “elucidation of their functional motivation does not excuse us from actually describing grammatical constructions in full and explicit detail” (Langacker 2012: 7). Humboldt’s problem (‘What is knowledge of language?’) turns out to be, on closer inspection, several problems at once: a problem about the nature of the object (I-language or natural language?), about the form of the explanation (domain-specific or domain-general?), and about the standards of evidence (acceptability judgments or corpus patterns?).

A CLN account of Humboldt’s problem extends, *mutatis mutandis*, beyond syntax: Samuels (2011) develops a substance-free, minimalist phonology in which the computational system governing phonological form is innate, universal, and modality-independent—indifferent to whether language is externalized in spoken (auditory-vocal) or signed (visual-manual) modalities. That a single computational system should underlie both speech and sign is itself a clear prediction of the generative model (Berwick et al. 2013): If externalization is secondary, then the modality of expression is not where the core of language resides.

The next chapter takes up Plato’s problem, how that knowledge is acquired, and the chapters beyond approach the remaining foundational questions with these divisions in mind, asking at each turn not only what the answers look like from within CLN but how they might be approached, and perhaps partially shared, across the CLN–CLB divide.

Plato's Problem

Abstract This chapter presents the second foundational question: *How is that knowledge of language acquired?* In essence, this amounts to the logical problem of language acquisition. The brief survey of language acquisition studies focuses on the poverty-of-stimulus argument, which for CLN invariably leads to the assumption that humans are genetically endowed with an innate capacity for language. CLB approaches to the acquisition of language take a different path; they, too, need to address Plato's problem, though. Some possible avenues are sketched here. We will end up with a biolinguistic perspective on language acquisition that takes seriously both the nativist argument from poverty of the stimulus and the usage-based emphasis on input, interaction, and general cognitive mechanisms asking, in each case, what an inclusive biolinguistics requires us to say about how children come to know their language.

Keywords Logical problem of language acquisition • Poverty of the stimulus • Genetic endowment • Critical period • Niche construction • Usage-based acquisition

Plato (429?–347 BCE) was one of the most influential thinkers in the history of Western civilization. A student of Socrates, he was an Athenian philosopher of the Classical period who founded the Academy, the first institution of higher learning in the Western world. While perhaps less amenable to Cartesian philosophy than, say, his student Aristotle (which may be debatable), his impact on the history of philosophy was nevertheless enormous. Plato's most important contribution is the theory of Forms, according to which abstract objects exist independently of space and time ('Platonism').

4.1 Language Acquisition and Development

Where CLN starts from the poverty of the stimulus and works towards an innate endowment, CLB starts from the richness of the input and works towards the structures it can sustain (cf. Sect. 2.4). Within CLN approaches, it is Plato's problem, "explaining how we can know so much given that we have such limited evidence" (Chomsky 1986: xxv), which lies at the heart of early generative conceptualizations of human language (or the faculty of language), and to a large extent also directly builds on Eric Lenneberg's pioneering work.

This chapter will present the most relevant empirical findings from (very early) first language acquisition, touching on bilingualism and the more recent focus on non-canonical contexts of language acquisition, from creolization and heritage language grammars to sign language emergence, all the way to atypical and/or impaired language. Emphasis will be put on the poverty-of-stimulus argument, both conceptually and empirically. This leads to a presentation of different theories of learnability. While I couch the issues in the wider perspective of Chomsky (2005), we will see that especially so-called third-factor considerations are closer to common CLB assumptions than many may think. I will explore what this could mean for a unified biolinguistic enterprise from both a strong generative CLN and a non-nativist CLB perspective.

The philosophical roots of Plato's problem run deep. In the context of linguistics, the problem has been given both a Platonist interpretation (language as an abstract object existing independently of any mind, as per Katz 1984) and a biolinguistic one, where the emphasis falls not on abstract objects but on the biological endowment that makes linguistic knowledge possible in the first place. Postal (2009) offers a pointed critique of the biolinguistic framing, arguing that Chomsky's ontology is "incoherent" (his characterization), yet a note of internal-generativist caution may indeed be in order. For example, Atkinson and Al-Mutairi (2012) question how much weight the appeal to 'conceptual necessity' can bear. Whether Merge is a property forced on any computational system or a contingent biological fact about humans is left unresolved by that rhetoric, as is the prior question of which properties belong to UG and which merely follow from it. The point is raised from within the minimalist program rather than against it, but it complicates a reading on which the lean endowment of UG (cf. Sect. 3.3) is simply handed over by conceptual necessity.

Going back a rather long time, the problem has its roots in Plato's dialogue between Socrates, Meno, and the slave boy. It was originally taken from Russell (1948): "How comes it that human beings, whose contacts with the world are brief and personal and limited, are able to know as much as they do know?" (Chomsky 1986: xxv, Chomsky 1988: 3–4). With reference to language, it relates to the above-quoted "problem of explaining how we can know so much given that we have such limited evidence." The elegance of this formulation is that it makes Plato's problem a logical problem, not merely an empirical one. It is not simply that children happen to acquire language quickly; it is that the richness and systematicity of the resulting grammar cannot, in principle, be derived from the available input alone—unless the child brings substantial prior knowledge to the acquisition task. The uniformity of outcomes across variable input, and the speed with which they are attained, point towards a further, independent argument: The input itself is too impoverished to explain the resulting grammatical competence.

Plato's problem refers to the paradox of rapid, uniform language acquisition by the infant child despite limited and imperfect input (as per Chomsky 1986), an observation closely related to the '*poverty of (the) stimulus*' (Chomsky 1980). Chomsky (1965) introduced the idea of the *Language Acquisition Device* as a hypothetical, innate mechanism that enables infants to acquire language in the first few years of their lives. In the generative tradition, this subsequently developed into the full-blown *Innateness Hypothesis*: Humans are biologically endowed with the means to allow for "language growth" (Chomsky 1975: 11).

UG pertains to the set of innate principles and parameters that constitute the initial state of the language faculty (or FLN, see Chap. 3). UG thus provides the structural foundation for all human languages, explaining not only the rapidity but also the uniformity of language acquisition in children as well as the diversity of languages across cultures. The underlying issue for cognitive linguistics is whether research aims at detailing the acquisition of 'language' (through interaction, use, etc.) or the role of 'knowledge of language' in the acquisition process.

4.2 THE LOGICAL PROBLEM OF LANGUAGE ACQUISITION

Essentially, the *logical problem of language acquisition* (Baker 1979) can be expressed through the poverty of stimulus. Such an approach

clearly falls within a CLN nativist purview (for responses to a range of criticisms, see e.g. Berwick et al. 2012). It can be asked in other words: “How do we come to have such rich and specific knowledge, or such intricate systems of belief and understanding, when the evidence available to us is so meager?” (Chomsky 1987: 33). The problem is *logical* in the sense that it concerns what can, in principle, be learned from the available evidence—not what is, as a matter of fact, learned by any particular child. Even an idealized learner with unlimited time and perfect memory could not, on the basis of positive evidence alone, converge on the grammar that speakers actually attain. The inference from input to grammar requires something beyond the input: And that something, for CLN, is UG.

The classical formulation of Plato’s problem assumes an idealized monolingual learner acquiring a uniform target grammar from consistent input. Aboh (2020), for example, challenges this idealization on empirical grounds: The great majority of the world’s language learners are multilingual, exposed from birth to multiple grammars, from registers and dialects to typologically distinct languages. The argument from the poverty of the stimulus is best understood as applying to the initial state; that is, whatever UG contributes must be sufficient to enable grammar construction from any distribution of input, not just from idealized monolingual data.

Rephrasing de Vos’ (2015) two-pronged approach to answering Humboldt’s problem, Plato’s problem can also be approached in two ways: (i) what we know to empirically constitute the logical problem of language acquisition and (ii) what theoretical tools, systems, and structures are required to explain those facts. Applied to Plato’s problem, this approach yields the following questions. On the empirical side, what does the child’s acquisition trajectory actually look like, and how does the child’s knowledge outstrip what the input provides? On a theoretical level, what mechanism could, in principle, account for the transition from infant to competent speaker? The straightforward CLN answer to the second question is UG; CLB offers a different answer, invoking, for example, statistical learning, social cognition, and the gradual entrenchment of usage-based schemas.

This is where the apparent opposition between innate structure and statistical learning begins to dissolve. As Pearl (2021) argues, the two are complementary rather than competing: UG can be understood as structuring the hypothesis space the child brings to the data, while

statistical learning does the work of searching that space and converging on the target grammar (see Pearl & Lidz 2013 and Pearl & Goldwater 2016). On this view, the poverty of the stimulus and the power of distributional learning are not rival answers to Plato's problem but two halves of a single account.

A staple in the history of generative grammar is the argument from the poverty of the stimulus (Chomsky 1978, 1980), that "our experience far underdetermines our knowledge and hence that our biological endowment is responsible for much of the derived state" (Lasnik & Lidz 2017: 221). The phrase 'derived state' is important here: It refers to the steady-state grammar, the mature linguistic competence which the child attains as the outcome of language acquisition. The argument is that this derived state is systematically richer than anything the input could have warranted, and that the gap between input and output is precisely the measure of what the biological endowment contributes.

The input to the child is degenerate in two senses (Chomsky 1965), in *scope* and in *quality*: "[T]he input cannot provide evidence about all possible sentences (sentence–meaning pairs, sentence structures, etc.) that the child will encounter [and] the input itself does not contain information about the kinds of representations that should be used in building a generative grammar of the language" (Lasnik & Lidz 2017: 222). Yet, except for pathological conditions, all children develop and master their native language(s) at a young age.

Here a useful analytical ladder comes in the form of three degrees of adequacy (Chomsky 1964). A grammar is *observationally adequate* if it correctly lists the sentences of a language. It is *descriptively adequate* if it correctly characterizes what native speakers tacitly know. And it is *explanatorily adequate* if it provides a principled basis for how that knowledge could have been acquired. CLB approaches have made their strongest contributions at the observational and descriptive levels; the argument from the poverty of stimulus is precisely the claim that these levels are insufficient; any theory which cannot explain acquisition has not explained language knowledge (see also Chap. 7). The debate between CLN and CLB can, at its core, once again be framed as a debate about whether explanatory adequacy requires a domain-specific endowment or whether the above-mentioned gap can be closed by domain-general mechanisms.

That is, the argument from the poverty of the stimulus is "based on the observation that there is a significant 'gap' between what seems to

be the experience facilitating first-language acquisition and the nature of the linguistic knowledge which results” (Roberts 2017a: 28). Roberts continues with “[a]n early statement of this argument”:

A consideration of the character of the grammar that is acquired, the degenerate quality and narrowly limited extent of the available data, the striking uniformity of the resulting grammars, and their independence of intelligence, motivation, and emotional state, over wide ranges of variation, leave little hope that much of the structure of the language can be learned by an organism initially uninformed as to its general character.

(Chomsky 1965: 58)

But the connection between the problem of language acquisition and the formalization of linguistic theory goes back further. Lasnik and Lidz (2017: 221) suggest that “a minimum explanatory criterion for a theory of linguistic knowledge and its acquisition” may already be traced back to at least a decade prior:

A speaker of a language has observed a certain limited set of utterances in his language. On the basis of this finite linguistic experience he can produce an indefinite number of new utterances which are immediately acceptable to other members of his speech community. He can also distinguish a certain set of ‘grammatical’ utterances, among utterances that he has never heard and might never produce. He thus projects his past linguistic experience to include certain new strings while excluding others.

(Chomsky 1955: 61)

The first passage already identifies the key features of the acquisition phenomenon that demand explanation: Language acquisition is rapid, uniform across children and across input conditions, and surprisingly independent of general cognitive ability.

The quality degeneracy point is particularly important. It is not merely that children receive too little data, but that the data they receive is of the wrong kind to determine the abstract representations that the grammar requires. No amount of exposure to surface strings, however extensive, will tell the child that syntactic dependencies are computed over hierarchical structure rather than linear order; yet children systematically ‘know’ this, even for constructions they have never encountered.

A concrete case makes the logical point vivid. To form a yes/no-question, English pairs a declarative clause (say, *The man is tall.*) with an interrogative in which an auxiliary appears at the front (*Is the man tall?*). A learner exposed only to such simple cases is not thereby told whether the rule is structure-dependent (“front the auxiliary of the main clause”) or structure-independent (“front the first auxiliary in the string”). The two diverge on sentences such as *The man who is tall is happy*, where the structure-independent rule yields the ungrammatical **Is the man who tall is happy?* Children do not make this error, even though the disambiguating sentences are vanishingly rare in their input (Legate & Yang 2002). What rules the structure-independent hypothesis out cannot be the input; it must be a prior constraint confining the child to structure-dependent rules.

Over time, this has turned into more general learnability issues, questions of the nature of the learning problem for human languages. As cited by Roberts (2017a), “Niyogi (2006) points out that, although children’s learning resources are finite, the target is not completely ‘unknown’: it falls within a limited hypothesis space” and “it is an accepted result of learning theory that this must be the case: the alternative of a tabula rasa imposes an insuperable learning problem on the child” (p. 28). The learnability argument thus generalizes beyond the specific examples debated in the poverty-of-stimulus literature. Whatever the specific content of UG, some prior constraint on the hypothesis space is mathematically necessary for acquisition to converge. The question is not whether such constraints exist (they must!), but what their nature is: whether they are language-specific (CLN) or derivable from domain-general learning principles (CLB).

The debate over the argument from the poverty of the stimulus has been unusually productive for the field, not because it has been resolved (for many, especially in CLB, it has not), but because it has forced both sides to be more precise about what they are claiming. CLN researchers have moved from global arguments about the impossibility of learning without innate constraints to specific, testable claims about particular constructions; at the same time, CLB researchers have developed increasingly sophisticated statistical models of the input.

The distance between the two camps narrows once the minimalist reduction is taken seriously. Hinzen (2012) observes that a UG account, pushed to its limit, becomes in effect an account of how little need be specified in the initial state for language acquisition to succeed. This is

thus a question of how little structure one can get away with, rather than a catalogue of innate rules. As Pleyer and Hartmann (2019) admit, “in line with our reasoning in this paper,” that even committed minimalists “state that in principle they share similar goals with usage-based, emergentist, and general cognitive approaches” (p. 7). It is precisely this move that brings the CLN position into contact with the emergent, usage-based emphasis on general learning (in the ‘lean’ sense).

Other approaches exist, of course, such as the Competition Model spanning over five decades, which originally led to emergentism (see MacWhinney 2021 and MacWhinney et al. 2022 for recent reflections and comprehensive references). While generative frameworks emphasize innate linguistic structures, usage-based theories stress children’s social cognitive abilities and their language acquisition as input-driven pattern recognition through language communication and language use (e.g., Bybee 2001, Pullum & Scholz 2002, Tomasello 2003; for a textbook introduction, see Saxton 2010).

Alternative explanations involve Bayesian learning models and statistical inference, for example. From a nativist-critical stance, arguably widely shared by the CLB community, learning algorithms have been exploited to account for child language acquisition (e.g., Tenenbaum & Griffiths 2001, Regier & Gahl 2003, Perfors et al. 2011). Among others, these are critically reviewed by Lasnik & Lidz (2017) and especially Yang (2016); see also Pearl and Goldwater (2016) for a handbook overview. As an interesting side note, a new collaborative research agenda may emerge from what Brasoveanu and Dotlačil (2020: v) originally planned as the subtitle of their book: “Integrating generative grammars, cognitive architectures and Bayesian methods.”

However, a distinct CLB objection comes in the form of the ‘input-richness argument’, sometimes conflated with the Bayesian approach (see above for references)—but importantly different: Child-directed speech is considerably richer in relevant structure than classical poverty-of-stimulus arguments assume, precisely because it is highly repetitive, socially scaffolded, and tuned to the child’s developmental level (Tomasello 2003, Goldberg 2006). On this view, the gap between input and output is not a logical gap requiring innate structure to fill; it is an empirical gap that disappears once the input is characterized accurately. CLN researchers have responded to this by developing more specific, construction-level tests of the poverty of stimulus (beyond the classics like auxiliary-fronting or constraints on *wh*-question formation); CLB

researchers, in turn, have developed more rigorous corpus analyses of child-directed speech. The debate is empirical, and progress has come from both sides.

In addition, much ongoing work explores Large Language Models (LLMs) and their purported—and intensely contested—relevance for language acquisition. Contemporary research on LLMs addresses possible connections to linguistic theory, especially regarding the ‘creative aspect of language use’ (Chomsky 1988; see Chap. 5), its importance for biolinguistics, and the role LLMs may (or may not) play.

A particular challenge to the poverty-of-stimulus argument comes from the success of LLMs in acquiring grammatical competence by statistical learning over text, with no domain-specific innate constraints. Piantadosi (2024) argues that this refutes the nativist account: If grammatical competence can be induced from data by a domain-general learner, the inference from impoverished input to an innate UG would be undercut. Among others, the claim has been contested by response papers in *Biolinguistics*—that these models are trained on data orders of magnitude beyond a child’s input, that engineering success is not in itself a theory of the human capacity, and that the relevant explanandum is I-language rather than next-token prediction. This primer takes no side here, but I will return to the debate in Sect. 8.3.

4.3 THE GENETIC ENDOWMENT FOR LANGUAGE

Concerning the theoretical tools, systems, and structures required to explain poverty of stimulus-style arguments, the most important one is the biological endowment of human language. UG, as the first factor, specifies the initial state of the language faculty, what the child brings to the acquisition task before any linguistic experience begins. On the most minimal current conception, this amounts to the operation Merge, the Labeling Algorithm (or third-factor Minimal Search; cf. Sect. 3.3), and the interface conditions that regulate the mapping between the computational system and the sensorimotor and conceptual-intentional systems.

Even this lean endowment is sufficient to explain the core poverty-of-stimulus observations because it constrains the hypothesis space in exactly the right ways: It rules out non-structure-dependent rules, ensures that dependencies are computed over hierarchical rather than

linear representations, and provides the recursive mechanism that underlies the discrete infinity of language. This picture can be pushed further still. If, as on Hornstein's (2024) Extended Merge Hypothesis (Sect. 3.3), the grammatically significant relations all reduce to Merge, then what must be innately specified shrinks, in the limit, to Merge itself. The leaner the endowment, the sharper Plato's problem becomes, since, in direct relation, less is given in advance and more must be recovered from the input. Bouchard (2021) proposes a complementary reframing: The force of the poverty-of-stimulus argument lies less in the sparseness of the evidence than in the way generative principles express the relevant generalizations. If correct, this would relocate the generalization, and the gap between stimulus and competence narrows.

Epstein (2016) sharpens the point from the other side. On the one hand, what can count as experience for an organism is fixed by its biology; after all, a waggle dance witnessed a million times is simply not data for a human as it is for a bee. On the other hand, the 'environment' is not an organism-independent second factor but is itself constituted by the innate endowment; *nurture*, on this view, "is then itself definable only in terms of *nature*" (Epstein 2016: 199, emphasis added). Even language variation is biologically constrained, where parameters form part of the endowment that delimits the range of attainable grammars. What a learner can extract from experience is therefore bounded in advance, so that the generalizations a speaker arrives at are limited by the genetic endowment and, Epstein argues, in part by third-factor properties of the organism. A corollary is that the very "data" of 'primary linguistic data' is not organism-neutral; the same external signal has different effects on different learners according to their internal constitution.

The CLB tradition focuses on the complementary point about the second factor. On the cognitive-developmental niche-construction view (Stotz 2010, Griffiths & Stotz 2013), humans (or rather organisms more generally) actively build the developmental environments that shape the next generation, transmitting a reliably inherited niche of social, material, and epistemic resources alongside their genes. Extending niche-construction theory from ecology to cognition (Odling-Smee et al. 2003; cf. Laland 2017), this reframes the poverty of the stimulus: The child does not confront a neutral, impoverished input but develops within a richly structured *niche* that its forebears constructed (such as joint attention, caregiving routines, and cultural scaffolding, but also

language) and that channels acquisition. On this reading, much of the work CLN assigns to an innate grammar is borne instead by the inherited cognitive niche. It is the mirror image of Epstein's point.

However, also in the interest of 'explanatory adequacy' (Chomsky 1964) introduced in Sect. 4.2, "given the speed and accuracy with which language acquirers converge on grammatical systems despite insufficient (and widely varying) input, whatever is responsible must be 'biased or constrained in certain ways' (Hauser et al. 2002: 1577) to be able to account for it" (Mobbs 2015: 2). In this context, Rizzi's (2017) discussion of explanatory adequacy is particularly relevant. After all, the Principles-and-Parameters framework (Chomsky 1981b, Chomsky & Lasnik 1993), built on Government and Binding theory (Chomsky 1981a), was designed specifically for providing explanatory adequacy (see e.g. Rizzi 1989 for early and Cipriani 2021 for recent discussion).

The basic idea is that a finite number of parameters would account for (i) variation across languages and (ii) as the compass for children to navigate through the many options in their early development (for a comprehensive overview, see Huang & Roberts 2017; for brief remarks, see also Leivada 2020). The key insight of principles-and-parameters approaches is that linguistic variation is not open-ended but parametric: Languages differ from one another along a finite number of dimensions, each determined by a setting that the child must acquire from the input. This makes the acquisition problem tractable; rather than learning each language-specific property independently, the child sets a small number of parameters, and the rest follows from universal principles.

At the molecular level, the genetic contribution to language is real but diffuse. The largest genome-wide analyses to date (see Dediu 2015 for the earlier synthesis) put the SNP-based heritability of reading- and language-related skills at roughly 13–26%, with the earlier candidate-gene findings largely failing to replicate at scale (Eising et al. 2022). This is best read as methodological progress rather than as a deflation of the genetic endowment: It locates that endowment in many genes of small effect operating through development, not in any 'gene for grammar' or 'language gene'.

Biases and constraints as mentioned by Hauser et al. (2002) come in various guises. In the wake of their view of the faculty of language (FLN vs. FLB, see Sect. 3.1), they make up the second and third factors, that is, language-specific and non-specific systems, structures, and processes. This three-factor decomposition is significant for the CLN–CLB

dialogue. The second factor (primary linguistic data) is something both traditions take seriously; they differ in how much weight they give it. The third factor (general computational and cognitive principles) is the terrain where CLN and CLB come closest, since both invoke principles of parsimony, analogy, and pattern generalization to explain aspects of language structure.

Biolinguistics, as Boeckx and Grohmann (2013) introduce their handbook, takes as its focus of inquiry the remarkable ability, present in all children barring severe pathologies (see also Sect. 6.2), to develop a grammatical system that is both incredibly *plastic* (if it were not, we would not witness the linguistic variation we find among human languages) and surprisingly *constrained* (so many logically possible rules do not appear to be part of any human language). The dual observation is precisely what one would predict if language were the product of a biological endowment interacting with environmental input.

Although the environment in which the child grows up, as said above, certainly plays a role in triggering the use of this capacity for language, it is equally obvious that the human child must be biologically equipped in a way distinct from other species if we are to explain how humans turn some of the noise around them into news. The nativist position sees this through the first factor (UG). For CLB, the answer does not invoke a domain-specific first factor but instead emphasizes the remarkable general learning capacities the human child brings to the acquisition task. This is the infant's and young child's ability to track statistical regularities in the input, to infer communicative intentions, and to organize experience into categories and schemas. These capacities are, on the CLB view, sufficient to explain the transition from pre-linguistic infant to competent speaker.

A stark empirical datum reinforces this point. Selinker (1972) estimated that only approximately 5% of second language (L2) learners attain native-like competence, regardless of instruction or motivation. This asymmetry is not explained by general learning ability: Many highly intelligent, motivated adults fail to reach native competence in an L2 acquired after childhood. What it points to is a developmentally sensitive window that closes not because the learner stops trying but because the neural substrate that supported first language (L1) acquisition is no longer available in the same form—a conclusion that the myelination account of Sect. 4.4 right below makes precise.

4.4 DEVELOPMENTAL AND NEURAL EVIDENCE

This book is not dedicated to language acquisition, so space does not permit extensive discussion or even illustration. What this section can do is highlight the empirical phenomena that any theory of language acquisition must address, and sketch how CLN and CLB approaches have responded to them—from first language acquisition in typical children through bilingual acquisition, non-standard acquisition contexts, and atypical or even impaired development.

The core findings of first language acquisition research are among the most robust in cognitive science. Children acquire the phonological, morphological, and syntactic properties of their native language by age 5–6, with remarkably little variation across children and across input conditions. They are sensitive to structural dependencies, including ones for which the input provides no direct evidence, from the earliest stages. These findings are naturally accommodated within CLN, by which they follow from the assumption that UG provides the structural templates that organize the child's intake of the input. CLB accounts must explain them differently, by invoking the statistical richness of child-directed speech, the role of social cues in establishing reference, or gradual abstraction of constructional schemas from specific usage instances.

Bilingual acquisition raises a set of questions that sit at the intersection of CLN and CLB concerns. For CLN, the core question is whether bi-/multilingual children acquire two (or more) separate grammars, each conforming to UG, or a single merged system—and the evidence strongly favors the former. For CLB, bi-/ multilingual acquisition provides a natural laboratory for investigating the role of frequency, input distribution, and social context in shaping language knowledge.

Tomasello's (2003) approach to child language acquisition centers on intention-reading as the driver of lexical learning, and on the gradual abstraction of constructions from specific instances of use. Children begin with item-specific 'verb island' constructions (Tomasello 2000), patterns tied to individual verbs rather than generalized grammatical rules. They then abstract over these progressively to form more schematic constructions as their exposure increases. This account does not deny that children find general patterns; it relocates the source of those patterns from a pre-specified UG to a domain-general capacity for intention-reading combined with distributional learning. The CLN

response is that this account cannot explain why the patterns children abstract are invariably hierarchical, structure-dependent, and poverty of stimulus-resistant—properties that would require exactly the kind of prior constraint that CLN posits.

From the usage-based side, Laland (2017) reframes the input itself. He argued that the child's evidence is not 'degenerate' but socially structured; it is shaped by the social-learning strategies and high-fidelity transmission that make cumulative culture possible. This means that some of the regularities a learner must recover are delivered by a structured learning environment rather than supplied in advance.

A CLB answer to Plato's problem also has an experimental analogue in implicit learning, particularly the *artificial grammar learning* (AGL) paradigm (Reber 1967): Exposed to strings from a finite-state grammar with no mention of rules, participants later classify novel strings above chance while remaining unable to state what they are tracking (for an overview, see Davies 2015). This looks like the learning-from-use of Table 2.5 (fifth hypothesis)—but what exactly is acquired is contested. Reber inferred implicit knowledge of abstract rules; others argue that subjects learn only frequent fragments and recombine them (Perruchet & Pacteau 1990, Shanks 2005), while amnesia studies suggest that both exemplar-specific and abstract information are picked up (Knowlton & Squire 1994, 1996). AGL thus operationalizes the CLN/CLB question without resolving it, the more so as the grammars at issue are finite-state rather than supra-regular (Fitch & Friederici 2012; cf. Sect. 3.3). Finer-grained work further separates conscious from unconscious structural and judgment knowledge (e.g., Dienes & Scott 2005).

The question of why the critical period ends has a neurobiological answer, as Navarro (2024) reviews. The dorsal white-matter tract connecting the frontal and temporal language areas, the arcuate fasciculus, is not fully myelinated until well into childhood, maturing only after puberty, and its maturation predicts the ability to process syntactically complex sentences (Skeide et al. 2016, Friederici 2017a). Lenneberg (1967) had already located such a window on behavioral grounds; as the tract myelinates and the architecture supporting the first language is laid down, the plasticity that made early acquisition possible declines.

Friederici (2017b) argues that myelination of white matter tracts is the key mechanism: Once the structural architecture supporting L1 grammar is laid down in white matter, the neural resources available for acquiring the computational properties of a second language differ from

those that supported L1. Earlier L2 acquisition produces more overlapping activation of Broca's area; later acquisition relies on different, less optimal pathways. The myelination account moves Lenneberg's critical period from a behavioral observation to a mechanistic claim. It provides the kind of bridge between Plato's and Broca's problems (Chap. 6) that biolinguistics is positioned to make (Zaccarella & Trettenbrein 2021).

A more thoroughly developmental challenge to the nativist picture comes from ecological developmental biology (eco-devo). Balari et al. (2020) take aim at what is usually treated as a matter of *virtual conceptual necessity*, the pairing of a lexicon with a computational system (Chomsky 1995). They argue that the trajectory from words to grammar is an emergent property of a developing system in its niche, rather than the unfolding of a pre-specified program. In other words, even this core interface is a contingent developmental achievement rather than something given in advance. On their account, a proto-lexicon that builds from the first words (around 1;0 years;months) scaffolds an incipient, linearly organized computational capacity, which in turn bootstraps the growth of the full lexicon and the structure-dependent syntax of mature language; the two subsystems scaffold one another in a developmental loop, fed throughout by the culturally supplied stream of words. The empirical hook is the 'critical mass effect' (Marchman & Bates 1994, Bates et al. 1988): Productive morphology and syntax do not surge on a maturational timetable but once the vocabulary crosses a threshold of around 500 words (around age 2;6), with syntactic frames in turn accelerating lexical learning (Gleitman 1990). On this reading, the developmental 'clock' is an effect of the dynamics, not a cause.

Seen this way, the critical period is itself an eco-devo phenomenon, a window of experience-dependent plasticity whose closure is governed by molecular 'brakes' rather than by the mere exhaustion of a learning program (see e.g. Kuhl 2010 on neural signatures of early acquisition and statistical learning in a social context or Werker & Hensch 2015 on speech-perception critical periods and molecular brakes on plasticity). This is the point at which the CLB tradition pushes hardest. Carried into linguistics, this yields the claim that the language faculty is a computational phenotype that is grown from a genetic specification (Balari & Lorenzo 2009, 2013), so that acquisition is development, not the triggering of a pre-specified grammar (Lorenzo & Longa 2009).

This is where the eco-devo position thrusts hardest against the myelination account presented above: Where Friederici (2017b) locates the

closing of the window in a maturational neurobiological mechanism, Balari et al. (2020) deny that maturation plays any causal role at all, reading the same timetable as the downstream signature of a constructive developmental process. The two are not wholly incompatible; a plasticity window can be at once experience-dependent and biologically constrained (Werker & Hensch 2015). However, the disagreement marks where a CLN mechanism and a CLB-leaning developmentalism give different answers to the same question.

Increasingly, this deflationary impulse is pursued within CLN itself—the attempt to ‘approach UG from below’ (Chomsky 2007), reducing the language-specific endowment in favour of Merge, third-factor principles (Chomsky 2005), and interface legibility, with the deeper question deferred to Chap. 7.

Non-standard acquisition contexts provide some of the most compelling evidence for the biolinguistic perspective. Creolization, the emergence of a new language from a pidgin contact situation within a single generation, has long been taken as evidence for the contribution of innate structure to language creation (Bickerton 1984, 1990). On Bickerton’s Language Bioprogram Hypothesis, the children of a pidgin-speaking community go beyond their variable and impoverished input, supplying grammatical machinery that the pidgin itself lacks, such as stable word order, embedding, or tense–mood–aspect marking. The hypothesis is contested, especially from the CLB-friendly side. Gradualist and ecological accounts treat creole formation as incremental rather than the work of a single generation (Mufwene 2001); others question whether creoles are structurally exceptional in the first place (DeGraff 2003). Even so, the speed and structural regularity with which creoles stabilize keep the question of the learner’s contribution squarely in play.

Evidence from sign language acquisition points the same way, but two cases must be kept apart. The first is community emergence: When a group of deaf signers comes together, a full language can develop across generations, as most famously documented for Nicaraguan Sign Language (Senghas & Coppola 2001). The same trajectory has also been traced for Cena, a village sign language that arose over three generations among deaf signers in an isolated community in Piauí, Brazil. Cena emerged under conditions still more spontaneous than those of Nicaraguan Sign Language, with no schooling to bring signers together and no contact with the national sign language, yet yielding a system with a stable lexicon and complex grammar (Almeida-Silva et al. 2025).

The second case is that of deaf children with no conventional model at all, who nonetheless develop “home sign” systems that reproduce many, though not all, of the properties of a natural language, a pattern that recurs across deaf children in unrelated cultures, from the United States to China (Goldin-Meadow 2003). On a stringent statistical test, such a system proves genuinely productive and combinatorial despite the absence of any adult model (Goldin-Meadow & Yang 2017), and recent work finds even phonological structure emerging the same way (Kita et al. 2025).

Cutting across both cases is the modality-independence of the acquisition trajectory. Children acquiring a signed language from birth pass through the same developmental stages as speaking children (babbling, first signs, etc.), with comparable error types and sensitive periods (e.g., Petitto 1987, Petitto & Marentette 1991, Lillo-Martin 1999). In other words, the acquisition mechanism does not appear keyed to a specific modality but to the computational system itself. For CLN, this convergence points to a domain-specific, modality-neutral endowment; for CLB, it sets the task of explaining how domain-general capacities yield the same trajectory irrespective of input modality. Cases of language deprivation mark the limiting condition on all of this.

A cross-domain convergence is worth flagging here. Bickerton (1990) observed four distinct data sources to exhibit the same profile (early language of children under two, first-generation pidgins, and post-critical-period language, as touched on above, and productions of ape language-trained primates; cf. Chap. 7): relatively free word order, no hierarchical embedding, and rearrangement of elements without a change in meaning. Regardless of whether one accepts Bickerton's evolutionary interpretation, the empirical convergence offers a picture of what language use looks like in the absence of a mature syntactic system—namely, one that turns out to be relevant for theorizing about the evolutionary precursors of modern language (see more in Chap. 7).

Atypical language acquisition, in children with Specific Language Impairment (or Developmental Language Disorder), Down syndrome, Williams syndrome, or autism spectrum conditions, provides a different kind of evidence. The selective impairment of specific grammatical properties in the context of otherwise intact cognition, as in SLI/DLD, has been taken as evidence for the modularity of the language faculty, though the profiles of linguistic strength and weakness in various clinical populations do not always conform to what a simple modularity

hypothesis would predict, and the debate over what these profiles reveal about the architecture of language remains productive for both CLN and CLB.

The dissociations seen in developmental disorders bear directly on this. The double dissociation between grammar and general cognition—relatively spared structure alongside impaired cognition in Williams syndrome, and the reverse profile in SLI/DLD—has been interpreted as evidence for the modular character of the language faculty (Curtiss 2013, arguing for ‘little modularity’ in opposition to ‘big modularity’; cf. Sect. 3.6), though the picture is one of specialization rather than strict domain-specificity (cf. Sect. 3.3). Kambanaros and Grohmann (2017) investigate “language and cognition under special genetic circumstances (22q11 deletion syndrome), also providing further evidence that a ‘pathologically lowered IQ [...] does not result in bizarre use of language’ (Lenneberg 1967: 326)” (p. 57). The fuller neural treatment is taken up in Chap. 6.

4.5 SUMMARY

In sum, to address Plato’s problem from a nativist biolinguistic stance (CLN), all three factors from Table 2.3 (Chap. 2) play a role: The child is equipped with an innate genetic endowment for language, receives language input in the form of primary linguistic data from the environment to construct and revise hypotheses about the grammar of the language acquired, and applies computational and cognitive processes not specific to language. The resulting picture is neither purely nativist nor purely empiricist; it is interactionist in a principled way, grounded in biology but sensitive to the environment.

Perhaps a clearer connection to Plato’s problem from a CLB perspective would be Reboul’s (2017) reformulation of Tinbergen’s four questions for language (Table 2.6, Chap. 2), concerning the ontogeny of language, though it does not directly delve into underlying concepts or structures. Reboul asks for Plato’s problem: What is the ontogeny of language? Posed this way, the question is broad enough to accommodate CLB answers—accounts of how language develops in the individual through interaction, use, and social scaffolding without presupposing the theoretical apparatus of UG. Both CLN and CLB traditions study language ontogeny; they differ in what they take the

developmental trajectory to reveal about the underlying biological organization. From a CLB angle, it has been argued that “language acquisition often involves mechanisms and constraints that are not specific to natural language” (but see Yang 2016) and “to the extent that there do exist constraints on learning that are restricted to natural language acquisition, that these will at least to some extent draw on ‘informational substance’ supplied by cognitive systems other than the linguistic” (Geeraerts & Cuyckens 2007: 6).

Plato’s problem (‘How do children come to know so much from so little?’) remains one of the most productive questions in the scientific study of language largely because it forces any theory of language to commit to a specific account of the acquisition process. CLN answers that commitment with the poverty-of-stimulus argument and the Innateness Hypothesis (UG); CLB addresses it with statistical learning, social cognition, and usage-based abstraction. The next chapter turns to Descartes’s problem, how that knowledge is put to use, and the chapters that follow will show that these commitments have different consequences for the remaining foundational questions. They will also demonstrate that the dialogue between the approaches (CLN and CLB), far from being resolved, grows richer as the questions become harder.

Descartes's Problem

Abstract This chapter presents the third foundational question: *How is that knowledge of language put to use?* In a sense, this is the bread and butter of theoretical linguists—the analysis of linguistic atoms and structures. The generative enterprise has found it useful to tease apart a speaker's competence in their language from their performance, thus often ignoring, or abstracting away from, processing and production issues. This allowed the detailed considerations of a transformational component which have played a leading role since the late 1950s. But it stands in opposition to approaches that view language use as inseparable from linguistic knowledge itself. For usage-based and emergentist traditions, there is no competence that can be cleanly isolated from performance: The creative use of language is not a problem to be set aside but the very phenomenon that any account of language must explain.

Keywords Cartesian linguistics • Creative aspect of language use • Production problem • Transformational component • Processing • Now-or-never bottleneck

René Descartes (1596–1650) was a French natural philosopher who spent most of his working life in the Dutch Republic. A key figure in the Scientific Revolution, he contributed greatly to mathematics, the natural sciences, and metaphysics. He is widely regarded as one of the founders of modern philosophy. With relevance to the study of language, among many other fields, Descartes formulated a modern version of the mind–body problem. One aspect of this dualism is the ability to use language, a capacity that distinguishes human beings from other animals.

5.1 Linguistic Performance and Processing

Here the usual contrast inverts: This is the foundational question where CLB is most at home; rather than asking how a competence grammar is deployed in real time, CLB asks how real-time use constitutes the grammar in the first place (cf. Sect. 2.4). The term ‘Cartesian linguistics’ originates with Chomsky (1966), whose historical reconstruction of the rationalist tradition was itself controversial. Bracken (1970) offered an early and searching critique of Chomsky’s reading of Descartes, questioning whether the Cartesian framework genuinely supports the nativist conclusions Chomsky draws from it; a more damning review can be found in Aarsleff (1970). The historical debate was revisited and partially surveyed by Thomas (2020), though see also, for example, Hinzen and Sheehan (2013) or Rey (2020). Whatever the merits of the historical argument, the philosophical core of Descartes’s problem (i.e. the creative, stimulus-independent use of language as a mark of mind) remains a genuine and productive challenge for biolinguistics.

Descartes’s problem addresses the *creative aspect of language use* (e.g., Chomsky 1964, 1966, 1988), which is neither strictly determined by external stimuli nor fully explainable by formal linguistic competence; in fact, when introducing it, Chomsky admitted that Descartes’s problem “still remains unsolved” (Chomsky 1988: 147) and “there seems to be little to say about [the production part of] it” (Chomsky 1991: 18). Its linguistic relevance encompasses two issues, the ‘production problem’ (creativity of language use) and the ‘perception problem’ or ‘parsing problem’ (language comprehension and production), where parsing and production are human performance tasks.

The foundations of ‘Cartesian linguistics’ (Chomsky 1966, after the *Discours* of Descartes 1637/1649) lie in the mind–body dualism. One aspect of this is the ability to use language, distinguishing “man” from “machines” or “beasts” (which even “men born deaf and dumb, and thus deprived of speech organs” can). Descartes’s problem refers to “the ordinary use of language in everyday life, with its distinctive properties of novelty, freedom from control by external stimuli and inner states, coherence and appropriateness to situations, and its capacity to evoke appropriate thoughts in the listener” (Chomsky 1988: 138). It is “the problem of accounting for ‘the creative aspect of language use’, one aspect of the more general problem of accounting for choice of action

that is evoked by situations but not determined by them, as in ordinary, normal human behavior, linguistic behavior included” Chomsky (1987: 35).

The minimalist program can be read as a direct continuation of this Cartesian project. Hinzen (2012) argues that minimalism is “nothing other than the project of a ‘natural philosophy’ begun in the early seventeenth century by Descartes” (p. 116). It is the attempt to bring the mind, and language as its most distinctive expression, within the scope of natural science. On this reading, Descartes’s problem is not a historical curiosity but the methodological commitment of the field. It also carries the further Cartesian notion that language is in the first instance an instrument of thought rather than of communication (touched on in Sect. 3.6 already and developed further in Sect. 6.3).

More recently, Hinzen (2014) approached the topic as ‘un-Cartesian linguistics’ (see also Hinzen & Sheehan 2013). As Narita (2019) recaps, they “propose the term *un-Cartesian linguistics* to refer to the research program that pursues the radical co-linearity of language and thought in the sense defined here” (p. 41, fn. 4). He cites the hypothesis that “the cognitive mechanism generating human-specific thought and those generating language should be the same” and that “language thereby becomes a principle of cognitive organization and no separate theory of human-specific thought is required” (Hinzen 2014: 227). This stands in opposition to the “Cartesian hypothesis” which for Hinzen and Sheehan (2013) is that “thought is universal and immutable in this species, while language is merely a contingent way of expressing it in a physical medium” (p. xvi).

A sharper version of this challenge comes from Baker (2007), in a discussion devoted precisely to the creative aspect of language use (CALU). Adopting Chomsky’s (1957 *et seq.*) factoring of the faculty into lexicon, grammar, and the CALU, Baker argues that the creative aspect is as plausibly innate as syntax. He shows that the standard poverty-of-stimulus arguments transpose onto it largely intact—children are never in a position to learn from the data that those around them are neither automata nor random word-generators, yet they reliably come to use language in the unbounded, stimulus-free, situation-appropriate way that defines the CALU.

What makes the argument pointed for Descartes’s problem is the second step. Baker holds that the CALU is innate without being ‘narrowly biological’: No aphasic syndrome selectively abolishes the

creative aspect while sparing lexicon and grammar; likewise, no developmental disorder isolates it, and no ape displays it. For Baker this is not a gap waiting to be closed but a symptom: The CALU is an intrinsically abductive capacity, and abduction (inference to the best explanation, with no advance specification of what is relevant) lies outside the computational theory of mind (Fodor 2000b). On this view, the creative aspect is exactly the residue of Descartes's problem that resists the biolinguistic reduction: real, innate, and pervasive in ordinary use, yet not, on current understanding, reducible to circuitry, genes, or computation. The CALU is the point at which the CLN program's confidence in an ultimately biological account is most exposed.

Adding to this, CLB has a lot to offer as well by putting a strong explanatory emphasis on semantic and functional explanations (cf. Sects. 2.4.2 and 2.4.3), which will be compared in this chapter. On the one hand, Descartes's problem can be seen as the core domain of modern theoretical linguistics, especially the interplay of lexicon/morphology, syntax, and semantics that lead to the creative aspect of language. On the other hand, empirical research over the past three decades, with a multitude of methodological tools and tasks, has shed considerable light on the ultimate distinction between production and parsing, especially since Colin Phillips' (1996) seminal dissertation ("the parser is the grammar"). Once again, the plethora of findings lends itself to treatments from macro- (nativist, generative, minimalist) and micro-cognitive perspectives alike (non-nativist, functional, cognitive).

Since Cognitive Linguistics sees language as emergent from general cognitive processes, CLB approaches may contribute more to this area. According to Wen & Taylor (2021a: 1), "Cognitive Linguistics has two main purposes: (1) to study how cognitive mechanisms like memory, categorization, metaphor, metonymy, attention, and imagery are used during language behavior; and (2) to develop psychologically viable models of language that cover the broadest possible range of linguistic phenomena, including idioms and figurative language."

Either way, Moreno & Mendiivil-Giró (2014: 30) rightly remark that "E-language as a heterogeneous inconsistent object" is "irrelevant to the [macro-cognitive] 'biolinguistic' point of view, which of course does not imply that [it is] also irrelevant for, perfectly licit approaches" such as micro-cognitive ones.

Before turning to the production problem proper, a comparative perspective is instructive. Bolhuis et al. (2014) identified four striking

parallels between birdsong and human language—both require external input during learning, both exhibit sensitive developmental periods, both show hemispheric lateralization, and both involve motor-auditory rehearsal; the parallel between birdsong subsong and human babbling is particularly direct. The crucial asymmetry is equally instructive: Birdsong entirely lacks lexical items, that is, there is no pairing of an acoustic unit with a conceptual content. This single asymmetry maps onto the FLN/FLB distinction. The sensorimotor infrastructure of vocalization is shared across species, but the interface with the conceptual-intentional system is not. The production problem is therefore not only a problem about syntax but about the integration of sound and meaning that appears to be unique to human language.

5.2 THE PRODUCTION PROBLEM

In its short form, we can express Descartes's problem as "How is that knowledge put to use?" (cf. Table 1.1 in Chap. 1). The method applied by de Vos (2015) for approaching Humboldt's problem, which I then also did for Plato's problem, may not work so obviously here: The empirical nature of Descartes's problem is not as clear-cut, which then makes explanatory implementations more difficult as well. Part of the difficulty is that production and comprehension, as performance tasks, involve a complex interaction between the grammar proper and a range of non-linguistic systems: working memory, pragmatic inference, attentional control, and social cognition. Disentangling the contribution of the grammar from those of the performance systems is methodologically demanding. In addition, the results look different depending on whether one starts from a CLN or CLB theoretical commitment.

On the one hand, Descartes's problem is 'the production problem', as Chomsky put it, the problem of accounting for the creative aspect of language use, that is, the creativity aspect of language which is tied to a speaker's ability to use language with "coherence and appropriateness to situations" (Chomsky 1988: 138). What makes this aspect of language use philosophically significant is precisely that it resists mechanistic explanation. A machine, Descartes argued, could be designed to produce sounds or even words in response to stimuli; but it could not produce language that is appropriate to all situations, as appropriateness is not stimulus-driven. For Chomsky, the creative use of language is

evidence of a mentalistic capacity that is not accounted for straightforwardly by formal grammar.

On the other hand, Chomsky stated, “having nothing to say about it, I will put it aside, keeping just to the perception problem, or what is sometimes called ‘the parsing problem’” (Chomsky 1997: 22), even though it presents itself as “[t]he central fact to which any significant linguistic theory must address itself” (Chomsky 1964: 7). This perception or parsing problem refers to how words are put together to form larger expressions, with the meaning they have, and the reverse (i.e. producing and comprehending linguistic expressions, in any modality).

The two halves of Descartes’s problem thus require different theoretical resources and different empirical methodologies. The parsing problem has generated a rich experimental literature, including studies of reading times, eye movements, ERP signatures, and cross-linguistic variation in parsing preferences. The production problem has generated a different body of literature, centered on, for example, speech errors, hesitation phenomena, priming effects, or the coordination of linguistic and non-linguistic communicative resources.

While Chomsky is less concerned with the first, he does sketch a route for the second, familiar from generative theorizing, though slightly differing across the frameworks over the past several decades: In short, “the hearer identifies words, and on the basis of their lexical properties, projects a syntactic structure as determined by principles of universal grammar and the values of the parameters” (Chomsky 1997: 26). This route, from lexical properties through structural projection to interpretation, is broadly confirmed by psycholinguistic evidence. Readers and listeners use lexical information to project syntactic structure incrementally, and the resulting structural representations are closely related to those posited by generative grammar. What the evidence also shows, however, is that the parser is sensitive to a range of non-structural factors that the grammar proper does not encode. These include frequency, plausibility, discourse context, and prosody.

Once again, this ties to Humboldt’s question and at least the first factor, namely what is there to begin with. And it forms part and parcel of linguistic theory (mostly viewed from a syntactic perspective, though the other domains play a role, too, including psycholinguistics)—the generative grammarian’s ‘bread and butter’ once more. The empirical study of language production has developed considerably since Chomsky’s early dismissal of the problem. The most influential framework

remains Levelt's (1989) model of speaking, which decomposes production into three broad stages: conceptualization (the construction of a pre-verbal message), formulation (mapping onto linguistic form, including lexical selection and syntactic encoding), and articulation (execution of the motor program for speech). This architecture is broadly compatible with the CLN picture; the formulation stage involves the construction of a syntactic structure guided by the properties of selected lexical items.

The study of speech errors has been the primary empirical window into production planning. Errors such as anticipations, perseverations, exchanges, and spoonerisms reveal that production involves planning ahead. Units from later in the intended utterance are activated before they are articulated, and this pre-activation can interfere with the units being currently produced (going back to Fromkin 1971 and Dell 1986). The units involved in errors are linguistic units, such as phonemes, morphemes, words, and syntactic phrases, confirming that the grammar plays a direct role in production planning.

This direct role should not be confused with the assumption, encouraged by CLB emphasis on real-time processing limits (Christiansen & Chater 2016a, 2016b), that CLN requires whole sentences to be planned in advance. Generative grammar is a model of competence, not a processing theory; it specifies which structures are well-formed, not how much structure must be assembled before articulation begins, and the competence/performance distinction (cf. Chap. 3) exists precisely to keep these apart. The speech-error evidence in fact cuts the other way, indicating that planning operates over a limited, locally bounded window—just what one expects when a structure-building grammar is deployed incrementally under the resource limits of the performance systems.

A closely related phenomenon is self-monitoring, a speaker's ability to detect errors, actual or anticipated, and repair them mid-utterance. Levelt's (1989) model includes a monitor that compares the pre-articulatory phonological plan against a comprehension check, and a further monitor that evaluates the acoustic output. The self-monitoring loop is a productive CLN–CLB dialogue point: CLN approaches locate the monitor's sensitivity to grammatical violations in the grammar itself, the same system that generates structures also recognizes their deviance; CLB approaches explain it through frequency-sensitive expectations about likely continuations, predicting that what gets caught and

repaired reflects distributional probability as much as abstract grammatical constraints. The two predictions are empirically distinguishable, and the psycholinguistic literature has not fully resolved the debate.

What speech errors also reveal, however, is that production is sensitive to frequency and lexical familiarity in ways that the grammar proper does not predict. High-frequency words are retrieved faster; familiar collocations and formulaic sequences are produced as chunks (Wray 2002). There is also evidence that formulaic retrieval and novel grammatical generation recruit partially distinct neural pathways, with formulaic processing showing greater bilateral and subcortical engagement (Conklin & Schmitt 2012). This is terrain where CLB approaches have made their most important contribution to Descartes's problem. Usage-based frameworks treat the lexicon as a structured inventory of units at multiple levels of abstraction, and they predict that the frequency and entrenchment of these units will directly affect production fluency.

The third-factor perspective offers a way of integrating these CLN and CLB observations. If the production system is subject to general principles of cognitive efficiency (minimizing working memory load, reusing recently activated structures, preferring shorter encodings under time pressure), then the frequency effects and chunking phenomena documented by CLB researchers are not evidence against a grammar-based account of production but evidence for the operation of third-factor constraints on the grammar-performance interface.

5.3 FROM TRANSFORMATIONS TO MERGE

The technical details of the generative-transformational component of the grammar have evolved in many details since the original conception. The history of the transformational component tracks the evolution of generative grammar from its earliest form with a rich inventory of construction-specific transformational rules—through the principles-and-parameters framework, which replaced many specific transformations with general principles and parametric variation, to the minimalist program, which aims to reduce the transformational component to a small number of highly general operations, most notably Merge.

For one, there is now a much improved approach to the old distinction between D- and S-structure in the minimalist architecture of the

computational system. In earlier versions of generative grammar, D-structure was the level at which thematic relations and argument structure were represented, and S-structure the level at which movement operations had applied. The minimalist program eliminated both levels as theoretically motivated objects, replacing them with a derivational architecture in which the computational system operates cyclically, building structures that are periodically transferred to the interface systems.

This distinction goes back to the suggestion, relevant to the parsing problem, that “we can formulate a second fundamental conclusion of Cartesian linguistics, namely, that deep and surface structures need not be identical. The underlying organization of a sentence relevant to semantic interpretation is not necessarily revealed by the actual arrangement and phrasing of its given components” (Chomsky 1966: 33). This insight, that the syntactically relevant organization of a sentence is not identical to its surface form, remains one of the most important contributions of generative grammar to the study of language use. It explains why sentences that look identical on the surface can have different interpretations (structural ambiguity), and why sentences that look different can mean the same thing (paraphrase).

In addition, the predominant view is that a dedicated computational system centers around Merge, the binary operation which applies iteratively to the syntactic building blocks of the grammar (e.g., words, affixes, categories, features), contributing to the recursive computation of human language (see also Sect. 3.2, 3.3., 4.3). The same recursive operation has been argued to underlie more than syntax. Mendivil-Giró (2025a) relates syntax and arithmetic through shared recursive computational capacity, on which the number sense and the language faculty draw on a common combinatorial resource rather than arithmetic being parasitic on language. This represents a CLN-adjacent reading that locates the link in computation rather than in language itself (cf. Hinzen 2012, who derives both from Merge applied at different arities). This is becoming a richer line of investigation; for early work, see e.g. Pica et al. (2004) for CLN approaches and Gordon (2004) for a CLB-take.

A crucial aspect of Merge, which makes it different from a mere combinatorial operation that simply concatenates relevant objects (as in, say, arithmetic or music structures), is that it also labels the syntactic objects it puts together. This so-called ‘Labeling Algorithm’ may thus form the unique ingredient of the human language capacity, as discussed in some detail above (Sect. 3.3).

The relevance of this reduction for Descartes's problem is direct. If the computational system consists of nothing more than Merge, Labeling, and the interface conditions, then the creative use of language—the ability to produce and understand an unlimited range of novel expressions with determinate interpretations—follows from a strikingly simple biological endowment interacting with a rich lexicon and general principles of efficient computation.

For CLB, this minimalist reduction is double-edged. On the one hand, a grammar reduced to Merge and Labeling is less obviously language-specific than the rich rule systems of earlier generative frameworks, and a less language-specific grammar may be easier for CLB to accommodate. On the other hand, even a minimal Merge remains a dedicated computational operation whose domain-specificity CLB must explain away.

5.4 FROM COMPETENCE TO PERFORMANCE

But where things get more interesting, both for biolinguistics at large and for micro-cognitive linguistics, is what goes on beyond. What goes on beyond the grammar proper is the domain of performance—the real-time deployment of grammatical knowledge in parsing, production, and use.

Bear in mind that generative grammar was set up as a model of language competence, though the computational system has been claimed to be optimally designed for performance systems (cf. Chomsky 1998, 1999). This claim, that the grammar is designed with the interfaces in mind, is characteristic of the minimalist approach to Descartes's problem. If the computational system is optimally designed for the performance systems it subserves, then the properties of the grammar should be explainable, at least in part, by the demands of the interfaces—providing a principled connection between competence and performance that earlier versions of generative grammar did not explicitly pursue.

Arguably the most prominent implementation of this view is Phillips's (1996) proposal that "the parser is the grammar" (p. 15, see also Phillips 2003): Syntactic structures are built from the left to the right, as opposed to top-down or bottom-up, and this incremental left-to-right system is possibly the only structure-building system that humans have.

Phillips's proposal has been enormously influential because it collapses the competence/performance distinction in a specific way that can be empirically encoded: If the parser directly instantiates the grammatical computation, evidence about parsing preferences is evidence about the grammar, and vice versa. The extensive literature on incremental syntactic processing that has developed since Phillips (1996) is largely organized around the question of how far this proposal holds, and where the grammar and the parser come apart.

One crucial aspect of Descartes's problem is the generativity and recursion of language (but see Christiansen & Chater 1999, Christiansen & MacDonald 2009, and many efforts from both sides of the macro-/micro-line). This can be approached theoretically with a focus on syntactic (macro/CLN) or semantic properties (micro/CLB), yet it inevitably refers back to the competence–performance distinction (Chap. 3).

Among many other specific proposals, see e.g. Knott's (2014) linking theoretical syntax with neural mechanisms in sensorimotor processing or Trotzke et al. (2013), who show “that systematic properties of performance systems can play an important role within the biolinguistic perspective on language by providing third-factor explanations for crucial design features” (p. 1). Construction Grammar, the most fully developed CLB framework for addressing Descartes's problem, dissolves the competence/performance distinction from the bottom up rather than the top down. Construction Grammar treats constructions (form–meaning pairings at all levels of complexity and abstraction) as simultaneously units of grammatical knowledge and units of processing. There is no level of ‘pure competence’ behind the constructions; what speakers know is precisely what they use, and what they use is a structured inventory of constructions built up through experience and shaped by frequency.

For Descartes's problem specifically, Construction Grammar offers a principled account of the production problem that CLN struggles to provide. On the constructionist picture, speakers do not assemble utterances from atomic lexical items using abstract syntactic rules; they select from a network of partially schematic constructions that are fitted to communicative situations through the combined effects of semantic appropriateness, frequency, and contextual priming. The creative aspect of language use, on this account, is the product of a rich, structured, frequency-sensitive network of constructions that can be combined and adapted in response to communicative need.

What CLN finds unsatisfying in this picture is the absence of a principled account of why certain combinations are possible and others are not. Construction Grammar can describe the constructions that occur and model their frequency-sensitive properties; it is less successful at explaining the systematic gaps, such as the constructions that do not occur or the structures that speakers uniformly reject. These gaps are what abstract principles and constraints in CLN are designed to explain; they thus constitute the strongest argument for the continued necessity of something like a grammar, even within a broadly usage-based account of language use.

While this may deal with the parsing problem, Chomsky did mention “problems that can be formulated in terms of such notions as determinacy, randomness, probability distributions and input-output systems”, but also held the view that “there is nothing very serious to say [at the level of theoretical understanding] about the production problem, except in very limited areas (e.g., the study of speech errors)” (Chomsky 1987: 35, 36). This may have changed, even in CLN; just think of third-factor considerations within the three-factors model (Table 2.3).

For an explicit biolinguistic treatment, though with focus on communicative/pragmatic aspects and no direct reference to the three factors, see Reboul (2017). And it most certainly plays a prominent role in micro-cognitive linguistics (see several chapters in e.g. Wen & Taylor 2021b or Wen & Sinha 2027 on relevant treatments). The CLB perspective on Descartes’s problem is, in many respects, richer than the CLN perspective—not because CLB has solved the production problem but because it never set it aside in the first place. Usage-based frameworks treat production and comprehension as the primary data for linguistic inquiry, not as peripheral performance phenomena to be abstracted away from.

Micro-cognitive linguistics may have more to add to this, however. By putting a strong explanatory emphasis on semantic and functional explanations, whether through focus on “language as a means of communication and how this communicative function has structured grammar” or “the relationship between language and cognition, how humans conceptualize the world around them, how they create and represent meaning and how this is reflected in language” (Siewierska 2013: 485). See also Christiansen and Chater (2016a, 2016b), who develop this perspective in a particularly systematic way (though a generative framework exists as well, as laid out by Pietroski 2008). They argue that the

form of language (including its syntactic properties) is shaped by the demands of sequential processing in production and comprehension. On their account, many of the properties CLN attributes to UG are better explained as adaptations of the language system to the constraints of the cognitive architectures through which language is processed.

Christiansen and Chater (2016b) argue that language processing faces a fundamental memory bottleneck; acoustic information is largely gone within 50–100ms. To cope with this ‘now-or-never’ bottleneck, the brain must compress and recode linguistic input immediately into progressively more abstract representations, from phonological chunks through words to discourse-level representations. The implication is that syntactic properties of language may be adaptations to this processing constraint rather than reflections of an autonomous computational system.

5.5 SUMMARY

In sum, the generative-minimalist framework that embraces third-factor considerations seems to clash here with a ‘third factor-only’ view familiar from micro-cognitive linguistics. The trajectory of this chapter has traced Descartes’s problem from its philosophical origins in Cartesian dualism through the formal architecture of the transformational-generative tradition to the contemporary interface between grammar and performance systems. Two themes have emerged, and I will summarize them here in more detail than other chapter summaries.

The first is *reduction*. The progressive reduction of the transformational component in the minimalist program is a research program for making the grammar more feasible as a biological object. A grammar reducible to Merge and interface conditions is one whose biological basis is easier to identify, whose acquisition is more explicable, and whose evolutionary emergence is less puzzling. The reduction of the grammar thus connects Descartes’s problem directly from Humboldt’s (Chap. 3) to Broca’s (Chap. 6) and Wallace/Darwin’s problems (Chap. 7).

The second theme is the interface. Descartes’s problem is, at its core, a problem about the interface between the grammar and the performance systems. This interface is where CLN and CLB come into closest contact, and where the most productive empirical collaboration is currently taking place. Third-factor considerations provide the

theoretical vocabulary for a cross-framework conversation about the production and parsing problems that neither tradition can fully address on its own.

Descartes's problem is perhaps more directly relevant for CLB if we refer to Reboul's (2017) reformulation of Tinbergen's four questions for language instead (Table 2.6, Chap. 2): "What are the functions of language?" Framed this way, Descartes's problem becomes tractable from both CLN and CLB perspectives. CLN can address the function of language at the level of the computational system; its function is to generate structured representations interpretable by the conceptual-intentional and sensorimotor systems. CLB can address it at the level of communicative use; language functions to coordinate attention, express intentions, manage social relationships, and transmit cultural knowledge. These are not competing answers but answers to different aspects of what 'function' means when applied to language.

Future biolinguistic collaboration should thus shed new light on what I take to be not as black and white as the authors would like us to have it: "Any time a particular phenomenon turns out to involve cognitive functioning rather than just formal syntax, the need to posit genetically given formal constraints on possible syntactic constructions diminishes" (Geeraerts and Cuyckens 2007: 6). The force of this observation depends on what one takes 'cognitive functioning' to mean. CLN does not deny that language use involves cognitive functioning—it insists on it; the grammar interfaces with the conceptual-intentional and sensorimotor systems precisely because language use requires cognitive functioning.

The disagreement is over whether that functioning is distinctive enough to require a domain-specific computational system of human language, or whether it can be fully explained by general cognitive principles. And Descartes's problem makes the stakes of that disagreement vivid. What it ultimately reveals is that the study of language use is not just peripheral to biolinguistics; it is central to it. Answering it requires both the formal precision CLN brings and the functional and contextual sensitivity CLB contributes. Broca's problem, addressed in the next chapter, moves the explanation from the computational to the neural level, asking whether a dedicated system for language leaves any trace in the brain itself.

Broca's Problem

Abstract This chapter presents the fourth foundational question: *How is that knowledge of language implemented in the brain?* After a brief review of the main results from research on language in the brain, including evidence from language pathologies, the main problem will be highlighted for relating language and the brain in a direct fashion: The granularity of neuroscience is too coarse, while linguistic interests are still too finely grained. There may be ways to approach this incommensurability, relevant for both nativist, generative (CLN) and emergentist, usage-based (CLB) approaches to human language, but this is still very much work in progress. A closely related issue addressed in this chapter, which at the same time may also guide such investigations, has to do with the relationship between language and thought.

Keywords Language in the brain • Language and thought • Language pathologies • Aphasia • Granularity mismatch • Incommensurability

Paul Broca (1824–1880) was a French physician and anatomist who also contributed to the development of physical anthropology. His research on the brains of aphasic patients led to the identification of a left frontal region now known as Broca's area. He showed that these brains contained lesions in the left frontal region of the cortex. Apart from many other research activities, he is credited with the first scientific explanation for the loss of speech production due to a left-hemispheric frontal lobe lesion, yet with language comprehension still intact.

6.1 Neural and Cognitive Bases of Language

Where CLN looks to the brain for evidence of a dedicated language faculty, CLB looks to the same evidence for signs that language recruits general-purpose neural resources shaped by use (cf. Sect. 2.4). With Chap. 6, we are moving from fundamental issues to more exploratory ones, or, as per Sect. 2.4.1, from biolinguistics in the narrow sense to biolinguistics in the broad sense. Broca’s problem ultimately concerns *language in the brain*, or *language and the brain*—what are the neural underpinnings of language, where and how is language computed, and so on. There is a lot of confusion, especially concerning the (often over-estimated) localization of brain areas and the (yet emerging) neuronal activities. In the old days, this was the fallacy of dualism; more recent takes follow up on David Poeppel’s characterization of the ‘granularity mismatch problem’ and the concomitant ‘ontological incommensurability problem’ that “the primitives of linguistic theory cannot be reduced to the fundamental units currently identified by neuroscience” (Poeppel 2017: 158). Not surprisingly, CLN couches this within the faculty of language; CLB considerations explore different directions (inspired by both Wacewicz et al. 2020 and Reboul 2017, for example).

In a nutshell, Broca’s problem refers to the question how (knowledge of) language is implemented in the brain (cf. Table 2.1). Cedric Boeckx, its name-giver, is more cautious, however—raising an important point we will pick up in the summary:

It is about the brain–mind connection, and very few people have had good intuitions as to how to go about solving that mystery. You could call it Broca’s Problem or Gall’s Problem [after the German neuroanatomist and physiologist Franz Joseph Gall (1758–1828), founder of phrenology and pioneer in the study of the localization of mental functions in the brain], but it is very difficult to find insightful antecedents for this issue.

(Boeckx 2009: 45)

The restraint Boeckx expresses is epistemically important. Broca’s problem is not simply a question about the localization of language in the brain. It is a question about the relationship between the theoretical constructs of linguistics on the one hand and the physical structures and processes of the brain on the other—and this relationship is far from transparent. What counts as a satisfactory answer to Broca’s problem

depends heavily on what one takes the object of linguistic inquiry to be. And this is where CLN and CLB diverge in ways that are directly consequential for neurolinguistic research design. The levels-of-analysis framing of that relation is developed in Sect. 6.3, and the granularity-mismatch and ontological-incommensurability problems that constrain it in Sect. 6.4.

Broca's problem, the name Boeckx (2009) suggested (alongside the alternative "Gall's problem") addresses the fundamental question of how language is represented and ultimately processed in the brain. Contemporary advances in neurolinguistics demonstrate that language involves multiple interacting neural regions that extend well beyond the traditional Broca's and Wernicke's areas or even just the cortex. Central to current debates is the 'granularity mismatch problem' (Poeppel & Embick 2005), which highlights critical discrepancies between the theoretical constructs of linguistic theory and available neural data.

Language pathology or breakdown provides another window into the biological basis of language. Comparative studies of typical and atypical populations (e.g., developmental impairments and dyslexias, or aphasias) offer insights into neural and genetic constraints on language. Investigating these pathologies helps refine theories of linguistic competence and performance, bridging generative and cognitive approaches (as sketched in Grohmann & Kambanaros 2021; for recent perspectives, see Leivada et al. 2017 and Tsimpli et al. 2017).

6.2 THE LANGUAGE NETWORK

At first glance, there may be a trivial way to understand Broca's problem as not specifically macro-cognitive (CLN), or even generative: An answer to how (knowledge of) language might be implemented in the brain requires one to look more closely at and into the brain, beyond the philosophical armchair investigations (whether, say, rationalist-dualist, Platonist, or empiricist)—important, though, they have been.

The empirical study of language in the brain has a history that predates modern neuroscience by more than a century. The classical era of aphasiology, from Broca's (1861) identification of a left frontal lesion in a patient who had lost the ability to produce speech to Wernicke's (1874) description of a posterior temporal lesion associated with impaired comprehension, established the foundational observation that

language is lateralized to the left hemisphere and that different aspects of language processing are associated with different neural regions; the brain was subsequently mapped more finely into originally 52 areas, the so-called Brodmann Areas (BA; e.g., Brodmann 1909, Zilles 2018). This lesion-based approach dominated the field for over a century and gave rise to the classical model: Broca's area (or BA44–45, including the opercular part and triangular part of the inferior frontal gyrus) for production and syntactic processing, Wernicke's area (or BA22, part of the superior temporal gyrus, included in Wernicke's area) for comprehension and lexical access, connected by the arcuate fasciculus (AF).

If Broca's problem is the problem of how linguistic computation is realized in neural tissue, then the white-matter connections of the language network matter as much as its grey-matter nodes. AF, the long-range dorsal fiber tract linking posterior temporal and inferior parietal cortex to the inferior frontal gyrus, has emerged as the structural backbone of syntactic processing.

On the account from Angela Friederici's research lab, it is the dorsal connection to BA 44, rather than the ventral route that carries semantic and local information, which supports the computation of complex hierarchical structure (e.g., Friederici 2017a). This tract matures late; its myelination continues through childhood (see Sect. 4.4), and the strengthening of the dorsal connection tracks the developmental emergence of complex syntax (Skeide et al. 2016).

That the pathway is built rather than merely switched on is shown most sharply by early language deprivation. Deaf individuals who acquire a first language only in adolescence (having had no accessible language in early childhood) display a measurably weaker AF than deaf peers who acquired a sign language from birth (Cheng et al. 2019). Their neural processing of language is correspondingly atypical, with altered engagement of the canonical left-hemisphere network (Ramírez et al. 2014, Mayberry et al. 2018). And while their earliest steps into a first language can resemble those of much younger children—late, in this respect, yet looking early (Ramírez et al. 2013)—they tend to retain lasting difficulty with exactly the complex, hierarchically organized syntax that the dorsal pathway subserves. The convergence of three measures (tract integrity, processing profile, and syntactic competence) on the same early window is, though correlational, a strong indication that experience within that window is a condition on the normal assembly of the syntactic substrate itself.

This sharpens Broca's problem in a particular direction. Localization of syntax within the fronto-temporal language network (Zaccarella & Trettenbrein 2021, Fedorenko et al. 2024a) is not localization of an innately wired, fixed module but of a substrate whose very connectivity is shaped by early linguistic experience; *nature* furnishing the maturational window and the broad architecture, *nurture* building the tract within it. The acquisition side of this evidence is introduced in Sect. 4.4; here it bears on the neural realization of syntax, and, through the resemblance between language-deprived profiles and the broader picture of language without a mature syntactic system, it connects forward to the evolutionary discussion (Chap. 7).

Before surveying what neuroimaging has revealed, a result from comparative neuroscience deserves attention because it reframes the standard question. Herculano-Houzel (2016) demonstrated that brain mass is an unreliable predictor of cognitive capacity across species because brains of similar mass contain very different numbers of neurons. The relevant metric is cortical neuron count. The implication for Broca's problem is direct: What is neurologically special about the *H. sapiens* language-ready brain is better framed in terms of cortical neuron density and connectivity than in terms of brain volume.

Nóbrega (2022) further asks: What enabled the evolution of this unusually dense cortex? Wrangham (2009) argues that cooking is the key innovation. Raw food provides insufficient caloric density to sustain a metabolically expensive large cortex; cooking dramatically increases caloric yield, providing the energetic surplus needed to build and maintain 16 billion cortical neurons. The argument is an exaptation narrative—cooking did not evolve for language, yet it created the biological conditions without which the neural infrastructure for language could not have been sustained. This is a concrete illustration of how Broca's and Wallace/Darwin's problems (Chap. 7) are linked by causal chains running through traits entirely remote from language itself.

The advent of modern neuroimaging transformed the field by making it possible to image language processing in the intact, living brain. The results both confirmed and substantially complicated the classical model. Language processing engages a far more extensive network of regions than Broca's and Wernicke's areas alone, including bilateral temporal regions, inferior parietal cortex, prefrontal cortex, and subcortical structures. The picture that has emerged is one of a distributed, hierarchically organized language network—a picture that may sit more

comfortably with the CLB emphasis on distributed, emergent processing than with the classical modular view.

An interesting case study illustrates both the power and the limits of the localization approach. Damasio et al. (1996) combined lesion analysis with PET neuroimaging to localize lexical retrieval deficits, simultaneously revealing category-specific storage effects. The convergence of two methods with different spatial and temporal resolutions on the same result is the most reliable form of evidence in cognitive neurolinguistics. At the same time, the study illustrates the granularity problem that Sect. 6.4 addresses: The neural regions implicated are broader than any single linguistic primitive, and the relationship between the anatomical finding and the theoretical construct remains indirect.

Cross-linguistic aphasia data strengthen this conclusion while also complicating the standard picture. Dąbrowska (2004) notes that the inflection deficits typically associated with Broca's aphasia in English speakers are English-specific: Polish and Turkish speakers with comparable lesions show different patterns, because the relevant morphological computations are distributed differently across those languages. This variation is not easily accommodated by strong localizationist accounts. It is important, however, to note what distributed networks do and do not show: They refute strict anatomical localization but do not refute domain-specificity. Language may be processed across a distributed network while still being processed by a network dedicated to language. This is arguably where a dialogue between CLN and CLB at the neural level turns most productive.

Cognitive linguists are typically not the first group of scientists that come to mind (though see the closing reference to Piattelli-Palmarini 2013 in Chap. 8), neurolinguists are often better equipped to doing so, and collaboration with neuroscientists is imperative. The questions that arise at the interface between linguistics and neuroscience cannot be answered from within either discipline alone: How is hierarchical syntactic structure computed in neural circuits? What neural mechanisms underlie rapid incremental language processing? How does the brain represent the abstract distinctions that native speakers make effortlessly? Linguistics supplies the theoretical constructs that specify what the brain must be doing; neuroscience supplies the methods for determining how those computations are physically realized.

One domain in which theorizing can play a role is in the interpretation of linguistic abilities directly linked to brain activity, structures,

stimuli, and so on—especially in the context of language pathology. Beyond that, theoretical linguistics contributes to neurolinguistics by specifying the contrasts that neuroimaging and electrophysiological experiments need to be sensitive to. A study of syntactic processing that uses stimuli designed without reference to current syntactic theory will inevitably conflate phenomena that the theory distinguishes and fail to detect effects that depend on theoretically motivated contrasts.

The dual-stream model proposed by Hickok and Poeppel (2000, 2007) represents the most influential current framework for organizing the neuroimaging findings on language. The model distinguishes two processing pathways extending from the primary auditory cortex: a dorsal stream, projecting to posterior parietal and inferior frontal regions (including Broca's area), and a ventral stream, projecting to inferior temporal and anterior temporal regions.

The dorsal stream is associated with the mapping between sound and articulation, the sensorimotor integration underlying speech production and the processing of complex syntactic structures. From a CLN perspective, this stream is the natural neural locus of the structure-building operations posited by generative grammar. The ventral stream is associated with the mapping between sound and meaning, the rapid, bilateral access to lexical and conceptual knowledge underlying normal comprehension. From a CLB perspective, this stream is more naturally at home; it is the pathway through which words activate their rich networks of semantic associations and contextual information.

Benítez-Burraco (2012) draws a minimalist implication from this body of work: A lean biological endowment (first factor) should correspond to a small number of developmental regulatory genes with broad effects, not a rich inventory of language-specific modules encoded in dedicated genes. The fact that FOXP2 is a regulatory hub rather than a language-specific gene is precisely what the minimalist reduction of UG predicts.

A direct bridge between the molecular and neural levels is provided by FOXP2 research, which Chap. 7 treats in full. KE family members carrying the disrupted FOXP2 variant show grey matter reduction in the inferior frontal gyrus and reduced left-hemisphere activation in phonological processing regions—deficits that map the genetic lesion onto precisely the neural territory most consistently implicated in syntactic and phonological processing (Spiteri et al. 2007). The FOXP2 findings illustrate what the granularity mismatch looks like in practice: The

causal chain runs from gene expression through neural organization to linguistic capacity, and neither level alone fully predicts what is found at the other.

The broader developmental language disorder (DLD/SLI) literature extends the picture. Twin studies estimate heritability at approximately 0.7 (Bishop et al. 1995), and genome-wide association studies have identified multiple genetic risk loci beyond FOXP2, including CNTNAP2 and ATP2C2. At the neural level, DLD is associated with structural differences in the same frontal and temporal regions implicated in the FOXP2 findings, suggesting that the language-ready brain is a developmental continuum, with genetic architecture influencing where on that continuum an individual falls. For CLN, this continuity is consistent with a biological endowment that varies quantitatively; for CLB, it illustrates that language development is shaped by genetic and environmental co-determinants throughout, not triggered by a single innate switch (the constructional side of which Sect. 6.4 below takes up).

What makes the dual-stream model productive for the CLN–CLB dialogue is that it suggests the two traditions may be emphasizing different, complementary aspects of the same neural system. The CLN emphasis on hierarchical structure building corresponds to the dorsal stream’s computational profile; CLB’s emphasis on meaning, context, and use corresponds to the ventral stream’s. A complete neurolinguistics requires both.

A crucial methodological complement to neuroimaging is EEG/ERP, which trades spatial precision for millisecond-level temporal resolution. The most theoretically significant ERP component for the CLN–CLB dialogue is the early left anterior negativity (ELAN), which arises within 150–200ms of a phrase-structure violation—before semantic context can exert influence (Friederici 2002). The early latency of ELAN has been interpreted as evidence for an automatic, early syntactic processing stage that operates independently of meaning: precisely the kind of dedicated syntactic mechanism that CLN predicts and CLB contests. CLB responses have argued that ELAN reflects a general prediction-error signal rather than a syntax-specific module. Friederici’s (2017b) framework mapping syntactic, semantic, and prosodic processing onto distinct ERP components with distinct neural generators represents the most comprehensive attempt to integrate the ERP and fMRI evidence.

6.3 LANGUAGE AND THOUGHT

But the real crux of Broca's problem is the interplay of thought, mind, brain, and analysis. Fodor's (1975) Language of Thought hypothesis represents the most explicit attempt to characterize the relationship between language and thought at the computational level (see Sect. 3.6). The core claim is that thought itself has a language-like structure: Mental representations are syntactically structured, compositionally interpretable, and causally efficacious in virtue of their formal properties. This 'Mentalese' is not identical to any natural language; it is a more abstract, universal representational format that natural languages express and make publicly available.

For CLN, the Language of Thought hypothesis is a natural companion. If thought is already language-like in structure, the relationship between the computational system of human language and the conceptual-intentional system it interfaces with is not mysterious: The faculty of language generates hierarchically structured expressions that are interpreted by a conceptual system which is itself hierarchically structured and compositionally organized.

For CLB, the Language of Thought hypothesis is much less congenial. CLB approaches tend to favor instead a view in which thought and language are not separable systems but mutually constitutive; linguistic categories shape the categories of thought, and the structure of thought reflects the structure of the language through which it has been organized. The language-thought question has direct consequences for Broca's problem: If language and thought share neural substrates, then the neural implementation of language is simultaneously the neural implementation of a significant portion of human cognition.

The most comprehensive recent synthesis of the language network literature can be found in Fedorenko et al. (2024a, 2024b), which also provides a direct challenge to the language-as-thought position. This work demonstrates that the left-hemisphere fronto-temporal network is strongly interconnected, independent of input and output modalities, causally important for language, and language-selective. Fedorenko et al. (2024b) review neuroimaging and behavioral evidence for a double dissociation between language and thought. People with severe aphasia retain the ability to perform algebra and solve spatial reasoning problems, demonstrating that complex thought does not require intact language; and the language network is selective for language rather than

co-opted for non-linguistic reasoning. Their conclusion that language is primarily a tool for communication rather than thought directly conflicts with the CLN position developed in Chap. 3.

The paper has generated a response by Watumull (2024), who argues that Fedorenko et al. conflate E-language and I-language; the aphasia evidence concerns externalization, not internal computation. This reply has been put most directly by MENDÍVIL-GIRÓ (2019): Evidence that language and thought are dissociable bears on externalized language (E-language) and its communicative use, not on the internal computational system (I-language) that the language-as-thought thesis is about; the dissociation is real but aimed at the wrong target.

Two findings from Fedorenko et al. (2024a) are particularly relevant. First, the network is selective for language across tasks involving logic, arithmetic, music, and motor planning, supporting CLN modularity claims. The universality of this network topology is itself striking; it is recovered across 45 languages from 12 families, isolates included, with no more variability among speakers of typologically diverse languages than among English speakers alone (Malik-Moraleda et al. 2022), a cross-linguistic uniformity that itself calls for explanation in terms of a shared biological endowment. Second, however, it does not preferentially support syntactic over semantic processing, which would challenge Friederici's (2017b) localization of hierarchical syntax to BA 44. The primer leaves this as an open empirical question.

Note that the selectivity result has also not gone unanswered. On Anderson's (2010) neural-reuse account, cortical circuits are redeployed across cognitively disparate tasks over both evolutionary and developmental time, so a network that responds preferentially to language need not be *dedicated* to it. It may be language-*recruited* general-purpose circuitry, shared with tool use and action sequencing (Kolodny & Edelman 2018). On this reading, the selectivity Fedorenko et al. document is the signature of a network tuned by use rather than reserved by innate design; this constitutes the CLB counter to the CLN modularity inference, and the hinge of the dialogue Sect. 6.4 below develops.

Classic keywords would be *mind-brain dualism* (Chomsky 1966), *language of thought* (Fodor 1975), the *levels of analysis* in cognitive systems (Marr 1982), and their modern counterparts (very selectively, this includes Fodor 2008, Gallistel & King 2009, Ludlow 2011, Berwick & Chomsky 2016, Kemmerer 2020, and many more). This interplay is not a problem unique to linguistics or even to cognitive science;

it is the mind–body problem, one of the oldest and most intractable in philosophy. What makes it acute for biolinguistics is that the study of language sits precisely at the intersection of the mental and the physical—language is simultaneously a system of abstract representations and a set of physical events in the brain and body. Broca's problem is the demand that these two descriptions be reconciled.

For starters, according to Berwick and Chomsky's (2016) understanding of the Basic Property of human language, "each language yields a digitally infinite array of hierarchically structured expressions with systematic interpretations at interfaces with two other internal systems, the sensorimotor system for externalization and the conceptual system for inference, interpretation, planning, organization of action, and other elements of what is informally called 'thought'" (pp. 89–90). The relationship between language and thought that Berwick and Chomsky invoke here is central to Broca's problem. If language interfaces with 'thought' (the conceptual-intentional system), then understanding how language is implemented in the brain requires understanding how the neural substrates of language relate to the neural substrates of thought more generally. A CLN account that treats the language–thought interface as a formal, compositional mapping predicts a different neural architecture than a CLB account that treats meaning as grounded in embodied experience.

More general than a specific linguistic implementation (whether on the macro- or on the micro-cognitive level) are Marr's (1982) three levels of analysis for any information-processing system. Unfortunately, "[o]ur general problem is that we understand very little about how even the most basic computational operations might be carried out in neural 'wetware'" (Berwick & Chomsky 2016: 51). Marr's framework is useful for organizing the relationship between linguistics and neuroscience because it distinguishes three specific levels at which any information-processing system can be described: the *computational level* (what is being computed, and why), the *algorithmic level* (how the computation is carried out), and the *implementational level* (how the algorithm is physically realized). Linguistics operates primarily at the computational level, neuroscience primarily at the implementational level. The algorithmic level (processing models, parsing algorithms, production architectures) is where the two disciplines most productively meet.

One direction to explore further could be Gallistel and King (2009), but see, among others, Fitch (2017a) for critical reflections and further

references. Whether the relevant computational substrate must involve a read–write memory system capable of storing and retrieving structured representations, as Gallistel and King argue, or whether connectionist and oscillatory models can accommodate the symbolic, compositional computations that cognition requires, remains genuinely contested. What is clear is that the representations linguistic theory posits are not obviously the kind of thing that current neural models can straightforwardly accommodate.

6.4 THE INCOMMENSURABILITY ISSUE

At the root of specific neurocognitive implementations lies the observation that “[t]he formal structures of linguistics and neurophysiology are disjoint” (Uriagereka 2019)—the ‘Granularity Mismatch Problem’, “a serious problem for those aiming to find brain correlates for the primitives of [the faculty of language]” (Hornstein 2009: 7, fn. 14):

Linguistic and neuroscientific studies of language operate with objects of different granularity. In particular, linguistic computation involves a number of fine-grained distinctions and explicit computational operations. Neuroscientific approaches to language operate in terms of broader conceptual distinctions.

(Poeppel & Embick 2005: 104–105)

The granularity mismatch is not merely a practical problem of measurement resolution; it is a conceptual problem about the relationship between the theoretical vocabulary of linguistics and the theoretical vocabulary of neuroscience. A neural region is not the same kind of thing as a syntactic operation, and a frequency band is not the same kind of thing as a level of representation. Bridging the two requires not just more data but a more principled account of what the relationship between them should look like.

Coupled with the Ontological Incommensurability Problem, “the primitives of linguistic theory cannot be reduced to the fundamental units currently identified by neuroscience” (Poeppel 2017: 158). For CLB, the incommensurability problem looks different. If linguistic knowledge consists of usage-based schemas and constructions rather than abstract syntactic representations, then the relevant neural correlates are those associated with frequency-sensitive pattern learning and

prototype formation—processes that are more directly approachable by current neural models. This is one reason CLB approaches have been more easily assimilated into mainstream cognitive neuroscience: Their theoretical constructs are closer to the vocabulary of neural modeling.

The ontological incommensurability problem goes deeper than the granularity mismatch. Even if we had neural measurements at any resolution we chose, it is not clear that the resulting data would directly confirm or disconfirm the existence of the constructs that linguistic theory posits because those constructs are defined at the computational level, not the implementational level.

Arguably, “[w]hatever operations grammarians propose must ultimately be embedded in brain circuitry” (Hornstein 2009: 3), that is, “brain structures that compute simple operations” about which he adds (p. 116): “A useful step in the direction of bridging the granularity problem would be for grammatical accounts to ‘make use of computational analyses that appeal to generic computational subroutines’ [(Poeppel & Monahan 2008: 83)].” Hornstein’s suggestion points towards a productive research strategy. Rather than trying to map the specific constructs of current linguistic theory directly onto neural structures, develop computational accounts of the grammar that appeal to operations simple enough to be plausibly implemented in known neural mechanisms. Merge, on this strategy, is a better candidate for neural implementation than the complex transformational rules of earlier generative frameworks; not because we know what the neural implementation of Merge looks like, but because a binary, structure-building operation is more likely to correspond to a basic neural subroutine.

Developments within the generative framework in the last decades have concentrated on the precise characterization of the mechanism enabling hierarchical structures to be generated. In this respect, the hypothesis is put forward that human beings must be endowed with some biologically determined, species-specific, universal computational mechanism (Lenneberg 1969). This mechanism, now called *Merge*, generates all possible hierarchical expressions of human language, by recursively assembling words into more complex syntactic objects (Chomsky 1995; Everaert et al. 2015; Friederici et al. 2017; Hauser et al. 2002).

(Zaccarella & Trettenbrein 2021: 326)

This computational claim has a neural counterpart. On the dominant view, the operation Merge is implemented in the most ventral portion

of Broca's area (BA 44): A single application of Merge to two semantically empty items selectively activates BA 44, whereas a non-hierarchical two-word sequence activates instead the phylogenetically older frontal operculum (Friederici 2017a). What this isolates is hierarchy-building as such, independent of meaning and of recursive depth, which is the neural form of the CLN claim that BA 44 is dedicated to structure-building rather than to combination in general.

One line of this work bears directly on whether Merge has a distinct neural signature. Tanaka et al. (2019) introduce the notion of Merge-generability and test it with fMRI: Dependencies that are fully Merge-generable selectively engage the left frontal grammar network with left temporal and angular regions, whereas non-Merge-generable dependencies recruit dorsal premotor and bilateral executive networks. A related strand asks whether what BA 44 isolates is genuine recursion or fast iteration. Dehaene et al. (2015) model the encoding of nested sequences as an iterative parser that builds tree-like representations, which sharpens a question reminiscent of Lobina (2017): Does the neural signature pick out recursion in the structure-generating sense or is it merely iterated embedding? See Sect. 3.3 for the same three-way distinction among senses of 'recursion'. The results are suggestive rather than decisive; as Hornstein (2009) cautions, current methods are too coarse to separate the properties of Merge from those of any hierarchical combination operation (also Hornstein 2017, 2024). But it establishes a testable complexity metric with consistent neural correlates.

One of the most promising recent directions for bridging the granularity gap is the proposal that neural oscillations (rhythmic fluctuations in the electrical activity of neural populations at different frequencies) provide the implementational basis for the hierarchical structures that linguistic theory posits. Murphy (2020) develops this proposal in detail, building on earlier work by Giraud and Poeppel (2012).

Construction Grammar has developed a growing neuroscience literature that maps onto the dual-stream architecture in a productive way. If the ventral stream is associated with meaning-based, frequency-sensitive processing linking sound to meaning via the anterior temporal lobe, then the CLB emphasis on distributional and contextual factors maps naturally onto that stream. Frequent constructions are processed faster and activate stronger networks than rare or novel constructions (Goldberg 2019). This creates exactly the complementary rather than competing relationship between CLN and CLB that the dual-stream

model supports. The dorsal stream tracks syntactic hierarchy (the primary concern of CLN), while the ventral stream tracks lexical-constructive frequency (the primary concern of CLB).

The core observation is that neural oscillations operate at multiple timescales simultaneously, and that these timescales correspond, at least approximately, to different levels of linguistic structure. High-frequency gamma oscillations track the processing of individual phonemes and words; lower-frequency beta oscillations track phrase-level structure; and low-frequency delta oscillations track sentence-level structure. If this correspondence is not accidental (i.e. if the hierarchical organization of neural oscillations reflects the hierarchical organization of linguistic structure), then neural oscillations may provide the physical implementation of the abstract hierarchical representations that CLN posits.

Murphy (2020) proposes that the operation of Merge is implemented in the synchronization of neural oscillations across different frequency bands. From a CLB perspective, if the hierarchical organization of neural oscillations reflects the hierarchical organization of linguistic structure, then this hierarchy is implemented in the same neural systems that process other temporally structured stimuli (music, action sequences, social interaction) consistent with the CLB emphasis on domain-generality.

6.5 SUMMARY

In sum, theories of UG and the faculty of language (both FLB and FLN) continue to evolve (e.g., Friederici et al. 2017, Roberts 2017, Mendivil-Giró 2018)—and they will likely undergo further development; for some recent examples, see e.g. Balari & Lorenzo (2015), Waciewicz et al. (2020), or Murphy (2025). These theoretical frameworks “provide just one part of an encompassing ‘theory of language’, [so] we must assume that something is special about human language (and thus genome and brain), at the very least the way in which it conciliated a multitude of cognitive functions” (Trettenbrein 2015: 6).

Trettenbrein's observation captures the epistemic humility that Broca's problem demands. Even a fully adequate generative grammar would not by itself constitute a solution to Broca's problem—it would constitute a specification of what the brain must be implementing, not

an account of how it implements it. The gap between specification and implementation is precisely the granularity mismatch and ontological incommensurability that Poeppel and colleagues have identified, and it is a gap that neither CLN nor CLB can close unilaterally. Such a perspective might also align better with Reboul's (2017) address of the functions of language (Table 2.6), advancing a framework that would more readily integrate CLB insights.

Reboul's reformulation is particularly apt for Broca's problem because it separates the question of mechanisms from the question of functions, a distinction that maps naturally onto the CLN–CLB divide. CLN approaches have concentrated on the mechanisms of language: the computational operations that generate syntactic structures, the algorithms that parse them, the neural circuits that implement them. CLB approaches have concentrated on the functions of language: what language does for its users, how it serves communicative, social, and cognitive purposes. For Broca's problem, these are complementary rather than competing questions. The challenge of defining Broca's problem more precisely may itself be instructive, as Boeckx (2009) observes: "[T]here is a lesson to be learned from the fact that we cannot really name that question, despite the fact that nowadays [Broca's problem] is taken in many circles to be the one on which the future of linguistics depends" (p. 45).

This chapter has traced Broca's problem from its origins in classical aphasiology through the neuroimaging revolution to the current theoretical debates about granularity and incommensurability. What has emerged is a picture of a field in productive tension: between the abstract precision of linguistic theory and the biological concreteness of neuroscience, between the domain-specific computations posited by CLN and the distributed, functionally organized processing documented by CLB-influenced neurolinguistics. That tension is not a failure of the field but its most productive feature; and it points directly towards Wallace/Darwin's problem, where the biological and evolutionary dimensions of language come fully into view.

Wallace/Darwin's Problem

Abstract This chapter presents the fifth foundational question: *How did that knowledge of language emerge in the species?* The topic of the evolution of language is arguably one that emergentist linguists have pursued for much longer than generative grammarians. It does not come as a surprise then that, not assuming a domain-specific language faculty, they have found very different answers. Yet, both types of approaches need to say something about the possible origins of the language-ready brain in *H. sapiens*. The following will return to the three factors but also address evolutionary developmental biology, review evolutionary evidence, weighing gradualist against saltationist accounts, and lastly discuss the role of genetics in human language, focusing on FOXP2.

Keywords Language evolution • Adaptation vs. exaptation • Gradualism vs. saltation • Third factors • Genotype/phenotype • Evolutionary developmental biology • Genetics • Language-ready brain

Charles Robert Darwin FRS FRGS FLS FZS (1809–1882) was an English naturalist, geologist, and biologist, arguably one of the most influential figures in human history. His enduring legacy is the theory of evolution by natural selection, which he extended to human evolution and even to the origins of a musical proto-language. His fellow countryman **Alfred Russel Wallace** OM FRS (1823–1913) was a naturalist, explorer, geographer, anthropologist, biologist, and illustrator. He was the first to identify difficulties for any straightforwardly adaptationist account of human language, arguing that no biological function he could identify required language that a species without it could not already meet.

7.1 Evolution of the Human Language Faculty

Where CLN asks what biological mutation made language suddenly possible, CLB asks how gradual changes in cognition, culture, and communication could have assembled something that looks, from the outside, like a faculty (cf. Sect. 2.4). For the longest time, addressing the evolution of human language in the species has been all but ignored in the generative tradition. This state of affairs is hardly due to the notorious ‘Paris ban’ (Sect. 2.5)—after all, functional and cognitive approaches to the study of language have done a lot more in this domain. With some notable exceptions from the early 1990s, generative-inspired biolinguistics has really started taking Darwin’s (or better: Wallace’s) problem seriously only in the wake of the FLB/FLN distinction, with the publication of Hauser et al. (2002).

On analogy with Plato’s problem, there is “the logical problem of language evolution” (Fujita 2007, 2009; cf. Boeckx 2009, Hornstein 2009, and, in a very different context, Christiansen & Chater 2008 and the response by Fitch 2008). This could be dubbed ‘Darwin’s problem’, an identification popularized by Boeckx (2009, 2011a) but first used, as far as I know, by Fujita (2002: 166). A more recent shift refers to it as ‘Wallace’s problem’ (Chomsky 2010, Bickerton 2014; see Levinson 2014, Berwick & Chomsky 2016, Hackett 2018, Terrace 2019), named “after the codiscoverer of evolution by natural selection, Alfred Russel Wallace,” as Berwick and Chomsky (2016: 3) explain:

Wallace was the first to call attention to the difficulties for any conventional Darwinian, adaptationist account of human language, since he could perceive no biological function that could not already be met by a species without language. [...] Wallace 1869 is generally regarded as one of his first public accounts of this difficulty, and presents his resolution of placing the origin of language and mind outside the realm of conventional biological Darwinism (though he envisioned a trans-Darwinian solution to the problem). This thread is picked up in Bickerton 2014; chapter 1 of this book is titled “Wallace’s Problem.”

(Berwick & Chomsky 2016: 3, 167, n. 1).

Wallace’s problem can be decomposed into two sub-questions (reviewed by Nóbrega 2022). The first is the ‘How’ question: *Is human language a product of natural selection?* CLN approaches tend to

answer ‘partly not’; the exaptation account holds that natural selection shaped components of FLB, while the computational capacity of FLN emerged through a different route. CLB approaches tend to answer ‘yes, through selection for communication’. The second is the ‘Why’ question: *What adaptive advantages do language, art, and music confer?*

These are different questions, and conflating them generates pseudo-debates. The disagreement is primarily about the How, not the Why. Darwin’s problem, conversely, concerns the difficulty of explaining language origins through natural selection, particularly the emergence of complex grammatical systems from simpler communicative forms. This challenge is amplified by the observation that no other animal species possesses a comparable communication system. And if it asks why humans developed language and complex cognitive abilities that appear disproportionate to their survival requirements, Premack’s (1985) old assessment makes sense, that human language presents “an embarrassment for evolutionary theory” (p. 282), since it is too powerful to be accounted for by selective fitness.

The extensive literature on language evolution cannot be comprehensively reviewed here, but it is notable that evolutionary work within Generative Grammar has historically lagged behind other approaches. Non-generative approaches have long predominated, including significant contributions from Cognitive Linguistics, with Tomasello (1999) representing one prominent example. For a comprehensive overview, see Györi (2021), who also identifies the purported reason: “Cognitive Linguistics does not have its own philosophically committed conception about language to which it wants to adjust the account of how language emerged, whereas the generative approach pursues an explanation of the emergence of language in order to validate its own—philosophically biased—conception of language” (p. 653). Pleyer and Hartmann (2019) provide a valuable discussion of how what is here termed macro- vs. micro-cognitive linguistic approaches to language evolution might differ, but also how the two might be brought closer together.

Evolutionary developmental biology (evo-devo) offers an important framework for understanding language evolution by emphasizing the complex interplay between genetic, developmental, and environmental factors in shaping biological traits (for relevant discussion and references, see Chomsky 2010, Balari et al. 2013, and Boeckx & Benítez-Burraco 2014a but also the exchange between Corballis 2017a, Everaert et al. 2017, and Corballis 2017b).

The treatment of the fifth and final foundational question or problem allows us to pick up on the notion of the “language-ready brain” from the beginning and put things together that have been expounded on in the previous chapter—and go beyond by exploring what four decades of research on FOXP2 and beyond has unearthed on the genetic underpinnings of language, what conclusions this body of work allows for evolutionary developments, and what the major debates are on evo-devo with respect to language and different evolutionary theories with relevance to both CLN and CLB perspectives.

7.2 LANGUAGE EVOLUTION

Wallace/Darwin’s problem addresses the question how (knowledge of) language emerged in the species. The question is deceptively simple. It asks not merely how language came to be used by *H. sapiens*, a question about cultural and behavioral evolution, but how the biological capacity for language, the language-ready brain, came to exist in our lineage. These are different questions with different answers, operating over different timescales and requiring different evidential sources.

Two attitudes frame the history of research on language evolution (Nóbrega 2022). One is skeptical: “We know essentially nothing about the evolution of our cognitive capabilities and there is a strong possibility that we will never know much about it” (Lewontin 1998: 109). The other is cautiously optimistic. There is now “an ongoing transition of scientific research on language evolution from one dominated by speculation and pet hypotheses to ‘normal’ science, marked by attempts to empirically evaluate multiple plausible hypotheses” (Fitch 2017c: 3). This chapter navigates between those two poles.

Pleyer and Hartmann (2019) draw out the general moral. They note that once development is taken seriously, few of the oppositions around which the field organizes itself—I-/E-language, nature/nurture, FLN/FLB, gradualism/saltationism, even adaptation/exaptation—“make perfect sense” (Martins et al. 2016: 161) as clean dichotomies. The commitment spectrum proposed here is the methodological consequence: If the dichotomies do not separate cleanly, approaches cannot be sorted by them into sealed camps.

The debate between gradualist and saltationist accounts of language evolution is one of the most fundamental in the field. Pinker and

Bloom's (1990) adaptationist account, the most influential gradualist position, argues that language evolved through natural selection for communication, just as the eye evolved through natural selection for vision. The argument is explicitly Darwinian and broadly compatible with CLB: If language evolved for communication, then its communicative function is not secondary or derivative but constitutive of what language is. Fodor and Piattelli-Palmarini (2010) take this further still, arguing that adaptationism fails not merely empirically but logically, leaving *natural history* as the narrative of what actually happened rather than *natural selection* as the explanatory vehicle for phenotypic evolution. Kolodny & Edelman (2018) provide a gradualist exaptation account on which language arose by coupling pre-existing action-sequencing and communication circuits, grounded in neural reuse rather than a dedicated mutation.

Chomsky's response has been consistently skeptical, and the saltationist account associated with Berwick and Chomsky (2016) represents the sharpest possible contrast. On their account, Merge did not evolve through gradual selection for communication but emerged, possibly through a single genetic event (the 'Basic Property'), as an internal computational capacity. Its recruitment for external communication was secondary. This package is stated most compactly by all four of its architects (Bolhuis et al. 2014): the Strong Minimalist Thesis, Merge as the single key innovation, 'no faculty of language before Merge', recent and sudden emergence, and language as primarily an instrument of thought. It is also where 'Darwin's problem' is coined for the sudden-origin puzzle. The same position is advanced across disciplines in Berwick et al. (2013), signed jointly by a computational linguist, a neuroscientist, a linguist, and a birdsong biologist, who date the faculty to roughly 100 kya (shorthand for '100,000 years ago'), with no significant change since the African exodus.

Against this, the weak saltationist alternative of Miyagawa et al. (2013) proposes that human language arose from the integration of two independently evolved systems: an expression system (E-system, linked to sequential combination in birdsong) and a lexical system (L-system, linked to primate referential communication). This preserves the saltationist intuition (the key step is combinatorial), while grounding it in precursor capacities.

One influential model elaborates the saltationist account in two stages (Nóbrega 2022). The computational system of thought emerges

(FLN combined with the conceptual-intentional interface), enabling complex internal representations without any requirement for external expression in the first stage (~175–200 kya). In the second stage (~80–100 kya), this capacity is coupled to the sensorimotor externalization system. On this account, the Upper Paleolithic Revolution (~50 kya) is not evidence for the first appearance of language but for the externalization of a capacity that already existed—a distinction with direct consequences for how the archaeological record should be interpreted.

Fitch (2010b) occupies a productive middle ground, arguing that the evolution of language involved multiple, distinct evolutionary events (voluntary breath control for speech, lowering of the larynx, vocal learning, and theory of mind), assembled over a long evolutionary time-scale, with natural selection playing a role at each step. This picture is consistent with emphasis on both the distinctiveness of the final product (CLN) and its continuity with precursor capacities (CLB).

It is fair to say that non-generative work on language evolution has prevailed for the longest time (see Györi 2021 for an overview). The reasons for the CLB dominance in language evolution research reflect theoretical alignment: If language is continuous with other cognitive and communicative capacities, its evolution is continuous with the evolution of those capacities, and the tools of comparative cognition, comparative ethology, and cultural evolution are directly applicable; the cultural origins framework is exemplary (Tomasello 1999). The capacity for language acquisition is traced to a species-specific form of social cognition enabling the understanding of others as intentional agents.

7.3 THIRD FACTORS

Interest in evolutionary questions within the generative tradition emerged more recently, particularly following the FLB–FLN distinction (Hauser et al. 2002) and related work (e.g., Freidin & Vergnaud 2001). By now, there are a number of book-length treatments (Fitch 2010b, Balari & Lorenzo 2013, Berwick & Chomsky 2016, Progovac 2019); for diverse perspectives on language evolution, see e.g. Gibson and Tallerman (2012), Balari et al. (2013), Lieberman (2015), Pennisi and Falzone (2016), and Reboul (2017).

Within the three-factors approach to the design of human language (Table 2.3), the most promising route to bring macro- and micro-

cognitive linguistic concerns together is arguably the third factor. Evo-devo provides the kind of bridge between CLN and CLB that this book has pursued throughout. Incidentally, in formulating it, Chomsky (2005: 6) explicitly makes reference to canalization: “principles of structural architecture and developmental constraints that enter into canalization, organic form, and action over a wide range, including principles of efficient computation.” For a broader biological context, see Lohndal and Uriagereka (2017).

The concept of canalization (Waddington 1942) refers to the tendency of developmental processes to produce consistent, stable outcomes despite variation in the genetic and environmental inputs that drive them. Applied to language, canalization could be argued to capture the well-known generalization: Children acquire the language of their community reliably, rapidly, and with a high degree of uniformity, despite enormous variation in input. This uniformity of outcome from variable input is precisely the explanandum of Plato’s problem (Chap. 4)—and evo-devo reframes it as a developmental question.

Developmental plasticity is the complementary concept, namely the capacity of the developing organism to adjust its developmental trajectory in response to environmental input. Language exhibits a productive combination of both. On the one hand, the core computational properties of language are robustly canalized and appear universally across human languages. On the other hand, the surface properties are developmentally plastic, shaped by the primary linguistic data of the child’s environment. This combination is neatly predicted by the three-factor framework.

The Baldwin effect, the process by which learned behaviors can, over evolutionary time, become genetically assimilated, offers a further evo-devo mechanism relevant to language evolution. If early hominins developed language-like communicative behaviors through cultural learning, and if individuals with genetic variants that made those behaviors easier to acquire had higher fitness, then selection would over time favor the genetic variants that facilitate language.

Evo-devo is compatible with a wide range of background assumptions in biolinguistics. For CLN, the evo-devo perspective is largely welcome: It provides a biological mechanism for the robustness of UG-constrained properties. For CLB, evo-devo is equally congenial: The emphasis on organism–environment interaction fits naturally with the usage-based perspective on language.

The genotype/phenotype distinction from developmental biology offers a productive reframing of the relationship between UG and the faculty of language for evolutionary purposes (Navarro 2024). If UG is the language genotype, the genetic endowment providing developmental instructions for building the neuroanatomical substrate of the faculty of language, and the faculty of language the language phenotype, the mature faculty resulting from the interaction of that endowment with the epigenetic environment, then a key qualification follows: UG is necessary but not sufficient. The phenotype is determined by the genotype in interaction with epigenetic and environmental co-determinants (Oyama 2000, Lorenzo & Longa 2009). Rather than asking what is the gene for language, we can ask which developmental pathway, involving genetic, epigenetic, and environmental factors, produces the language-ready brain. The pathway becomes the object of evolutionary inquiry.

Gene-culture co-evolution (Richerson & Boyd 2005, Laland 2017) provides a further evolutionary mechanism relevant to the origins of language. Cultural practices can create selective pressures that favor genetic variants which facilitate those practices, which in turn enable more elaborate cultural elaboration—a reciprocal causal loop. The cooking hypothesis discussed in Sect. 6.2 is a paradigm case: Cooking is a cultural practice that drove the evolution of cortical neuron density, which in turn enabled the cognitive elaboration that language requires. Gene-culture co-evolution is significant for the CLN–CLB dialogue as it dissolves the forced choice between genetic determinism and cultural constructivism; the biological endowment and the cultural environment are intertwined in a feedback loop that neither can describe unilaterally.

Lohndal and Uriagereka (2017) consider the three factors in a larger perspective, and while focusing on the third, they observe that “[t]he three factors are more general than they might appear. Gould (2002) discusses three similar factors that hold for organisms more generally” (p. 118), what he refers to as the ‘adaptive triangle’. They continue:

While most contemporary scientists accept Dobzhansky’s [(1973)] dictum that ‘nothing in biology makes sense except in the light of evolution’, the issue is to what extent this amounts just to natural selection. [...] [T]he biolinguistic enterprise raises the question of the role of third factors, which Thompson [(1917/1942)] pointed out provide some fundamental explanations for the growth and form of biological entities [...].

(Lohndal & Uriagereka 2017: 118).

That is, the third factor is not merely a feature of the biolinguistic framework but an instance of a more general biological pattern, one that any account of language evolution must grapple with.

In the context of 'knowledge' of language and the human language faculty, Ferretti et al. (2018) put forth a suggestion that could bring macro- and micro-cognitive linguists one step closer as well: "[T]he idea of considering language as the broad FLB and not as FLN is a way to explain language functioning through more complex cognitive architectures than those postulated by classic biolinguistics"—especially as "many cognitive devices characterizing FLB are also present in nonhuman animals; the focus on FLB allows development of a continuistic view of language" (p. 221). This reframing has direct evolutionary consequences. If the relevant object of evolutionary inquiry is FLB rather than FLN, then the evolutionary question can be addressed in a way that the narrow FLN view makes difficult: Every capacity that language requires can in principle be studied in other species, and the story of language evolution is the story of how these capacities were assembled and integrated in the human lineage.

They continue that, "[a]s highlighted by Waciewicz and Żywicznyński (2014), indeed, anything important regarding language origins is ascribable to FLB" (*ibid.*) See also the aforementioned Waciewicz et al. (2020) for critical reflections on FLB/FLN from an evolutionary perspective. Waciewicz et al.'s pluralist position, that language evolution research should not be hostage to any single theoretical framework's conception of what language is, connects directly to the present book's broader agenda: A biolinguistics that takes CLN and CLB approaches equally seriously must, on the evolutionary question, be willing to investigate the origins of FLB capacities as well as whatever specifically linguistic endowment CLN posits.

Boeckx (2021) revisits Wallace/Darwin's problem from the perspective of what he calls 'pluralist biolinguistics', an approach that deliberately moves beyond the Chomsky-centric framing and draws on evo-devo biology, network neuroscience, and comparative cognition so as to address the evolutionary question from multiple angles. His central argument is that the minimalist emphasis on Merge qua Basic Property as the key evolutionary innovation, the 'single mutation' view, is too narrow: Language evolution is better understood as a process of rewiring existing neural circuits in ways that enabled new combinatorial and communicative capacities.

7.4 GENETICS AND THE LANGUAGE-READY BRAIN

The genetics of language provides one of the most direct empirical windows into the biological basis of the language-ready brain. The story of FOXP2, the first gene specifically associated with language, illustrates the promise, but also the limits, of genetic approaches to the evolution of human language.

FOXP2 was identified through the study of the KE family, a multi-generational British family in which half the members suffer from a severe developmental disorder affecting speech and language (Gopnik 1990, Lai et al. 2001). The affected family members show specific difficulties with the sequential motor control of speech (articulatory dyspraxia) as well as broader grammatical impairments. The gene responsible, FOXP2, encodes a transcription factor, and the KE family's disorder results from a point mutation that disrupts its function.

The evolutionary significance of FOXP2 lies in its comparative profile. The human version of the gene differs from the chimpanzee version at two amino acid positions, changes estimated to have occurred within the past 200,000 years (Enard et al. 2002). When the human version of FOXP2 is introduced into mice, the animals show altered vocalization patterns and changes in the neural circuits involved in motor learning.

These findings suggest that FOXP2 is not a 'language gene' in any simple sense but a gene involved in the development and function of the neural circuits that underlie learned vocal behavior. The human-specific changes to FOXP2 may have fine-tuned these circuits in ways that enabled the precise articulatory control required for human speech. What FOXP2 does not explain, however, is the syntactic and semantic complexity of human language: The gene affects the externalization of language more directly than its computational core.

Songbird data add a comparative dimension that experiments with mice alone cannot provide (Bolhuis et al. 2010). Note that Lenneberg (1972) already set out four parallels between human vocal learning and birdsong: a babbling/subsong stage, vocal development under auditory guidance, a post-pubertal decline, and lateralized neural control. In songbirds, FOXP2 is expressed in Area X of the basal ganglia (a region functionally analogous to the human striatum) and its expression levels rise during active song learning, falling when learning is complete (Scharff & Haesler 2005). There is an intriguing parallel to the role of basal ganglia circuits in human speech acquisition (Enard 2011); it

suggests that the function of FOXP2 in coordinating learned vocal behavior predates the hominid lineage and reflects a deep evolutionary conservation of the sensorimotor learning circuits that underlie vocal acquisition. The songbird model therefore supports the interpretation of FOXP2 not as a 'language gene' but as a regulator of the fine-tuning of sensorimotor circuits for learned vocalization, a function with a long evolutionary history that the human-specific amino acid changes then refined further.

Two further findings extend the significance of FOXP2 considerably. First, the human-specific FOXP2 variant is shared with Neanderthals and Denisovans (Krause et al. 2007), placing the relevant mutation in a common ancestor at least 300 kya. Second, the FOXP2 protein regulates a network of around 50 downstream genes that are differentially expressed in humans compared to chimpanzees (Spiteri et al. 2007).

FOXP2 is therefore not 'the gene for language' but a regulatory hub in a complex gene network—a distinction with direct implications for what kind of evolutionary change could produce the language-ready brain. The ≥ 300 kya date implied by the shared Neanderthal and Denisovan variant sits awkwardly with the 'recent emergence' timeline (Berwick & Chomsky 2016), a tension Berwick and Chomsky (2019) address, revising the estimate to roughly 300–200 kya, bringing molecular and linguistic datings into closer alignment.

A skeptical counter-voice on the genomics comes from DeSalle and Tattersall (2019), who caution against reading candidate genes (CNTNAP2, GLI3, and WDPCP) as a genetic account of the phenotype. To this argument, Navarro (2024) adds a particularly striking connection between morphological and molecular evolution presented by the relationship between RUNX2 and FOXP2 (Boeckx & Benítez-Burraco 2014a). RUNX2 is a transcription factor whose promoter underwent changes after the divergence of the human lineage from Neanderthals; changes that may have altered cranial suture sealing and the globularization of the modern human skull.

Crucially, the RUNX2 protein binds to the FOXP2 promoter: The two factors are connected by a direct regulatory relationship, not merely by both being relevant to language. This means that the biological innovations underlying the language-ready brain included morphological reorganization of the skull, which altered the developmental space available for brain growth, in a pathway linking skull shape, gene regulation, and linguistic capacity. Wallace/Darwin's problem, at this level

of analysis, involves causal chains running through traits that appear entirely remote from language.

The broader genetics of language confirms this picture of complexity. Genome-wide association studies have identified dozens of genetic variants associated with language and reading disorders, each of individually small effect, and the genetic architecture of normal language variation appears to be similarly polygenic and distributed (Deriziotis & Fisher 2017). No single gene determines language ability; rather, language emerges from the interaction of a large number of genes, each contributing to the development of the neural circuits and cognitive capacities that collectively constitute the language-ready brain.

7.5 THE MECHANISMS OF EVOLUTION

A large part of the discussion about the evolution of language thus touches on the theory of evolution itself, which comes out of virtually all sources mentioned in this chapter so far. The engagement with evolutionary theory that Wallace/Darwin's problem requires is more than a matter of selecting a preferred model of evolution and applying it to language. It involves genuine theoretical uncertainty about the mechanisms of evolution; uncertainty directly relevant to the language evolution debate because different evolutionary mechanisms yield different predictions about the form and trajectory of language evolution.

Relating to both Darwin and Wallace, "the question of how natural selection could possibly account for the seemingly excelling intellectual, linguistic, and technological abilities of humankind" (Trettenbrein 2015: 5) is indeed puzzling; moreover, it touches on more than Wallace/Darwin's problem. The adaptationist answer, that language evolved because it conferred survival and reproductive advantages, is intuitively appealing but theoretically problematic. A communication system far simpler than language would suffice for the coordination of hunting and the maintenance of social bonds. The excess cognitive capacity humans possess appears to go well beyond what selection for communication would produce—and this excess is precisely what makes Wallace's problem, rather than Darwin's, the more penetrating formulation.

Another research arena comes from relevant fossil records—from symbolic artefacts, for example. Any inference to purported linguistic capacity requires navigating a chain of steps, with each requiring

independent support. Botha (2010, 2016) formalizes the structure: from assumptions about a set of artefacts (say, pierced shells) to assumptions about who made them, to assumptions about their symbolic function (e.g., as ornaments), to the conclusion that their makers had linguistic capacity. Every link is a potential point of failure. We lack an agreed theory of what makes an object a symbol and how symbols can be identified in the material record (Coolidge & Wynn 2011), and we lack a precise characterization of what ‘fully syntactic language’ consists in, making the inference from symbolic behavior to syntax at risk of circularity. Botha’s critique does not establish that early *H. sapiens* or Neanderthals lacked language; it establishes that the argument as standardly presented does not establish that they had it. Notably, the critique is neutral between CLN and CLB—it applies to overconfident claims from both traditions about what the archaeological record shows.

A more formally rigorous approach to the same evidence exploits the formal language hierarchy (Sect. 3.3). Sauvet et al. (2018) analyzed engraving motifs across species and time: zigzag patterns (*H. erectus*, ~500 kya) require only finite-state computation (Type 3); crosshatched grids (*H. neanderthalensis*, ~40 kya) require push-down memory (Type 2); structured sequentiality of early *H. sapiens* engravings (~130–100 kya) suggests Type 1 capacity. Camps and Uriagereka (2006) applied the same logic to knot-making. This approach specifies what kind of computation is implicated and what its formal properties are—giving the inference from material complexity to cognitive capacity a precision that the standard symbolic-behavior approach lacks.

The question of Neanderthal language is best approached through three clearly distinguished positions. The first holds that Neanderthals had a fully-fledged linguistic capacity equivalent to that of modern humans (d’Errico & Vanhaeren 2009, Papagianni & Morse 2013). The second assumes that they had a restricted, non-modern form (Stringer 2011, Wynn & Coolidge 2012, Conard 2015). And the third claims that their communication was non-symbolic in kind, it was qualitatively distinct from modern language (Tattersall 2008, 2017, Mithen 2014). Tattersall’s formulation of the skeptical position expresses it most lucidly: Neanderthals “clearly reacted to stimuli from the surrounding social and external environments in very sophisticated ways; but they most probably did not remake those worlds in their heads” (Tattersall 2008: 106). This skeptical position has been complicated by more recent evidence: Cave art at La Pasiega, Maltravieso, and Ardales has been dated

to >65 kya, well before modern human arrival in Europe (Hoffmann et al. 2018), and the Bruniquel stalagmite structures in France to ~176 kya (Joordens et al. 2015).

Evans and Levinson (2009) challenge the assumption that language evolution must explain universal structures. If few structural properties hold universally, the evolutionary explanandum shifts: What needs to evolve is not a specific UG, but a capacity for reliable convergences across languages. This would sharpen Wallace/Darwin's problem from a CLB perspective by specifying what the biological endowment must minimally consist in. Klein's (2009) 'Great Leap Forward' hypothesis proposes that fully modern symbolic behavior arose suddenly around 50 kya from a neurological mutation. However, McBrearty and Brooks (2000) document many 'modern' behaviors in the Middle Stone Age well before 50 kya, and the two-stage model (Sect. 7.2) places the relevant biological event at 175–200 kya, earlier than Klein's proposal.

If FLN was not selected for communication, how did Merge emerge? Bouchard (2021) offers Exapted Language Nativism: Combinatorality arises 'for free' from a general capacity for detachment, not from a dedicated mutation. He advances a 'three mysteries' critique of Narrow-UG evolvability—the origins of atomic concepts, combinatoriality, and residual language-specific devices—arguing that minimalism, despite its reductionist power, still fails the evolvability test. FLN might have spread through genetic drift, or emerged as a spandrel of selection for other cognitive capacities (hierarchical planning, concept formation).

A response, already foreseen by Fitch (2010b), would be that "no sub-component of the language faculty is truly unique" (p. 22, emphasis removed), namely that FLN cannot be distinguished from FLB if FLN does not exist (see e.g. Pinker & Bloom 1990 for early and Pinker & Jackendoff 2005 for more recent discussion). The CLB response is not merely a negative claim; it is a positive research program: If no component of the language faculty is truly unique to humans, then language evolution is fully manageable by comparative methods. After all, every capacity that language requires can be studied in other species, and the story of language evolution is the story of how these capacities were assembled and integrated in the human lineage.

If the language faculty is no older than 200,000 years (Berwick & Chomsky 2016), human language is "a very recent achievement in evolutionary terms" (Rosselló & Martín 2012: 16). This recency cuts differently for CLN and CLB. For CLN, it supports single-mutation

emergence; for CLB, it suggests that sophisticated linguistic structures are recent cultural achievements built on older biological foundations—a complex interplay of genetic and cultural evolution.

7.6 SUMMARY

In sum, Wallace/Darwin's problem is of utmost importance for all kinds of cognitive approaches to linguistics. And while the road to an answer looks different again, it does not solely rely on the CLN–CLB divide or hinge on nativism. What makes Wallace/Darwin's problem distinctive among the five foundational questions is precisely its irreducibility to any single theoretical framework. Even within CLN there is no consensus on the evolutionary story: The discontinuist account of Berwick and Chomsky (2016) sits alongside the more gradualist accounts of Fitch (2010b) and the pluralist account of Boeckx (2021). CLB is similarly internally diverse: Adaptationist accounts coexist with cultural evolutionary accounts and evo-devo accounts.

A large role is played by theories of evolution (independently of language) and counts as relevant evidence for the emergence of language (communication, thought, cognition, computational abilities, and so on, as well as genetic evidence, fossil records, and other sources). What this means practically is that Wallace/Darwin's problem cannot be addressed from within linguistics alone. It requires genuine collaboration with evolutionary biologists, geneticists, paleontologists, comparative psychologists, and computational modelers.

Wallace/Darwin's problem brings the five foundational questions full circle. The question of how the language-ready brain evolved in the species is inseparable from the question of what that brain is ready for—which is Humboldt's problem. It is inseparable from the question of how the capacity it embodies is acquired by individual children—Plato's problem. It connects to how the resulting knowledge is put to use—Descartes's problem. And it connects most directly to Broca's problem, because understanding what evolved requires understanding what neural organization was established by that evolution.

The five problems are not independent questions but aspects of one (complex) question: What is the human capacity for language, and how did it come to be? The preceding chapters have traced the answers that CLN and CLB traditions have offered to each aspect of that question.

Outlook

Abstract This chapter draws the five foundational questions together and takes stock of the CLN–CLB spectrum that has organized the pivot, arguing that the dialogue between the two traditions is the field’s most productive resource. It revisits terminological conundrums that divide macro- and micro-cognitive linguistics and looks beyond the five questions to three arenas where a broad-church biolinguistics can contribute: clinical and patholinguistic research, the challenge posed by large language models, and an expanding conception of the faculty of language. It closes on some future directions, foremost the neural-implementation bottleneck, and a possible sixth question for biolinguistics.

Keywords Comparative biolinguistics • Terminological conundrums • Large language models • Biolinguistics and beyond • Future directions

8.1 RECAP

Biolinguistics is larger than any single tradition; our five foundational questions (Tables 1.1 and 2.1) can be addressed from both CLN (Table 2.3) and CLB perspectives (Table 2.5), and the dialogue between them is arguably the field’s most productive resource (Table 2.6). For this pivot I aimed at a reference guide to many issues, problems, and questions in biolinguistics, heading from a historical review to an appreciation for research in cognitive linguistics of any stripe. Throughout, I presented these stripes as sitting on a spectrum of cognitive approaches to the study of human language (Table 2.4).

I referred to the two poles as *cognitive linguistics in the narrow sense* (CLN), or ‘macro-cognitive linguistics’ (cf. Wen et al. 2014) focusing

on generative minimalism, and *cognitive linguistics in the broad sense* (CLB), or ‘micro-cognitive linguistics’ focusing on usage-based, emergentist approaches. The distinction has proven useful across the five questions, not necessarily by resolving disagreements between the two traditions but by making those disagreements more precise, and thereby more manageable.

Let me put these in one-sentence summaries:

1. On Humboldt’s problem, it clarified that the debate between CLN and CLB is not merely about formalism versus functionalism but about prior commitments to modularity, innateness, and the nature of linguistic knowledge.
2. On Plato’s problem, it showed that the argument from the poverty of the stimulus is not a knock-down argument for nativism but a logical observation that constrains what any theory of language acquisition must say.
3. On Descartes’s problem, it revealed that the production problem, long set aside within CLN, is where CLB has much to contribute.
4. On Broca’s problem, it suggested that the neural architecture of language may itself be organized in ways that reflect both CLN concerns (dorsal stream for structure-building) and CLB concerns (ventral stream for meaning-based processing).
5. And on Wallace/Darwin’s problem, it showed that evolutionary questions are not hostage to any single theoretical framework.

In this primer I focused on an exposition of the five foundational questions for biolinguistics and discussed each in considerable detail, if of varying length. The variation reflects not only my own bias and core areas of expertise (as a generative syntactician), but also the variation in how far each problem has been developed across the two traditions. Humboldt’s and Plato’s problems have been most extensively theorized within CLN, and many CLB responses to them are correspondingly well-developed. Descartes’s problem is really two-fold, with only one having received much attention in CLN (language comprehension and production under the ‘perception problem’ or ‘parsing problem’) and the other largely set aside to “remain unsolved” (CALU, the creative aspect of language use, under the ‘production problem’). Broca’s and Wallace/Darwin’s problems are, in important respects, the most open—empirical tools for addressing them are still developing, and theoretical frameworks for interpreting their results still being established.

It is worth marking a distinction that has hovered over the entire book. Following Chomsky (2000), Collins (2021) separates ‘problems’ from ‘mysteries’: questions that fall within the reach of our cognitive capacities and admit of progress as opposed to questions that may lie permanently beyond them. The five ‘problems’ organizing this primer are so called advisedly: Each has received sustained inquiry, whatever remains unresolved within it, and none has shown itself to be a mystery in Collins’ sense.

However, the uniquely interdisciplinary character of biolinguistics poses difficult problems of communication and misunderstandings. This can also be a chance, with a growing community of scientists from diverse backgrounds, including linguists, evolutionary and molecular biologists, neuroscientists, anthropologists, psychologists, computer scientists, speech–language pathologists, and so on, slowly overcoming these challenges. Picking up on Taylor’s (2007) theme of polemics and collaboration (cf. Sect. 1.1), “only collaboration and mutual respect will make this type of research possible, to clarify issues, unearth new data, and answer some of the questions that will help us understand the nature of language, and what it is that makes us human” (Boeckx & Grohmann 2007: 3–4). The danger with too strong convictions expressed by either side of the macro-/micro-cognitive divide on the CLN–CLB spectrum at this point lies in the perceived certainty. To quote from Robert Harris’ novel *Conclave* (p. 44 in the first edition published by Hutchinson in 2016): “Certainty is the great enemy of unity. Certainty is the deadly enemy of tolerance.”

8.2 TERMINOLOGICAL CONUNDRUMS

One difficulty talking across fields, even if they are close, as in macro- and micro-cognitive linguistics (studying natural language as a species-specific mental phenomenon), has to do with terminology. For instance, by employing long-standing generative concepts, considerations, and consensus, non-generative cognitive linguists may shy away right from the start. It is hence important to put all relevant notions into the right context and then regroup (see e.g. Adger 2019 for a recent example of doing this from an even wider angle); perhaps some ideas are not so bad after all or even find their counterparts across the fence upon closer inspection.

A case in point is the competence/performance distinction, which CLB has consistently rejected as a “false dichotomy”—even though it captures a genuine tension between systematic linguistic knowledge and its variable deployment in use. This is a tension that usage-based approaches must also acknowledge, even if they model it differently. Similarly, the concept of UG, when stripped of the specific technical apparatus of any particular generative framework and understood as the biological constraints that shape the space of possible human languages, is not obviously incompatible with a CLB perspective that acknowledges some degree of language-specific developmental constraint. The terminological barriers between CLN and CLB, though, are real; but they need not, optimistically speaking, be as deep as they appear.

Other linguists struggle with terminology as well, including such apparently basic notions as a definition of ‘language’ (argued to be neither possible nor desirable by Waciewicz et al. 2020) or, as Haspelmath (2020) recently did, finding the right label for linguists “who are interested in language structures” but who do “non-psycholinguistic, non-sociolinguistic research on languages” (suggesting ‘structuralist’ or ‘structural linguist’). Haspelmath’s terminological intervention is symptomatic of a broader challenge: The vocabulary of linguistics has been shaped by theoretical commitments that are not always made explicit, and terms that seem neutral (‘grammar’, ‘competence’, ‘structure’, ‘meaning’) carry different theoretical baggage depending on the tradition in which they are used. Making those disagreements visible is one of the contributions a book like the present one can make.

This is reminiscent of another trend in some biolinguistic corners. While “[t]he theory of each individual language is called its Generative Grammar” (Chomsky 2017: 3), many generative linguists are just not that interested in the kinds of structural properties of individual, actual languages. This internal tension within the generative tradition between the theoretical study of the language faculty and the descriptive study of individual languages is a microcosm of the broader tension between CLN and CLB. Both raise the same question: What is the proper object of linguistic inquiry? The answer one gives determines not only one’s theoretical commitments—but also one’s empirical program.

A relevant label that comes to mind is ‘biolinguists vs. philologists’ for practitioners with or without a strong commitment to the nature of the human language faculty (e.g., Boeckx 2011a, 2012), though it is not meant derogatively (Martins et al. 2016: 166, n. 1). Echoing this label

is the assessment that prior to minimalism, “generative grammar as practiced was largely philology on steroids” (Hornstein 2013b: 405), by which he meant that “[t]he fissure separates the philologically inclined (whose main interest lies in the (surface/observable) properties of languages) and the cognitively/biologically inclined (who, following Chomsky’s formulation of the Generative agenda, understand the object of study to be the mental capacities that underlie linguistic facility)” (Hornstein 2024: 4).

And it relates to Hornstein’s (2013a) proposal, not necessarily meant jokingly (Leivada 2015: 94, Hornstein 2024: 4), to distinguish by name “*linguistics* (or biolinguistics, with focus on [the faculty of language]) and *linguistics* (with focus on what Felix [(2010: 71)] calls ‘language data’)—a state of affairs reminiscent of the distinction between biolinguistics in the strong and biolinguistics in the weak sense (Boeckx & Grohmann 2007)” (Leivada 2015: 94, emphasis added; see also Leivada 2014). The ‘linguistics/linguistics’ distinction captures something important. There is a genuine difference between studying the language faculty as a biological object versus studying properties of individual languages as cultural and historical objects, and conflating the two generates confusion in both directions. Biolinguistics in the strong or broad sense is unambiguously concerned with the former; the present book has aimed to show that this concern need not be the exclusive property of the generative tradition.

What these terminological propositions share is an attempt to map the intellectual landscape of linguistics in a way that makes productive collaboration possible—by identifying which researchers are asking the same questions with different vocabularies, and which are genuinely asking different questions. My CLN–CLB distinction is itself such a proposal. On the one hand, it is a way of showing that the questions biolinguistics asks are not the exclusive property of generative grammar, and that the answers usage-based approaches offer are not irrelevant to the biolinguistic enterprise; on the other, it allows for a more embracing rather than dividing understanding of *cognitive* linguistics.

8.3 BIOLINGUISTICS AND BEYOND

A larger issue that looms in an adequate dealing with the five foundational problems is that biolinguists can, and therefore should, inform

research in different or even new domains. Three domains are worth highlighting here (or rather, that is what space allows). The first relates to patholinguistics, clinical linguistics, and speech-language pathology. A biolinguistics that takes seriously both the formal precision of CLN and the sensitivity of CLB to use and context is better positioned to contribute to the understanding of language disorders than either tradition alone. CLN provides the theoretical constructs that specify what the language faculty consists of, and therefore what it means for it to be impaired; CLB provides the functional and communicative perspective that matters most for clinical intervention. This is a program we have dubbed *comparative biolinguistics*, relating to Broca's problem in Chap. 6 and not to be confused with the 'comparative method' from Chap. 7. As reviewed in Grohmann and Kambanaros (2021), which also discusses additional domains of expansion, first attempts can be found in Leivada et al. (2017) and Tsimpli et al. (2017).

The second domain I want to highlight is artificial intelligence and language technology. Tying the fictional Pope from *Conclave* to Leo XIV (2026), the recent development of large language models (LLMs) has opened many fundamental questions—but also possibly reopened our foundational questions about the nature of linguistic knowledge in ways that biolinguistics is uniquely positioned to address (Mahowald et al. 2024, Piantadosi 2024, Watumull 2024, and many since). Do LLMs acquire something like grammatical competence, or only statistical regularities in the surface form of language? What do the successes and failures of LLMs tell us about the poverty of the stimulus argument? These are questions that neither CLN nor CLB can answer alone, but that both traditions have resources to address.

The biolinguistic response to LLMs requires two distinctions. The first is between data quantity and data quality. LLMs are trained on datasets orders of magnitude larger than any child's linguistic experience, which makes them poor simulations of poverty-of-the-stimulus acquisition—the relevant question is not whether a very large model can learn syntactic patterns, but whether a very small one can. The second is between syntactic sensitivity and syntactic knowledge. LLMs perform well on many syntactic benchmarks, but performance does not imply knowledge of the principles that generate correct responses. Mahowald et al. (2024) distinguish formal linguistic competence (sensitivity to distributional patterns in the signal) from functional linguistic competence (language for inference, communication, and action) and

find that LLMs excel at the former while showing systematic gaps in the latter, a pattern more consistent with sophisticated pattern matching than with the kind of knowledge that generative grammar posits. Whether this distinction is principled or merely reflects current LLM limitations is a genuinely open question that biolinguistics is positioned to address.

Viewed through the five foundational problems, the shortcomings of current LLMs might actually fall into a suggestive pattern. Their well-documented difficulty in following explicit rules, and their reliance on distributional statistics rather than an acquired generative system, bears on Humboldt's and Plato's problems, while their sometimes erratic, contextually inappropriate output bears on the creative aspect of use at the heart of Descartes's problem. Perhaps one even could go so far and relate the 'delusion', "defined as high belief hallucinations, incorrect outputs with abnormally high confidence" (Xu et al. 2025), to a quasi-psychiatric level, therefore falling under Broca's problem. In addition, since the confabulations or 'hallucinations' of today's LLMs are not grounded in any biologically implemented or evolved system, they could be read against Wallace/Darwin's problem as well. Some failure modes, moreover, appear resistant to scaling: efficiency, hallucination rates, and reliability all tend to improve with model size and training, yet sycophancy (the tendency to tell users what they want to hear) has so far admitted no optimal solution. Whether such limitations are principled or merely artifacts of current architectures is, once again, an open question—one that a biolinguistic perspective is well placed to sharpen.

Continuing on this note, the third domain explicitly goes to *beyond*. We may, for example, want to consider expanding conceptions of FLB. Beyond Hauser et al.'s (2002) exposition, what are its exact properties, and who has it? Gagliano and Grimonprez (2015) apply the concept to the network underlying plant intelligence. There is also the fast growing field of 'super linguistics' (e.g., Patel-Grosz & Schlenker 2018). So, there are many possible beyonds to explore. With this primer, and in the context of 'Biolinguistics 2.0' (Boeckx & Benítez-Burraco 2014b), we can now substantiate the earlier proclamation: "Biolinguistics is a fairly broad research program, and allows for the exploration of many avenues of research: formalist; functionalist; nativist and insisting on the uniqueness of the language faculty; nativist about general (human) cognition, but not about language per se; etc." (Di Sciullo & Boeckx 2011: 5).

8.4 WHERE TO FROM HERE?

The major theoretical agendas in biolinguistics are not mutually exclusive; they represent different emphases and explanatory strategies, as portrayed here with the CLN–CLB spectrum. Recent trends point towards greater integration and convergence; researchers increasingly recognize the need to account for biological but also experiential factors in language development and evolution. Also, technological advances in genetics, neuroimaging, and computational modeling are providing new tools for investigating the biological basis of language, though significant challenges remain.

Grohmann and Kambanaros (2021: 712) closed with a perhaps dissatisfying synthesis: “We acknowledge that the preceding pages may read a little bit like the manifesto of a United Church of Everything.” However, this need not be a bad thing. Language must be studied as a mental phenomenon alongside its social, cultural, ecological, and even historical dimensions. The point developed in this pivot has not been to dissolve the differences between CLN and CLB—those differences are real (and productive in many ways). Rather, I wanted to show that they share a subject matter serious enough to reward sustained, mutual engagement. A biolinguistics adequate to that subject matter is necessarily a broad church; it must be sufficiently rigorous to satisfy the formal demands of CLN and sufficiently flexible to incorporate functional and contextual sensitivities of CLB.

The mental side, however, is distinctly bio-cognitive—hence, biolinguistics must be cognitive linguistics. The same cannot be said for other approaches to language, so “cultural-historical factors are not within the purview of the biology of grammar.” This is a quote from Zaefferer and Poeppel, who rightly note immediately preceding:

Although the biolinguistic axiom that “nothing in language makes sense except in the context of the biology of grammar” (Boeckx & Grohmann 2007: 3) is not necessarily incompatible with explanations based on cultural-historical factors, given that culture and history need a biological basis to live on, such a view would trivialize the notion ‘biolinguistics’ (cf. Koster 2009: 92).

(Zaefferer & Poeppel 2010: 161)

The single greatest empirical bottleneck facing biolinguistics in the coming decade is not theoretical but one of bridging: We still lack a

detailed account of how the computational primitives posited by linguistic theory are implemented in neural tissue—the granularity mismatch between the units of grammar and the units of neuroscience (Poeppel 2017, discussed especially by Hornstein 2009; 2024). Closing that gap, rather than adjudicating between CLN and CLB at the level of theory, is where the decisive evidence will come from.

As Poeppel (2017) observes, this is not simply a matter of better measurement resolution; it is a conceptual challenge, as the theoretical vocabulary of generative linguistics and the theoretical vocabulary of neuroscience are not commensurate. The most productive direction for the foreseeable future is likely the development of intermediate-level processing theories (parsing models, neural oscillation frameworks, computational accounts of production) that are specific enough to make simultaneously testable predictions at both levels. This is where formal precision from CLN and sensitivity to real-time processing from CLB converge most directly.

One proposal worth registering as a horizon question is Navarro's (2021) suggestion that the five foundational questions be supplemented by a sixth: *What is the molecular nature of language?* This 'Mapping Problem' ("How does the genetic endowment map onto the neuroanatomical structures that support the faculty of language?") sits at the intersection of Broca's and Wallace/Darwin's problems. Whether it constitutes a genuinely separate question or a refinement of the existing five is itself a question about the field's self-definition.

It is fitting to end where the framework began. Each chapter has opened against the relevant row of Table 2.4; returning here to the table as a whole gives the book its concluding arc and its agenda for the decade ahead: asking which commitments have been most productively engaged, which remain most sharply contested, and where the two traditions have converged or merely talked past one another.

Let us close, then, with Massimo Piattelli-Palmarini, the scholar who gave the name 'biolinguistics' its modern meaning and one of the strongest advocates of interdisciplinarity in the field. The following words are from the preface to his enlightening volume *Language and learning: The debate between Jean Piaget and Noam Chomsky*, which captures one of the first truly biolinguistic events that brought together linguists, psychologists, philosophers, neurologists, and biologists to debate fundamental issues in, for, and about language:

[I]t has become evident that the great interest in the book is accompanied by some frustration among nonprofessionals. Some questions have recurred, such as: “In the end, where does all this lead?” “What is really at stake here?” and “How does that relate to problems in my field?” Quite often, this “field” is simply the human condition.

(Piattelli-Palmarini 1980: xvii)

Piattelli-Palmarini’s three questions are the right ones with which to leave for now. My own complex answer to the first is:

Biolinguistics leads towards a more complete science of language

- (i) that integrates the formal precision of CLN with the functional but also the contextual sensitivity of CLB,
- (ii) that connects the abstract constructs of linguistic theory to the biological structures of the brain and the evolutionary history of the species, and
- (iii) that respects the universality of the human language capacity as well as the diversity of its expression across all individuals, communities, and cultures.

An answer to the second question could be that what is at stake is no less than an understanding of what kind of creatures we are (Chomsky 2016)—for language is not merely one cognitive capacity among others but the capacity that, more than any other, defines the human form of life. And a reasonable answer to the third is that what is at stake here relates to every field that deals with human beings: education, medicine, law, philosophy, psychology, anthropology, artificial intelligence—and yes, as Piattelli-Palmarini put it, simply the human condition. Thus:

Biolinguistics is the scientific study of what makes that condition possible; and it is a field that, as this primer has tried to argue, is richer and more inclusive than any single theoretical tradition can contain.

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INDEX

A

- acquisition
- adaptationism
- adequacy
 - descriptive
 - explanatory *See also* beyond explanatory adequacy
 - observational
- analogy
- ape
- aphasia
- arcuate fasciculus (AF)
- Area X
- artificial grammar learning (AGL)
- autonomy of syntax

B

- basal ganglia
- Basic Property (of language)
- Bayesian
- beyond explanatory adequacy
- bilingualism
- Biolinguistics 2.0
- biolinguistics
- biology
 - comparative
 - ecological developmental (eco-devo)
 - evolutionary developmental (evo-devo)
- birdsong *See also* songbird
- brain
- Broca's area
- Broca's problem
- Brodman Area

C

- canalization
- Cartesian linguistics
- categorization
- Chomsky hierarchy
- cognition
 - social
- cognitive linguistics
- cognitive linguistics in the broad sense (CLB)
- cognitive linguistics in the narrow sense (CLN)
- cognitive system
- comparative method *See* biology, comparative
- competence
- competition model
- computational phenotype
- conceptual-intentional system
- Construction Grammar
- corpus
- creative aspect of language use (CALU)
- creoles
- critical period

D

- D-structure
- Denisovan
- Descartes's problem
- discrete infinity
- displacement
- domain-general
- domain-specialization
- domain-specificity
- dual-stream model

E

E-language
 eco-devo *See* biology, ecological developmental (eco-devo)
 EEG/ERP *See* neuroimaging
 emergentism
 entrenchment
 ethology
 evo-devo *See* biology, evolutionary developmental (evo-devo)
 evolution
 exaptation
 externalization

F

faculty of language
 in the broad sense (FLB)
 in the narrow sense (FLN)
 first factor
 first language
 fMRI *See* neuroimaging
 formal language hierarchy *See* Chomsky hierarchy
 foundational questions, the five
 FOXP2
 frequency effects

G

generative grammar
 gene
 genetics
 government-and-binding
 gradualism
 granularity mismatch problem

H

heritability
 heritage language
 hierarchical structure
 home sign
Homo sapiens (*H. sapiens*)
 human
 Humboldt's problem

I

I-language
 ideal speaker-listener
 imitation
 incommensurability problem
 innateness

input richness *See* poverty of stimulus; richness of the input
 interface conditions
 interfaces
 island constraints

J

joint attention
 judgements
 acceptability
 grammaticality

K

KE family *See also* FOXP2
 knowledge of language,
 tacit
 explicit

L

Labeling
 Labeling Algorithm
 Language Acquisition Device
 language evolution
 language faculty *See* faculty of language
 language of thought
 language-readiness
 Large Language Model (LLM)
 learnability
 lexical item
 lexicon
 Logical Form (LF)

M

macro-cognitive linguistics *See also* CLN
 Marr's levels of analysis
 MEG *See* neuroimaging
 Mentalese
 Merge
 micro-cognitive linguistics *See also* CLB
 Minimal Search
 minimalism
 Minimalist Program *See* minimalism
 modularity
 Modularity Hypothesis
 morphology
 multilingualism
 myelination

N

nativism
Neanderthals
neural oscillations
neural reuse
neuroimaging
niche construction
now-or-never bottleneck

O

ontogeny
oscillations

P

parameters
parsing
performance
Phonetic/Phonological Form (PF)
phonology
pidgin
phylogeny
Plato's problem
pluralist
poverty of stimulus
Primary Linguistic Data
principles
principles-and-parameters
production
productivity

Q

questions, foundational. *See* foundational questions, the five

R

reasoning, analogical *See* analogy
Reboul's reformulation
recursion
representations
 mental
richness of the input *See also* poverty of stimulus
rules
RUNX2

S

S-structure
saltationism
schemas
second factor
second language
self-monitoring
sensorimotor system
sign language *See also* home sign
social cognition *See* cognition, social
songbird *See also* birdsong
speech errors
statistical learning,
Strong Minimalist Thesis
strong sense *See also* biolinguistics
symbolic behavior
syndromes

T

theory of mind
third factor
thought
three factors in language design
Tinbergen's four questions
transformations
triggering

U

un-Cartesian linguistics
Universal Grammar (UG)
universals, language
usage-based approaches

V

variation, linguistic
vocal learning

W

Wallace/Darwin's problem
weak sense *See also* biolinguistics
Wernicke's area

Y

Y-model